Enterprise Credit Decisioning Strategy Simulator

Module 1: Synthetic Application & Risk Modeling Engine v2.0

Business Requirements Document (BRD) - Full System Design Artifact

Author	Andrew R. Goad
Document Type	Full System Specification
Project Intent	Portfolio-grade, governed decision system design artifact
Primary Purpose	Pre-production strategy testing without exposing PII
-- EXECUTIVE SNAPSHOT--	
What This Is	A governed synthetic credit decisioning engine; a pre-production strategy testing environment; and a portfolio-scale simulation framework that enables strategy design without exposing PII.
Why It Matters	This module makes it possible to test affordability, approval, and risk tradeoffs safely before implementation, while clearly showing how structure affects risk.
What This Demonstrates	Decision-system architecture thinking, deep credit and affordability intuition, and the ability to design governed, explainable, audit-ready logic.
Core Insight	Credit risk is not solely a function of borrower profile — it is materially shaped by loan structure, affordability, and macro environment.
Reader Takeaway	This document should read as a full-system design artifact: what will be built, why it is built this way, how it will be governed, and how the logic translates into downstream risk evaluation.

This document defines the business, data, and risk design requirements for a synthetic application engine intended to demonstrate senior-level credit decisioning architecture. It is intentionally written to show not only what will be built, but how the logic is reasoned, structured, bounded, and governed.

1. EXECUTIVE SUMMARY

Module 1 establishes the foundational synthetic application layer for the Enterprise Credit Decisioning Strategy Simulator. Its purpose is to create a **realistic, explainable, and portfolio-scale simulation environment** that enables controlled testing of credit policy, affordability boundaries, segmentation logic, and expected loss tradeoffs **prior to real-world deployment**.

The module is intentionally framed as **decision infrastructure—not a predictive model or reporting layer**. The deliverable is a **governed, modular simulation engine** capable of supporting scenario-based strategy evaluation across multiple lending products, without reliance on sensitive customer data.

Version 2.0 represents a **structured evolution of the core architecture**, introducing enhanced realism, improved risk sensitivity, and a formalized scenario framework that enables **direct, matched comparison of strategy outcomes under controlled conditions**.

The system is designed around a central economic principle:

Credit risk is not solely a function of borrower profile—it emerges from the interaction between borrower characteristics, loan structure, affordability, and macro conditions.

To operationalize this, the system explicitly models the transmission chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

ensuring that structural changes (e.g., pricing shifts, product mix adjustments) propagate through affordability into risk in a transparent and explainable manner. This chain is intentionally designed as a transparent synthetic risk framework, not as a calibrated production default model. The `estimated_pd` field functions as a governed risk proxy that supports scenario testing, segmentation, affordability analysis, and expected-loss comparison.

Key V2.0 design enhancements include:

- **Mortgage realism improvements**, introducing affordability-aware constraints to prevent unrealistic exposure scaling
- **Refined estimated-risk differentiation**, reducing artificial clustering near the estimated-PD proxy governance cap while preserving directional risk ordering.
- **Formal scenario framework**, enabling repeatable, population-controlled strategy comparisons
- **Revolving credit sensitivity enhancements**, restoring the relationship between APR, payment burden, and risk

The module continues to serve as both:

- A **production-aligned system design artifact**, documenting how a governed simulation environment should be structured
- A **teaching and transparency tool**, exposing the logic, constraints, and assumptions underlying synthetic credit modeling

This document defines the **business, data, and risk design architecture** of the system—what is built, how it operates, and why it is structured in this manner—while explicitly distinguishing the engine from:

- A production underwriting system
- A pricing optimization engine
- A behavioral time-series or forecasting model

Instead, it is designed as a **pre-production decisioning framework**, enabling institutions to evaluate strategic tradeoffs in a controlled, auditable, and fully explainable environment.

TABLE OF CONTENTS / DOCUMENT NAVIGATION

2. BUSINESS OBJECTIVE & STRATEGIC VALUE

2.1 PRIMARY OBJECTIVE

Create a **deterministic, portfolio-scale synthetic application simulation engine** that enables controlled, repeatable testing of credit strategy decisions—including approval logic, affordability constraints, product structure, and risk outcomes—across multiple lending products, without reliance on sensitive customer data.

The system is designed to produce **fully explainable, parameter-driven outputs** that support:

- Pre-production strategy evaluation
- Cross-scenario comparability using identical borrower populations
- Transparent interpretation of how structural and policy changes impact risk

2.2 STRATEGIC VALUE

The module delivers strategic value by enabling:

- **Scenario-Based Decision Testing**
  Evaluate how changes in pricing, product mix, affordability constraints, or risk tolerance impact portfolio outcomes under controlled conditions.
- **Structure-to-Risk Transparency**
  Provide clear visibility into how loan structure translates into affordability (PTI), estimated risk, and Expected Loss using a transparent estimated-PD proxy framework.
- **Portfolio Composition Sensitivity Analysis**
  Assess how shifts in product distribution or score segmentation affect portfolio-level performance and stability.
- **Pre-Production Strategy Validation**
  Allow institutions to test and refine credit strategies before deployment, reducing implementation risk.
- **Governed Simulation Environment**
  Ensure all assumptions are parameterized, bounded, and auditable, supporting regulatory and model governance expectations.
- **Reusable Decision Infrastructure**
  Serve as a modular foundation for future extensions, including policy layers, decision strategies, and advanced risk modeling components.

2.3 V2.0 STRATEGIC ENHANCEMENTS

Version 2.0 introduces targeted enhancements that materially expand the system’s strategic utility:

- **Scenario Framework Formalization (Workstream C)**
  Enables matched, row-level comparison across scenarios using consistent borrower populations, eliminating noise from stochastic variation.
- **Improved Estimated-Risk Differentiation (Workstream B)**
  Enhances estimated-PD proxy dispersion, reducing artificial clustering near the governance cap while preserving directional risk ordering in weaker segments.
- **Affordability-Driven Mortgage Realism (Workstream A)**
  Corrects prior exposure scaling behavior, improving the reliability of affordability and risk insights in high-balance products.
- **Revolving Credit Sensitivity (Workstream D)**
  Restores the relationship between APR, payment burden, and risk, enabling more realistic evaluation of pricing and utilization strategies.

These enhancements transition the system from a **static simulation tool** to a **scenario-capable decisioning framework**.

2.4 WHAT THIS MODULE DEMONSTRATES

1. The ability to design a **deterministic, auditable simulation environment** suitable for enterprise decision support
2. A deep understanding of how borrower attributes, product structure, and affordability interact to drive credit risk
3. The construction of a **transparent, forward-looking estimated-risk framework** using explainable, SQL-native logic
4. The ability to **govern, parameterize, and document** a complex system at a level appropriate for business leaders, model validators, and technical stakeholders
5. The evolution from **data generation** to **decision-system architecture**, with explicit support for scenario analysis and strategy testing

3. DESIGN PRINCIPLES

3.1 DETERMINISTIC REPRODUCIBILITY

The system is designed to produce **fully deterministic outputs**, ensuring that identical inputs yield identical results across all runs.

This is achieved through:

- Deterministic pseudo-random generation using hash-based methods (e.g., MD5 tied to application_id and population_id)
- Explicit seeding controls embedded within the parameter framework

This principle enables:

- **Matched, row-level scenario comparison** across different parameter configurations
- **Stable QA and validation outputs**, eliminating run-to-run variability
- **Traceable debugging and audit reproducibility**, supporting governance and model review requirements

Deterministic reproducibility is foundational to positioning the system as a **controlled simulation environment rather than a stochastic generator**.

3.2 PARAMETER GOVERNANCE

All key assumptions, distributions, and constraints are **centrally parameterized**, ensuring that the system can be configured, audited, and extended without modifying core logic.

Design requirements include:

- Parameter blocks located at the top of the module for visibility and control
- Logical validation constraints (e.g., portfolio mix sums to 1.0)
- Bounded ranges to prevent unrealistic or unstable configurations

This enables:

- **Transparent assumption management**
- **Scenario configuration without code changes**
- **Alignment with enterprise governance and audit expectations**

3.3 ECONOMIC COHERENCE

The system is designed such that variables interact in **economically intuitive and directionally consistent ways**, reflecting real-world credit dynamics.

Key relationships include:

- Income influencing exposure capacity and affordability
- Loan structure (APR, term) influencing payment burden
- Payment burden translating into affordability stress (PTI)
- Affordability acting as a primary bridge into the estimated-PD proxy and Expected Loss

The system explicitly models the transmission chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

ensuring that structural changes propagate through the system in a **coherent and explainable manner**.

3.4 BEHAVIORAL REALISM

The engine is designed to replicate **tendencies, not deterministic segmentation**.

Design expectations include:

- Stronger score bands generally exhibit better outcomes, but not universally
- Weaker segments exhibit higher risk, but retain overlap with stronger profiles
- Mixed-signal scenarios (e.g., high income + high PTI, low score + low utilization) must exist

This ensures:

- **Realistic portfolio overlap**
- Avoidance of artificial clustering or segmentation artifacts
- Preservation of meaningful variation for downstream strategy testing

3.5 SEPARATION OF CONCERNS

The system separates:

- **Borrower characteristics** (income, score, behavior)
- **Product structure** (amount, term, APR, payment mechanics)
- **Risk translation** (estimated-PD proxy, LGD, Expected Loss)

This modular design ensures that:

- Changes in one layer (e.g., pricing) propagate transparently through others
- Components remain independently interpretable and testable
- Scenario analysis can isolate the impact of specific decision levers

3.6 SCENARIO-BASED DESIGN

The system is explicitly designed to support **scenario-based analysis**, where different parameter configurations can be evaluated against identical borrower populations.

This requires:

- Consistent population generation anchored to deterministic seeds
- Scenario labeling and persistence (scenario_set_name, scenario_name)
- Comparable output structures for direct evaluation

This principle elevates the system from a **static simulation tool** to a **decision-support framework capable of controlled experimentation**.

3.7 EXPLAINABILITY & AUDITABILITY

All system behavior must be:

- **Fully explainable** through documented logic and parameter definitions
- **Traceable** from input assumptions to final outputs
- **Auditable** without requiring reverse engineering of the implementation

The system is designed such that a reviewer can answer:

- What assumptions drive this output?
- How are variables generated and constrained?
- Why do estimated risk and Expected Loss change under different scenarios?

This principle ensures alignment with:

- Model governance standards
- Regulatory expectations
- Enterprise audit requirements

4. SCOPE

4.1 INCLUDED IN MODULE 1

Module 1 includes the design, generation, and validation of a deterministic synthetic application-level simulation engine. Specifically, it includes:

- **Synthetic application-level population generation**
  Creation of one row per synthetic credit application using deterministic generation methods.
- **Multi-product `product_type` assignment**
  Assignment across Mortgage, Auto Loan, Revolving Line, Personal Loan, and HELOC populations according to governed portfolio mix parameters.
- **Borrower and application characteristic generation**
  Generation of synthetic borrower attributes such as credit score, income, DTI, utilization, delinquency indicators, file maturity, and related risk signals.
- **APR proxy generation with configurable rate environment**
  Generation of product-aware APR proxies using base rate assumptions, borrower risk adjustments, deterministic variation, and configurable `rate_shift_bps`.
- **Monthly payment proxy generation using product-specific methods**
  Calculation of installment-style payment proxies for amortizing products and enhanced APR-sensitive payment logic for revolving credit.

4.1 INCLUDED IN MODULE 1 - CONTINUED

- **Derived affordability metric generation, including PTI**
  Calculation of Payment-to-Income (PTI) as the primary affordability bridge between structure and risk.
- **Transparent estimated-PD proxy framework**
  Generation of explainable estimated_pd values, interpreted as estimated-PD proxy outputs, using score-band baselines, borrower risk signals, affordability effects, product effects, and V2.0 soft-saturation refinements.
- **LGD approximation and Expected Loss calculation**
  Product-aware LGD approximation and Expected Loss calculation using exposure, estimated-PD proxy, and LGD.
- **Scenario framework and comparison architecture**
  Scenario labeling through `scenario_set_name` and `scenario_name`, archive persistence, and matched row-level comparison across controlled scenario runs.
- **Documentation, validation requirements, and QA guidance**
  Embedded SQL documentation, teaching comments, validation queries, and board-ready documentation support.

4.2 EXCLUDED FROM MODULE 1

Module 1 is intentionally bounded. It does **not** include:

- **Production underwriting rules or final approval decisions by strategy**
  The system supports strategy testing but does not constitute an operational underwriting engine.
- **Actual account booking, servicing behavior, performance time series, or observed outcomes**
  The engine simulates application-level structure and risk proxies; it does not generate real booked-account performance histories.
- **Macroeconomic overlays beyond the global rate-shift parameter**
  The configurable rate environment is a sensitivity control, not a macroeconomic forecasting model.
- **Fair lending analysis, bias testing, or regulatory compliance treatment beyond general governance framing**
  The system is designed with governance discipline, but it does not perform protected-class analysis, disparate impact testing, or compliance adjudication.
- **Model estimation from historical data**
  Estimated-PD proxy, LGD, and Expected Loss are generated through transparent simulation logic rather than statistically estimated production models. The system does NOT estimate empirically calibrated default probabilities, perform model fitting, or produce production-ready PD forecasts.
- **Advanced lifetime transition curves, survival modeling, or macro scenario forecasting**
  The system does not model full account lifecycle transitions, vintage curves, loss timing, or survival/hazard-based risk paths.
- **Credit card billing engine functionality**
  Revolving payment behavior is represented through an APR-sensitive proxy designed for strategy simulation, not contractual billing-cycle calculation.
- **Pricing optimization or profitability maximization**
  APR is simulated as an explanatory proxy and scenario lever; the system does not optimize pricing or recommend price points.

4.3 INTERPRETATION BOUNDARIES

Module 1 should be interpreted as a **pre-production decision-support and strategy simulation framework**.

Valid uses include:

- Testing directional sensitivity of credit strategies
- Comparing scenario outcomes under controlled assumptions
- Evaluating product, affordability, and risk interactions
- Demonstrating governed synthetic data generation and explainable risk translation

Invalid uses include:

- Making real customer credit decisions
- Estimating production-grade default probabilities or treating estimated_pd as a calibrated production PD
- Substituting for regulatory model validation
- Drawing fair lending or compliance conclusions
- Treating APR, estimated-PD proxy, LGD, or Expected Loss as calibrated production estimates

The system is designed to answer:

“How would strategy outcomes change under controlled assumptions?”

It is not designed to answer:

“What exact decision should be made for a real applicant?”

5. SYSTEM ARCHITECTURE OVERVIEW

5.1 END-TO-END SYSTEM FLOW

The system follows a **staged, economically coherent architecture**, where synthetic applications are generated and progressively transformed into forward-looking risk outcomes.

The design explicitly mirrors the real-world transmission of credit risk:

Application → Product Structure → APR → Payment → PTI → estimated-PD proxy → Expected Loss

Each stage introduces controlled transformations that preserve explainability while enabling realistic interaction between borrower characteristics, loan structure, and affordability.

<table>
<tr><th colspan="3">Core Module 1 Data Flow</th></tr>
<tr><td>Step 1</td><td>Synthetic Population Generation</td><td>Generate one row per application using deterministic seeding tied to population_id, ensuring reproducible borrower populations.</td></tr>
<tr><td>Step 2</td><td>Product Assignment & Structural Initialization</td><td>Assign product type (Mortgage, Auto, Revolving, Personal Loan, HELOC) and initialize product-conditioned variables (amount ranges, term structures, payment mechanics).</td></tr>
<tr><td>Step 3</td><td>Borrower Attribute Generation</td><td>Generate credit and capacity variables (score, income, DTI, utilization, derogatory indicators) using correlated, bounded distributions.</td></tr>
<tr><td>Step 4</td><td>Pricing Layer (APR Proxy Generation)</td><td>Derive APR using:
• Product base rates
• Risk-based adjustments
• Deterministic noise
• Rate environment parameter (rate_shift_bps)</td></tr>
<tr><td>Step 5</td><td>Payment Translation Layer</td><td>Convert structure into monthly payment using product-specific logic:
• Amortizing formulas for installment products
• APR-sensitive payment proxy for revolving products (V2.0 enhancement)</td></tr>
</table>

Core Module 1 Data Flow		
Step 6	Affordability Layer (PTI Calculation)	Translate payment burden into affordability stress using PTI (Payment-to-Income), the central bridge variable in the system.
Step 7	Risk Translation Layer (Estimated-PD Proxy)	Estimate a synthetic estimated_pd proxy using: • Base score-driven risk • Behavioral multipliers • Structural and affordability-driven adjustments • V2.0 soft-saturation enhancements
Step 8	Loss Translation Layer (Expected Loss)	Calculate Expected Loss using: Expected Loss = Exposure × estimated-PD proxy × LGD

This staged design ensures that **risk emerges from structure and affordability**, not arbitrary assignment.

5.2 DETERMINISTIC GENERATION FRAMEWORK

All stochastic elements in the system are replaced with **deterministic pseudo-random generation**, derived from:

- application_id
- population_id
- deterministic seed labels

This ensures:

- Identical outputs for identical inputs
- Stability across runs
- Traceable variable generation

This framework is foundational for:

- QA reproducibility
- Debugging
- Scenario comparability

5.3 SCENARIO ARCHITECTURE

Version 2.0 introduces a **formal scenario framework**, enabling structured comparison of alternative strategy configurations.

The architecture introduces:

- **scenario_set_name**
  Groups related simulations (e.g., baseline vs stress scenarios)
- **scenario_name**
  Identifies individual parameter configurations within a scenario set
- **Population Anchoring**
  All scenarios within a set share the same deterministic borrower population (via population_id)
- **Output Persistence Layer**
  Results are stored in a way that supports direct row-level comparison across scenarios

This enables:

- **Matched, row-level comparisons** (same borrower, different conditions)
- Isolation of parameter-driven effects
- Elimination of stochastic noise in scenario analysis

The system therefore supports **controlled experimentation**, where outcome differences are attributable solely to structural or policy changes.

5.4 ARCHITECTURAL DESIGN PRINCIPLES IN PRACTICE

The system architecture operationalizes core design principles as follows:

- **Deterministic Reproducibility** → population anchoring and stable generation
- **Separation of Concerns** → staged transformation layers
- **Economic Coherence** → structured transmission from rate to affordability and estimated risk
- **Scenario-Based Design** → formal scenario framework with comparability

This ensures the system functions as a **governed simulation environment**, not a collection of independent calculations.

Key Insight: Structure, Affordability, and Interaction Effects Drive Risk ☆

The same borrower can produce materially different risk outcomes depending on product design, term structure, and pricing conditions. This engine explicitly models the economic transmission path — Rate Environment → APR → Payment → PTI → estimated-PD proxy — ensuring that structural decisions translate into affordability stress and downstream estimated risk in a transparent, explainable, and scenario-comparable manner.

6. PRODUCT BEHAVIOR MATRIX

6.1 PRODUCT AS A STRUCTURAL DRIVER OF RISK

Product design is a **primary structural determinant of affordability, estimated risk, and Expected Loss** within the simulation engine.

Requested amount, term, pricing behavior, and payment mechanics are **explicitly conditioned on product type**, ensuring that:

- Loan structure varies meaningfully across products
- Affordability (PTI) is product-sensitive
- Risk emerges through structural differences rather than borrower attributes alone

This prevents the system from collapsing into a **single-product abstraction**, preserving realism across retail lending domains.

6.2 PRODUCT BEHAVIOR MATRIX

Product	Structure Type	Requested Amount	Term Structure	Payment Behavior	APR Behavior	Affordability Sensitivity	Risk / Affordability Interaction Notes
UNSECURED_PERSONAL_LOAN	Installment / Unsecured	$1k – $50k	24–60 months	Fully amortizing	High (risk-sensitive)	Moderate	Higher APR drives PTI; affordability pressure emerges quickly
AUTO_LOAN	Installment / Secured	$5k – $80k	36–84 months	Fully amortizing	Moderate	Mild to Moderate	Term and amount strongly influence payment burden

Product	Structure Type	Requested Amount	Term Structure	Payment Behavior	APR Behavior	Affordability Sensitivity	Risk / Affordability Interaction Notes
REVOLVING_LINE	Revolving / Unsecured	$1k – $30k limit proxy	N/A	APR-sensitive minimum payment (V2.0)	High	Moderate	APR → payment → PTI → estimated-PD proxy linkage restored (Workstream D)
HELOC	Revolving / Secured	$10k – $250k	120–240 reference horizon	Interest-only proxy	Low-Moderate	Strong	Large exposure but lower immediate PTI; contrasts with mortgage
MORTGAGE	Installment / Secured	$75k – $1.5M+	180–360 months	Fully amortizing	Low	Strong	Exposure constrained by income + PTI (Workstream A)

6.3 V2.0 PRODUCT BEHAVIOR ENHANCEMENTS

Product behavior in V2.0 includes targeted refinements to improve realism and structural coherence:

- **Mortgage Realism (Workstream A)**
  - Exposure scaling constrained by affordability
  - Tail smoothing prevents unrealistic PTI outcomes
  - PTI acts as a governing constraint rather than a passive output
- **Revolving Credit Sensitivity (Workstream D)**
  - Payment proxy now responds to APR
  - Restores critical transmission: APR → Payment → PTI → estimated Risk
  - Enables realistic pricing sensitivity analysis
- **Improved Cross-Product Differentiation**
  - Stronger separation in affordability behavior
  - Reduced artificial overlap in structural characteristics

6.4 CROSS-PRODUCT INTERACTION INSIGHT

As established in the system architecture, risk emerges through the transmission of structure into affordability and affordability into risk.

Within this framework, product design plays a critical role:

- The same borrower profile can produce materially different outcomes across products
- Differences in term structure, pricing behavior, and exposure scaling drive variation in PTI
- These differences translate directly into variation in the estimated-PD proxy

This confirms that:

Product structure is not a secondary attribute—it is a primary driver of affordability, estimated-risk differentiation, and Expected Loss.

6.5 DESIGN IMPLICATIONS

Product-aware modeling enables:

- Realistic portfolio composition analysis
- Strategy testing across product mixes
- Clear attribution of risk differences to structural drivers

This is essential for:

- Scenario analysis
- Risk segmentation
- Pre-production decision evaluation

7A. CORE VARIABLE BEHAVIOR SPECIFICATIONS (BORROWER, STRUCTURAL, & RISK VARIABLES)

The following variable specifications define the intended behavior of the synthetic engine at the field level. Each variable is documented using a consistent structure to ensure the design remains reviewable, implementation-ready, and audit-ready.

In Version 2.0, variables are defined not only as standalone fields, but as components of an interconnected simulation system, where borrower attributes, product structure, pricing, affordability, and risk interact through a governed economic chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

This section therefore preserves the detailed V1.0 field-level design discipline while adding V2.0 enhancements for:

- Scenario metadata and matched comparison support
- Mortgage affordability and tail-smoothing diagnostics
- Revolving APR-sensitive payment behavior
- Estimated-PD proxy soft-saturation and cap-crowding reduction
- Deterministic population traceability

7A.1 PRODUCT TYPE

Business Purpose: Defines the lending domain and acts as the structural driver for requested amount, term, payment method, APR proxy, PTI interpretation, and downstream risk treatment.

Target Behavior: A categorical product assignment must be present on every synthetic application row.

Segment Relationship: Product is not driven by score band; rather, it is assigned using the configurable portfolio mix and then interacts with score and capacity variables downstream.

Distribution Design: Drawn from parameterized portfolio mix weights across Personal Loan, Auto, Revolving, HELOC, and Mortgage.

Outlier Design: Outliers are not created through impossible product assignments, but through within-product variation in requested amount, term, pricing, payment burden, and affordability.

Constraints: Enumerated string field; values restricted to supported product set.

Downstream Impact: Foundational driver for structure, payment burden, PTI, product-specific APR, LGD, estimated-PD proxy behavior, and later decision strategy variation.

7A.1 PRODUCT TYPE - CONTINUED

V2.0 Relevance: Product type remains one of the primary comparison dimensions in scenario analysis. It is also central to Workstream A and Workstream D because mortgage and revolving products received targeted behavioral enhancements.

7A.2 CREDIT SCORE

Business Purpose: Primary underwriting anchor variable and the starting point for segmentation, pricing pressure, and base estimated-PD proxy assignment.

Target Behavior: Clustered rather than uniform; should mimic broad bureau-style bands with meaningful population in prime and near-prime ranges.

Segment Relationship: Acts as the anchor signal that shifts the odds and central tendencies of several other variables, but does not deterministically assign them.

Distribution Design: Generated through weighted score bands with ranges such as 760–850, 700–759, 640–699, 580–639, and below 580. Within-band dispersion should still exist.

Outlier Design: Minimal extreme outlier treatment; the main realism comes from overlap in other variables rather than bizarre score behavior.

Constraints: Defined floor at 500 and ceiling at 850 for Module 1.

Downstream Impact: Drives base estimated-PD proxy, influences APR risk premium, supports segmentation, and shapes directional tendencies in delinquency, derogatory, utilization, and DTI.

V2.0 Relevance: Credit score remains the primary anchor for risk segmentation, but V2.0 strengthens downstream estimated-PD proxy behavior so that weaker score bands remain differentiated without excessive clustering at the estimated-PD proxy cap.

7A.3 ANNUAL INCOME

Business Purpose: Represents borrower earning capacity and supports requested amount sizing, affordability calculations, and manual-review realism.

Target Behavior: Right-skewed distribution with meaningful overlap across score segments.

Segment Relationship: Should have only a mild positive relationship with stronger score bands; high-income weak-credit and moderate-income strong-credit populations must both exist.

Distribution Design: Generated from weighted income bands or a deterministic skewed transform; defined broad ranges from roughly $20k to $300k+ for Module 1.

Outlier Design: High-income weaker-credit profiles and moderate-income prime borrowers should both appear to preserve realism.

Constraints: Positive numeric field; defined floor at approximately $20,000 and configurable high-end cap for Module 1.

Downstream Impact: Feeds DTI reasonableness, requested amount center, payment-to-income ratio, amount-to-income ratio, manual review logic, and portfolio mix realism.

V2.0 Relevance: Income becomes even more important in V2.0 because mortgage exposure generation now incorporates affordability-aware constraints. Income is therefore not just a descriptive borrower attribute; it is a governing capacity variable for high-balance products.

7A.4 DEBT-TO-INCOME RATIO (DTI)

Business Purpose: Measures capacity relative to obligations and serves as a core affordability and policy constraint variable.

Target Behavior: Broad, overlapped distribution rather than clean segmentation.

Segment Relationship: Higher on average in weaker credit segments, but not universally so. Strong-score applicants can still show high DTI; weaker applicants can still have moderate DTI.

Distribution Design: Generated within a bounded practical range, with lower average centers in stronger bands and higher average centers in weaker bands. Suggested global range: 0.05 to 0.70.

Outlier Design: Include a modest share of high-DTI strong borrowers and moderate-DTI weak borrowers to avoid synthetic rigidity.

Constraints: Continuous numeric variable; hard floor 0.05 and hard ceiling 0.70 for Module 1.

Downstream Impact: Used in later approval logic, estimated-PD proxy multiplier logic, and general capacity realism. Also complements PTI by reflecting existing debt burden rather than requested-loan burden alone.

V2.0 Relevance: DTI remains a borrower-capacity risk signal, while PTI remains the requested-credit affordability signal. Keeping both measures distinct supports clearer interpretation of existing obligations versus new requested payment burden.

7A.5 UTILIZATION RATE

Business Purpose: Captures revolving credit stress and serves as an important behavioral indicator of financial pressure.

Target Behavior: Broad range with overlap; should not simply mirror score.

Segment Relationship: Higher utilization is more common in weaker segments, but high utilization must still appear in some strong-credit cases and low utilization must still appear in some weaker-credit cases.

Distribution Design: Continuous bounded variable, typically from 0.00 to 0.95, with lower average centers in stronger segments and higher centers in weaker segments.

Outlier Design: Temporary-balance-style outliers should allow some prime borrowers to show very high utilization, while lower-risk-looking subprime cases should also appear.

Constraints: Continuous numeric field; floor 0.00, cap 0.95.

Downstream Impact: Strong estimated-PD proxy multiplier driver, especially for revolving products; also contributes to realism around mixed-signal applications.

V2.0 Relevance: Utilization remains a critical behavioral risk signal, but revolving product behavior is strengthened in V2.0 through APR-sensitive payment logic, allowing pricing and utilization stress to flow more realistically into PTI and estimated-PD proxy.

7A.6 DELINQUENCY COUNT (12 MONTHS)

Business Purpose: Captures recent payment-performance problems and acts as one of the strongest non-score risk amplifiers.

Target Behavior: Discrete count variable with zero most common overall, but higher counts increasingly concentrated in weaker populations.

7A.6 DELINQUENCY COUNT (12 MONTHS) - CONTINUED

Segment Relationship: Stronger segments designed to be dominated by zeros, while weaker segments designed to show a materially higher probability of one or more delinquencies.

Distribution Design: Defined integer design from 0 to 6, with probability mass shifting upward as score weakens, utilization rises, and derogatory signals worsen.

Outlier Design: Some subprime applications should still have zero recent delinquencies, and rare prime applications should still show isolated delinquency counts.

Constraints: Integer field with bounded minimum of 0 and capped maximum such as 6 for Module 1 simplification.

Downstream Impact: Large estimated-PD proxy multiplier driver, key ingredient in decline reason logic later, and one of the clearest variables for scenario sensitivity testing.

V2.0 Relevance: Delinquency count remains a strong estimated-PD proxy driver, but V2.0's soft-saturation approach improves how strong risk multipliers translate into final estimated-PD proxy without excessive upper-cap crowding.

7A.7 MAJOR DEROGATORY FLAG

Business Purpose: Represents severe historical credit distress that materially changes risk posture and many decision outcomes.

Target Behavior: Rare overall, but increasingly common in weaker credit populations.

Segment Relationship: Strongly correlated with lower score bands and higher-risk profiles, but not mathematically impossible in stronger segments.

Distribution Design: Binary flag. Approximate rates should rise materially from super-prime through deep subprime. Example directional pattern: around 0.5–1% in strongest bands and materially higher rates in weakest bands.

Outlier Design: A very small number of stronger-score derogatory cases should exist to preserve realism and to support mixed-signal manual review scenarios.

Constraints: Boolean field.

Downstream Impact: Strong estimated-PD proxy multiplier, future automatic-decline candidate, and critical scenario contrast variable.

V2.0 Relevance: Major derogatory remains one of the strongest risk-amplifier fields. V2.0 improves final estimated-PD proxy behavior so these cases can remain high-risk without forcing excessive uniformity at the estimated-PD proxy cap.

7A.8 BANKRUPTCY FLAG

Business Purpose: Represents an extreme adverse credit event indicating severe financial distress.

Target Behavior: Very rare across the total population.

Segment Relationship: Highly correlated with the weakest credit segments and generally nested within or near the major-derogatory population.

Distribution Design: Binary flag. Defined design: approximately 20–40% of the major-derogatory population in weaker segments, with very limited presence elsewhere.

7A.8 BANKRUPTCY FLAG - CONTINUED

Outlier Design: Minimal but non-zero presence in near-prime populations helps preserve realism without overstating frequency.

Constraints: Boolean field.

Downstream Impact: Produces significant estimated-PD proxy uplift, often triggers automatic declines in later strategy layers, and creates sharp scenario differentiation.

V2.0 Relevance: Bankruptcy remains an extreme risk signal. V2.0's estimated-PD proxy dispersion improvements help preserve differentiation among severe-risk cases rather than collapsing them all into the same capped estimated-PD proxy value.

7A.9 RECENT INQUIRY COUNT

Business Purpose: Represents recent credit-seeking behavior and adds nuance beyond traditional capacity and performance variables.

Target Behavior: Moderate, somewhat noisy count variable rather than a primary driver.

Segment Relationship: Slightly higher on average in weaker populations, but with enough noise to avoid predictable sorting.

Distribution Design: Discrete count, such as 0 to 6+, with mild upward drift in weaker segments or in more aggressive credit-seeking profiles.

Outlier Design: Strong borrowers with high inquiry counts should appear, especially where other characteristics remain reasonable.

Constraints: Integer field; defined cap of 6 or 7 for simplification.

Downstream Impact: Mild estimated-PD proxy multiplier, useful nuance for manual review design, and helpful for preventing the engine from feeling too mechanically score-driven.

V2.0 Relevance: Inquiry count continues to support mixed-signal realism and prevents the risk framework from relying too heavily on score, DTI, or utilization alone.

7A.10 TRADELINE COUNT

Business Purpose: Measures breadth and structure of credit exposure and serves as a proxy for both file depth and account complexity.

Target Behavior: Moderate tradeline counts designed to be most common; extremes designed to be less frequent.

Segment Relationship: Higher tradeline counts are somewhat more common in stronger and more seasoned populations, but unusually high counts may signal complexity rather than pure strength.

Distribution Design: Defined range from 1 to 25 trades for Module 1. The risk treatment is designed to be mild and U-shaped: very low counts suggest thin file; very high counts suggest complexity or credit appetite.

Outlier Design: Thin-file high-score cases and thick-file lower-score cases should both appear to avoid rigid profiling.

Constraints: Integer field with positive minimum and reasonable upper cap.

Downstream Impact: Contributes to the estimated-PD proxy framework via a mild U-shaped multiplier and adds structural realism not fully captured by DTI or utilization alone.

7A.10 TRADELINE COUNT - CONTINUED

V2.0 Relevance: Tradeline count remains part of the file-depth risk framework and supports richer borrower heterogeneity in matched scenario comparisons.

7A.11 MONTHS SINCE OLDEST TRADE

Business Purpose: Represents file maturity and the depth of observable credit history.

Target Behavior: Risk should decrease as the file matures and then plateau rather than continue improving forever.

Segment Relationship: Older files are more common in stronger and more seasoned populations, but old-file weak profiles and young-file strong profiles should both appear.

Distribution Design: Defined integer range from 6 to 360 months. The estimated-PD proxy impact is designed to be a decaying uplift: very young files carry extra uncertainty, then the effect stabilizes.

Outlier Design: Outliers should preserve realism by allowing young but otherwise strong files and older but still stressed files.

Constraints: Positive integer field.

Downstream Impact: Feeds a decaying file-age multiplier in the estimated-PD proxy framework and complements tradeline count as a second credit-depth signal.

V2.0 Relevance: File age remains a stabilizing risk signal and contributes to the engine's ability to model thin-file and mature-file behavior without deterministic segmentation.

7A.12 REQUESTED AMOUNT - [V2.0 UPDATED — Workstream A]

Business Purpose: Represents the size of the requested exposure and serves as a central bridge between product structure, affordability, estimated risk, and Expected Loss.

Target Behavior: Highly product-conditioned and only loosely centered by income rather than linearly tied to it.

Segment Relationship: Product determines the baseline range; income shifts the center of the range; risk can influence aggressiveness but should not completely determine amount.

Distribution Design: Requested amount is generated in stages:

1. A product-consistent base amount drawn from product-specific ranges
2. A borrower-aware refinement that incorporates:
    a. income capacity
    b. product structure
    c. moderate deterministic variation
3. For mortgage observations, a V2.0 affordability-aware refinement layer is applied to reduce unrealistic exposure scaling.

Outlier Design: High-income small loans, lower-income aggressive requests, and strong-score conservative borrowing should all exist to preserve realism. Mortgage tail observations should still exist, but they should be bounded by plausible affordability behavior.

Constraints: Positive numeric field; product-specific floors and ceilings apply.

Downstream Impact: Direct input to payment burden, PTI, amount-to-income ratio, expected loss, and later policy questions around exposure appetite.

7A.12 REQUESTED AMOUNT - [V2.0 UPDATED — Workstream A] - CONTINUED

V2.0 Enhancement: Mortgage exposure generation now incorporates tail smoothing and PTI-aware dampening. This ensures large mortgage balances remain tied to borrower capacity and do not produce structurally unrealistic PTI concentrations.

7A.13 LOAN TERM MONTHS

Business Purpose: Defines the contractual or modeled repayment horizon and materially influences monthly burden.

Target Behavior: Product-conditioned categorical or numeric term assignment rather than one universal range.

Segment Relationship: Product is the primary driver of term; weaker-credit profiles may tilt slightly toward longer terms in some products, but not deterministically.

Distribution Design: Examples: personal loan 24–60 months; auto 36–84; revolving N/A; HELOC reference horizon 120–240; mortgage 180–360.

Outlier Design: Some shorter-term weak-credit and longer-term stronger-credit cases should still exist within plausible product constraints.

Constraints: Product-specific allowable values; NULL or special handling for revolving products.

Downstream Impact: Critical input to payment calculation, PTI, affordability realism, and structure-driven estimated-PD proxy behavior.

V2.0 Relevance: Loan term remains part of the structural transmission path from product design into payment burden and affordability.

7A.14 BASE APR FOR PAYMENT CALCULATION - [V2.0 UPDATED]

Business Purpose: Represents a realistic pricing proxy used to estimate payment burden before downstream strategy layers are introduced.

Target Behavior: Environment-dependent and product-sensitive; should not be treated as a final production price.

Segment Relationship: Product determines the baseline rate level; score/risk determines the premium; macro rate environment shifts applicable product APRs up or down.

Distribution Design: APR is constructed from:

- product base rate
- risk premium
- deterministic noise
- rate-shift parameter

Then bounded by product-specific floors and ceilings.

Outlier Design: Within a product and score segment, small pricing dispersion should exist so applications do not cluster unrealistically at identical rates.

Constraints: Positive percent or decimal field; bounded by product-specific min/max rules.

Downstream Impact: Directly drives payment burden, PTI, affordability stress, and downstream estimated-risk behavior.

7A.14 BASE APR FOR PAYMENT CALCULATION - [V2.0 UPDATED] - CONTINUED

V2.0 Enhancement: The APR field now plays a more explicit role in scenario analysis because `rate_shift_bps` can be used to test broad pricing environment changes. It also has increased behavioral importance for revolving products due to the V2.0 APR-sensitive payment proxy.

7A.15 MONTHLY PAYMENT PROXY - [V2.0 UPDATED — Workstream D]

Business Purpose: Translates the requested structure into an affordability burden that can influence forward-looking risk.

Target Behavior: Must be product-specific rather than approximated by a crude one-size-fits-all formula.

Segment Relationship: The burden effect varies by product because products amortize differently, revolve differently, and carry different structural economics.

Distribution Design:

- Installment products use an amortizing payment formula.
- HELOC uses an interest-only proxy for the Module 1.
- Revolving products use a minimum-payment-style proxy.

V2.0 Enhancement: Revolving payment behavior now incorporates APR sensitivity. This restores the economic transmission path:

APR → Payment → PTI → estimated-PD proxy

This is a material improvement from a fixed minimum-payment assumption because it allows pricing changes to flow through affordability and into downstream risk.

Outlier Design: Large-balance but manageable-payment structures should coexist with smaller but high-payment structures to preserve structure realism.

Constraints: Positive numeric field; calculation method depends on product.

Downstream Impact: Feeds PTI directly and is the central operational bridge from product structure into estimated-PD proxy.

7A.16 PAYMENT-TO-INCOME RATIO (PTI) - [V2.0 UPDATED]

Business Purpose: Represents the affordability burden of the requested credit relative to borrower income and serves as a major forward-looking stress metric.

Target Behavior: Should vary widely by product, amount, term, APR, and income.

Segment Relationship: PTI is not directly score-driven; it is structure-driven, though score can influence it indirectly through APR and product outcomes.

Distribution Design: Calculated as monthly payment divided by monthly income. Strong structure choices, such as longer terms or lower rates, can materially improve PTI even at higher balances.

Outlier Design: Some large mortgage balances with reasonable PTI and some smaller unsecured exposures with aggressive PTI should both appear.

Constraints: Continuous ratio field, non-negative, with high-end values bounded by generation constraints.

7A.16 PAYMENT-TO-INCOME RATIO (PTI) - [V2.0 UPDATED] - CONTINUED

Downstream Impact: Major estimated-PD proxy multiplier driver and one of the strongest demonstrations of how this engine translates loan structure into forward-looking risk.

V2.0 Enhancement: PTI becomes a more explicitly governed bridge variable in V2.0. Mortgage generation now uses affordability-aware controls to prevent unrealistic PTI inflation, while revolving payment logic now allows APR movement to influence PTI more directly.

7A.17 AMOUNT-TO-INCOME RATIO

Business Purpose: Provides a simple cross-product view of requested exposure size relative to earning capacity.

Target Behavior: Should complement PTI rather than replace it.

Segment Relationship: Varies by product and income; higher ratios are more expected in mortgage and HELOC contexts than in unsecured products.

Distribution Design: Calculated as requested amount divided by annual income. Used mainly as a modest secondary stress signal rather than the primary affordability measure.

Outlier Design: A small number of aggressive high-ratio requests should exist to preserve mixed-signal realism.

Constraints: Continuous non-negative ratio field.

Downstream Impact: Useful secondary estimated-PD proxy input and helpful for exposure appetite interpretation across products.

V2.0 Relevance: Amount-to-income remains useful for interpreting exposure scale, especially after V2.0 mortgage refinements. It helps distinguish large exposure that is income-supported from exposure that creates structural affordability stress.

7A.18 ESTIMATED-PD PROXY - [V2.0 UPDATED — Workstream B]

Business Purpose: Represents a transparent synthetic estimated-risk proxy generated by the simulation engine for scenario testing, segmentation analysis, and Expected Loss calculation. It is not a production-calibrated probability-of-default model.

Target Behavior: Should rise directionally with weaker credit score bands, higher utilization, recent delinquency, major derogatory events, bankruptcy, higher DTI, thin-file behavior, high PTI, and other risk-amplifying characteristics.

Segment Relationship: Score band provides the base estimated-risk anchor, while borrower behavior, file depth, affordability, and product structure modify the final estimated-PD proxy. Stronger score bands should generally show lower estimated risk, but mixed-signal cases must exist.

Distribution Design: The estimated-PD proxy is generated through a transparent multiplier framework:

1. Base estimated-risk proxy assigned by score band
2. Behavioral multipliers applied for utilization, delinquency, derogatory, bankruptcy, DTI, and file depth
3. Structural multipliers applied for affordability and exposure-related stress
4. V2.0 soft-saturation logic applied before the final governance cap

Outlier Design: High-risk observations should exist, but the system should avoid excessive clustering at the maximum governance cap. Strong-score/high-stress and weak-score/low-stress observations should both remain visible.

7A.18 ESTIMATED-PD PROXY - [V2.0 UPDATED — Workstream B] - CONTINUED

Constraints: Positive ratio bounded between 0 and a governed upper cap. The cap is a simulation guardrail, not a calibrated production default-rate ceiling.

Downstream Impact: Core estimated-risk input used to calculate Expected Loss and support downstream strategy testing.

V2.0 Enhancement: V2.0 introduces soft saturation between raw multiplied estimated risk and the final governance cap. This reduces cap crowding, improves differentiation in weaker segments, and preserves monotonic risk behavior without flattening severe-risk observations into a single indistinguishable capped value.

7A.19 ESTIMATED LGD

Business Purpose: Represents the severity component of Expected Loss after default and allows expected loss to differ from estimated-PD proxy alone.

Target Behavior: Simplified and intentionally less complex than estimated-PD proxy in Module 1.

Segment Relationship: Primarily product-driven rather than score-driven in the initial module.

Distribution Design: In Module 1, LGD is assigned as a fixed product-level assumption rather than a row-level varying field. Secured products such as mortgage and HELOC carry lower LGD values, while unsecured products such as personal loan and revolving carry higher LGD values. This simplified structure is intentional to maintain clarity and explainability while still producing meaningful expected loss differentiation.

Outlier Design: Outlier treatment is limited in Module 1 to avoid unnecessary complexity.

Constraints: Positive ratio bounded between 0 and 1.

Downstream Impact: Combines with requested amount and estimated-PD proxy to produce Expected Loss and supports meaningful cross-product risk-return contrasts.

V2.0 Relevance: LGD remains intentionally stable in V2.0 so that improvements in scenario behavior, mortgage realism, revolving sensitivity, and estimated-PD proxy dispersion can be evaluated without introducing unnecessary loss-severity noise.

7A.20 EXPECTED LOSS AMOUNT

Business Purpose: Represents the forward-looking dollar loss proxy associated with the requested exposure.

Target Behavior: Must rise logically with larger exposures, higher estimated-PD proxy, and higher LGD.

Segment Relationship: Not directly segment-assigned; emerges from upstream structural and risk variables.

Distribution Design: Calculated as:

requested amount × estimated-PD proxy × LGD

Outlier Design: Large exposures with manageable risk and smaller exposures with extreme risk should both appear, preserving realistic contrasts.

Constraints: Positive numeric output field.

Downstream Impact: Key KPI for downstream strategy testing, tradeoff analysis, portfolio comparisons, and scenario evaluation.

7A.20 EXPECTED LOSS AMOUNT - CONTINUED

V2.0 Relevance: Expected Loss becomes more valuable in V2.0 because scenario outputs can be archived and compared on matched borrower populations, allowing changes in EL to be attributed to defined scenario levers.

7B. SYSTEM-LEVEL, SCENARIO, & DIAGNOSTIC VARIABLES (V2.0 ENHANCEMENTS)

The variables in Section 7A define the borrower-level and structural behavior of the simulation engine.

The variables in this section extend the system into a **scenario-capable, reproducible, and diagnostically transparent framework**.

These fields operate at the **system, scenario, or intermediate-calculation level** and are explicitly **not part of borrower-level modeling or underwriting logic**.

They function as **system controls, metadata, and diagnostic outputs**, and are essential for:

- Scenario comparison and attribution (7B.1.#)
- Deterministic reproducibility (7B.2.#)
- Validation transparency & workstream-level diagnostics (7B.3.#)

7B.1 SCENARIO COMPARISON & ATTRIBUTION

7B.1.1 SCENARIO SET NAME - [V2.0 NEW — Workstream C]

Business Purpose: Groups related simulation runs into a common comparison family.

Target Behavior: Constant across all rows within related scenarios that are intended to be compared.

Segment Relationship: Not borrower-dependent. This is a system-level metadata field.

Distribution Design: Static string assigned through the scenario configuration framework. Example values may identify a baseline-versus-stress family or a specific strategy-testing package.

Outlier Design: Not applicable.

Constraints: Non-null string field when scenario archive or comparison functionality is used. Should remain consistent across scenarios intended to be compared.

Downstream Impact: Enables grouped reporting, scenario archive organization, and matched comparison logic across multiple scenario runs.

7B.1.2 SCENARIO NAME - [V2.0 NEW — Workstream C]

Business Purpose: Identifies the specific parameter configuration applied to a given simulation run.

Target Behavior: Constant across all rows within a scenario; varies across scenarios within a scenario set.

Segment Relationship: Not borrower-dependent. This is a scenario label, not a borrower feature.

Distribution Design: Static string assigned at runtime. Examples may include `BASELINE`, `RATE_UP_100BP`, `TIGHTER_CREDIT_MIX`, or similar strategy names.

7B.1.2 SCENARIO NAME - [V2.0 NEW — Workstream C] - CONTINUED

Outlier Design: Not applicable.

Constraints: Should be unique within a scenario set so that comparison outputs remain interpretable.

Downstream Impact: Enables side-by-side scenario reporting, row-level matched comparison, and strategy outcome attribution.

7B.1.3 SCENARIO ARCHIVED TIMESTAMP - [V2.0 NEW — Archive Persistence]

Business Purpose: Captures when a scenario output was persisted to the scenario archive.

Target Behavior: Populated consistently for archived scenario outputs.

Segment Relationship: Not borrower-dependent. This is audit metadata.

Distribution Design: Timestamp assigned when archive persistence occurs.

Outlier Design: Not applicable.

Constraints: Should be non-null for archived records and should not be used as a modeling input.

Downstream Impact: Supports audit traceability, run lineage, and scenario archive review.

7B.2 DETERMINISTIC REPRODUCIBILITY

7B.2.1 POPULATION ID - [V2.0 NEW / ELEVATED — Deterministic Reproducibility]

Business Purpose:
Anchors deterministic population generation and enables identical borrower populations to be regenerated or compared across scenarios.

Target Behavior:
Constant for a given synthetic population.

Segment Relationship:
Not borrower-dependent; however, it governs the deterministic randomization stream that creates borrower and product attributes.

Distribution Design:
Static identifier incorporated into hash-based generation logic.

Outlier Design:
Not applicable.

Constraints:
Must remain unchanged when scenarios are intended to compare different parameter settings against the same borrower population. Should be intentionally changed only when the user wants to create a different synthetic population.

7B.2.1 POPULATION ID - [V2.0 NEW / ELEVATED — Deterministic Reproducibility] - CONTINUED

Downstream Impact:
Enables:

- Reproducibility
- Debugging
- QA stability
- Matched row-level scenario comparison

V2.0 Implementation Note (Deterministic Seed Framework):

Deterministic reproducibility in the system is governed not only by the `population_id`, but also by **run-level configuration inputs** such as seed labels or run identifiers used during execution.

These controls:

- Are **not persisted as row-level variables** in the final dataset
- Operate at the **execution / configuration layer** of the system
- Influence the deterministic hashing process that generates synthetic values

When used consistently, they ensure that:

Identical populations can be regenerated and compared across scenarios without stochastic drift

This design allows the system to maintain:

- Clean output datasets (no unnecessary metadata fields)
- Strong reproducibility guarantees
- Clear separation between **data outputs** and **execution controls**

7B.3 VALIDATION TRANSPARENCY & WORKSTREAM-LEVEL DIAGNOSTICS

7B.3.1 RAW ESTIMATED-PD PROXY BEFORE SATURATION - [V2.0 NEW — Workstream B Diagnostic]

Business Purpose: Captures the unsaturated estimated-PD proxy value before V2.0 soft-saturation logic is applied.

Target Behavior: May exceed the desired final dispersion range or approach/exceed the hard governance cap in high-risk cases.

Segment Relationship: Highest values should concentrate in weaker score bands and among borrowers with severe risk amplifiers, but some variation should remain across all bands.

Distribution Design: Derived from base estimated-PD proxy and risk multipliers before the saturation layer.

Outlier Design: Extreme raw values are expected and useful diagnostically; they should not necessarily become final estimated-PD proxy values one-for-one.

Constraints: Diagnostic field; should be interpreted as an intermediate risk signal, not the final estimated-PD proxy output.

Downstream Impact: Supports validation of the estimated-PD proxy dispersion improvement by showing how much risk pressure existed before soft saturation.

7B.3.2 ESTIMATED-PD PROXY SATURATION ZONE - [V2.0 NEW — Workstream B Diagnostic]

Business Purpose: Classifies whether an observation falls into the estimated-PD proxy soft-saturation region.

Target Behavior: Most low- and moderate-risk observations should not fall into the high-saturation zone. Higher-risk observations should increasingly enter saturation zones as raw estimated-PD proxy rises.

Segment Relationship: Saturation-zone membership should increase in weaker score bands and severe-risk populations.

Distribution Design: Derived categorically from the raw estimated-PD proxy level relative to saturation thresholds.

Outlier Design: Severe-risk cases may enter the highest saturation zone, but this should not collapse all high-risk records into identical final estimated-PD proxy values.

Constraints: Diagnostic categorical field.

Downstream Impact: Supports validation that cap crowding reduction is functioning as intended.

7B.3.3 ESTIMATED-PD PROXY SATURATION FACTOR - [V2.0 NEW — Workstream B Diagnostic]

Business Purpose: Represents the dampening or transformation factor applied during soft saturation.

Target Behavior: Minimal impact for observations below the saturation threshold and increasing impact for observations above the threshold.

Segment Relationship: Stronger impact should occur more often among weaker credit and severe-risk observations.

Distribution Design: Deterministic calculated factor based on raw estimated-PD proxy and saturation logic.

Outlier Design: The factor should prevent runaway raw estimated-PD proxy values from mechanically collapsing into the governance cap.

Constraints: Numeric diagnostic field; should remain within defined logical bounds.

Downstream Impact: Provides transparency into how the raw estimated-PD proxy was transformed into the post-saturation estimated-PD proxy.

7B.3.4 ESTIMATED-PD PROXY AFTER SATURATION - [V2.0 NEW — Workstream B Diagnostic]

Business Purpose: Captures the estimated-PD proxy value after soft saturation but before or as part of final bounded estimated-PD proxy treatment.

Target Behavior: Should remain directionally consistent with raw estimated-PD proxy while reducing excessive clustering near the cap.

Segment Relationship: Should preserve higher average values in weaker score bands, but with improved dispersion.

Distribution Design: Derived from raw estimated-PD proxy and saturation factor.

Outlier Design: High-risk observations should remain high risk, but not all severe-risk cases should become indistinguishable.

Constraints: Numeric diagnostic or intermediate output, bounded according to estimated-PD proxy framework rules.

Downstream Impact: Supports validation of Workstream B by demonstrating improved estimated-PD proxy differentiation.

7B.3.5 MORTGAGE AFFORDABILITY DIAGNOSTIC FIELDS - [V2.0 NEW — Workstream A Diagnostic Field Family]

This field family includes intermediate values used to validate mortgage tail smoothing and affordability-aware exposure behavior. These may include fields such as:

- mortgage candidate amount
- mortgage provisional monthly payment
- mortgage provisional PTI
- mortgage PTI dampener zone
- mortgage PTI dampener factor
- mortgage requested amount after PTI dampener

Business Purpose: Provide transparent diagnostics showing how mortgage requested amount is refined to reduce unrealistic high-PTI tail behavior.

Target Behavior: Mortgage observations with elevated provisional PTI should receive stronger dampening than observations with manageable provisional PTI.

Segment Relationship: Not score-driven directly. These diagnostics are driven primarily by the interaction of requested amount, income, APR, term, and provisional payment burden.

Distribution Design: Derived during the mortgage generation process as intermediate audit fields.

Outlier Design: Large mortgage candidates may still exist, but affordability-aware dampening should reduce structurally unrealistic tail behavior.

Constraints: Diagnostic fields; should not be interpreted as independent model inputs unless explicitly surfaced in final output design.

Downstream Impact: Supports validation of Workstream A by demonstrating that mortgage exposure realism is governed through affordability-aware logic rather than arbitrary hard clipping.

7B.3.6 REVOLVING APR-PAYMENT SENSITIVITY DIAGNOSTIC - [V2.0 NEW / UPDATED — Workstream D]

Business Purpose: Demonstrates that revolving payment behavior responds to APR movement rather than remaining fixed as a flat percentage-only proxy.

Target Behavior: For revolving products, higher APR should generally contribute to higher payment burden, all else equal.

Segment Relationship: Not directly score-driven, although score and risk influence APR, which then influences payment.

Distribution Design: Derived through APR-sensitive payment proxy logic.

Outlier Design: Some high-APR observations may still show manageable PTI if exposure or income offsets the burden; some lower-APR observations may show elevated PTI if balance relative to income is high.

Constraints: Product-specific; relevant primarily to Revolving Line observations.

Downstream Impact: Supports validation of the V2.0 transmission chain:

APR → Payment → PTI → estimated-PD proxy

7C. CROSS-VARIABLE INTERACTION FRAMEWORK

While each variable is defined independently above, the system is designed such that risk emerges through interaction effects across variables, not isolated inputs.

Key interaction pathways include:

- **Income → Requested Amount → Payment → PTI**
- **APR → Payment → PTI → estimated-PD proxy**
- **Product Type → Term / APR / Payment Logic → Affordability → Risk**
- **Score Band → Base estimated-Risk proxy / APR Premium / Behavioral Risk Signals → Final estimated-PD proxy**
- **Requested Amount × estimated-PD proxy × LGD → Expected Loss**

This interaction framework ensures that:

- Structural decisions translate into affordability stress
- Affordability influences forward-looking risk
- Product behavior creates meaningful portfolio differentiation
- Scenario changes propagate through a consistent and explainable chain
- Matched scenario comparisons isolate parameter-driven effects rather than population noise

This is the core V2.0 evolution of Section 7: variables are not merely generated fields; they are governed components of a scenario-capable decisioning simulation framework. This interaction framework is what enables the system to support **scenario-based decision testing with controlled, explainable outcome variation**.

8. PAYMENT, AFFORDABILITY, & RISK TRANSLATION FRAMEWORK

8.1 PURPOSE

This section defines how Module 1 translates product structure, pricing assumptions, borrower capacity, and payment burden into forward-looking affordability and risk outputs.

The purpose of this framework is to ensure that synthetic credit risk is not assigned arbitrarily. Instead, downstream risk outcomes are generated through an explainable economic chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

This design ensures that affordability, estimated risk, and Expected Loss emerge from the interaction of borrower characteristics and product structure, rather than from isolated or independent variable assignment.

This section builds directly on the variable behavior specifications in Section 7, where product type, requested amount, APR, payment proxy, PTI, estimated-PD proxy, LGD, and Expected Loss are defined as interconnected system components.

8.2 DESIGN OBJECTIVE

The payment and affordability framework is designed to support four objectives:

1. **Translate product structure into payment burden**
   Requested amount, term, APR, and product type should produce distinct payment profiles.
2. **Translate payment burden into affordability stress**
   Payment-to-Income ratio (PTI) should serve as the primary affordability bridge between product structure and borrower capacity.
3. **Translate affordability stress into estimated risk**
   PTI and related structural variables should influence the estimated-PD proxy through transparent, bounded multiplier logic.
4. **Translate estimated risk into loss exposure**
   Expected Loss should reflect the combined effect of exposure, estimated-PD proxy, and LGD.

This framework supports scenario testing by allowing controlled changes in pricing, product structure, rate environment, or affordability assumptions to propagate through the system in a transparent and reviewable manner.

8.3 CORE TRANSMISSION CHAIN

Module 1 is organized around a controlled economic transmission chain:

Stage	Description	Output
1. Product structure	Product type determines amount range, term behavior, payment method, APR treatment, and LGD assumption	Product-specific structural profile
2. Pricing layer	APR proxy is generated using product rate assumptions, borrower risk characteristics, deterministic noise, and rate-shift parameters	APR for payment calculation
3. Payment layer	Requested amount, APR, term, and product method are translated into monthly payment proxy	Monthly payment proxy
4. Affordability layer	Monthly payment is compared to monthly income	PTI
5. Estimated-risk layer	Score band, behavioral signals, product effects, and affordability stress are translated into estimated-PD proxy	Estimated-PD proxy
6. Loss layer	Exposure, estimated-PD proxy, and LGD are combined	Expected Loss

This staged structure preserves auditability. Each layer has a defined role, a clear input/output relationship, and a bounded interpretation.

8.4 PRODUCT-SPECIFIC PAYMENT DESIGN

Payment behavior must be product-aware. A single generic payment formula would distort affordability comparisons because consumer credit products do not impose payment burden in the same way.

Module 1 therefore uses product-specific payment logic:

Product	Payment Design	Business Rationale
Mortgage	Amortizing payment proxy using mortgage term and APR	Captures large-balance secured installment burden
Auto Loan	Amortizing payment proxy using shorter installment terms	Captures structured repayment on secured vehicle credit
Unsecured Personal Loan	Amortizing payment proxy using shorter installment terms and higher APR	Captures unsecured installment burden and higher pricing
HELOC	Interest-only proxy for Module 1	Captures lower immediate payment burden relative to amortizing mortgage
Revolving Line	APR-sensitive minimum-payment proxy	Captures simplified revolving payment burden without becoming a billing engine

The intent is not contractual precision. The intent is to create realistic, directionally correct payment burden suitable for strategy simulation.

8.5 REVOLVING PAYMENT SENSITIVITY – V2.0 ENHANCEMENT

In V2.0, revolving payment logic was enhanced to include APR sensitivity.

Prior versions represented revolving payment primarily as a fixed minimum-payment proxy. This produced stable behavior but limited the system’s ability to transmit rate changes through revolving payment burden.

V2.0 introduces a blended revolving payment proxy that includes:

- a base minimum-payment component;
- an APR-sensitive interest-pressure component;
- bounded behavior to avoid unrealistic payment spikes;
- preservation of simplified product treatment.

This enhancement restores the intended transmission path:

APR → Payment → PTI → estimated-PD proxy

Important Boundary

The revolving payment proxy is not a full credit-card billing or servicing model.

It does not model:

- statement balances;
- contractual minimum-payment hierarchies;
- transactor versus revolver behavior;
- fees;
- grace periods;
- promotional APRs;
- daily interest accrual;
- or full balance amortization.

This boundary is intentional. Module 1 is a pre-production strategy simulation engine, not a credit-card servicing system.

8.6 PTI AS THE CENTRAL AFFORDABILITY BRIDGE

Payment-to-Income ratio (PTI) is the primary affordability bridge in Module 1.

PTI is calculated conceptually as:

Monthly Payment Proxy ÷ Monthly Income

PTI is important because it converts product structure into borrower-level affordability stress.

A borrower's credit score alone should not determine affordability. Instead, affordability should emerge from the interaction of:

- income;
- requested amount;
- product type;
- APR;
- term;
- payment method;
- and borrower capacity.

This is why PTI is treated as a major explanatory link between loan structure and downstream estimated risk.

8.7 MORTGAGE AFFORDABILITY GOVERNANCE – WORKSTREAM A

Mortgage behavior receives specific V2.0 governance because mortgage is the largest-balance product and has the greatest potential to distort affordability tails if not controlled.

V2.0 mortgage realism includes two layered design concepts:

A2 — Amount-to-Income Tail Smoothing

This layer evaluates mortgage candidate amount relative to borrower income and applies deterministic tapering in the upper exposure tail.

Its purpose is to reduce unrealistic large-balance outcomes while preserving valid high-capacity mortgage borrowers.

A3 — PTI-Aware Affordability Overlay

This layer evaluates provisional mortgage payment and provisional PTI after tail smoothing.

Its purpose is to apply targeted dampening only where mortgage structure produces excessive affordability pressure.

Together, these layers ensure that mortgage exposure is:

- still product-appropriate;
- still capable of producing large balances;
- tied more clearly to income capacity;
- less likely to create structurally inflated PTI tails;
- and more transparent for validation and review.

The design goal is **governance, not clipping**. Large mortgage observations should remain possible, but they should be explainable.

8.8 ESTIMATED-PD PROXY TRANSLATION

The estimated-PD proxy is the synthetic risk translation output used by Module 1.

It is generated through a transparent multiplier framework rather than a fitted production model.

The estimated-PD proxy is influenced by:

- credit score band;
- utilization;
- delinquency count;
- major derogatory flag;
- bankruptcy flag;
- DTI;
- file depth;
- product type;
- PTI;
- amount-to-income ratio;
- and V2.0 saturation logic.

Boundary

The estimated-PD proxy is not a calibrated production probability-of-default model.

It is designed for:

- relative risk ordering;
- segmentation analysis;
- affordability sensitivity;
- scenario comparison;
- and Expected Loss simulation.

It should not be interpreted as an empirically calibrated default forecast.

8.9 ESTIMATED-PD PROXY GOVERNANCE – WORKSTREAM B

V2.0 introduces soft-saturation logic to improve estimated-PD proxy dispersion in high-risk segments.

The purpose of this layer is to reduce excessive clustering at the governance cap while preserving severe-risk differentiation.

The design requirements are:

- high-risk observations should remain high risk;
- weaker score bands should generally retain higher estimated-PD proxy values;
- severe-risk observations should not collapse unnecessarily into identical capped values;
- the governance cap should remain in place as a guardrail;
- the framework should remain explainable and auditable.

The cap functions as a **simulation guardrail**, not as a calibrated default-rate ceiling.

Workstream B therefore improves how risk pressure is translated into final estimated-PD proxy values without asserting that the upper bound represents an empirical production threshold.

8.10 EXPECTED LOSS TRANSLATION

Expected Loss is the final forward-looking dollar-loss proxy generated by Module 1.

It is calculated conceptually as:

Expected Loss = Exposure × estimated-PD proxy × LGD

Where:

- **Exposure** is represented by requested amount;
- **estimated-PD proxy** represents synthetic estimated risk;
- **LGD** represents product-level severity assumption.

Expected Loss is not assigned directly. It emerges from upstream borrower, product, affordability, and estimated-risk variables.

This design is important because Expected Loss should reflect the interaction of multiple components. A product with low LGD can still produce high Expected Loss if exposure is large. Conversely, a high-LGD product can produce lower average Expected Loss if balances are small.

Expected Loss is intended for:

- scenario comparison;
- product-level contrast;
- portfolio segmentation;
- strategy tradeoff review;
- and educational interpretation.

It is not a production loss forecast.

Expected Loss Interpretation Examples

Example	Exposure	Estimated-PD Proxy	LGD	Expected Loss Interpretation
Large secured mortgage	High	Moderate	Low	Expected Loss can still be high due to exposure
Small revolving line	Low	Moderate / high	High	Expected Loss may remain moderate due to low exposure
Unsecured personal loan	Moderate	Moderate	High	Loss driven by unsecured severity and borrower risk
HELOC	High	Moderate	Low / moderate	Secured exposure creates intermediate loss behavior

8.11 SCENARIO BEHAVIOR & RATE ENVIRONMENT

The payment and affordability framework is scenario-aware.

The configurable rate environment parameter allows scenarios to test how broad pricing shifts propagate through the system.

A rate-up scenario should affect:

- APR;
- monthly payment;
- PTI;
- estimated-PD proxy where affordability thresholds are impacted;
- and Expected Loss.

This supports controlled strategy testing because the same borrower population can be compared across different parameter configurations.

Scenario interpretation should focus on **attribution**:

“What changed because the scenario changed?”

not:

“What changed because a different random population was generated?”

This is why the payment and affordability framework depends on deterministic population anchoring and scenario metadata.

8.12 DESIGN REQUIREMENTS

The payment, affordability, and risk translation framework must satisfy the following requirements:

Requirement	Description
Product-aware payment logic	Payment proxy must vary by product structure rather than applying one generic method
Affordability linkage	Monthly payment must translate into PTI using borrower income
Scenario responsiveness	Rate environment changes must be able to propagate through APR, payment, PTI, estimated risk, and Expected Loss
Mortgage tail governance	Mortgage exposure and PTI tails must remain bounded and explainable
Revolving APR sensitivity	Revolving payment must respond directionally to APR movement without becoming a full billing model
Estimated-risk transparency	Estimated-PD proxy must be generated through explainable, bounded logic
Loss interpretability	Expected Loss must emerge from exposure, estimated-PD proxy, and LGD
Boundary discipline	Outputs must remain framed as synthetic simulation outputs, not production decisions or forecasts

8.13 BUSINESS INTERPRETATION

The payment and affordability framework is the point where the simulator becomes strategically useful.

It allows users to evaluate questions such as:

- How does product structure affect affordability?
- How does a rate shift affect payment burden?
- Which products are most sensitive to APR changes?
- How does PTI influence downstream estimated risk?
- How do exposure, estimated risk, and LGD interact to produce Expected Loss?
- How do scenario outcomes differ when the borrower population is held constant?

This enables controlled decision-system thinking before any production policy is designed or operationalized.

8.14 VALIDATION LINKAGE

Validation evidence for this framework is provided in the companion Validation Summary, including:

- product-level behavior validation;
- score-band behavior validation;
- affordability-to-risk relationship validation;
- workstream validation for mortgage realism, estimated-PD proxy dispersion, scenario framework design, and revolving sensitivity;
- mixed-signal realism validation;
- and edge-case / boundary behavior validation.

The BRD defines the intended design. The Validation Summary demonstrates that the design behaves as intended under the final V2.0 implementation.

8.15 CONCLUSION

The payment, affordability, and risk translation framework is central to Module 1.

It ensures that:

- product structure drives payment behavior;
- payment behavior drives affordability;
- affordability contributes to downstream estimated risk;
- estimated risk and LGD combine with exposure to produce Expected Loss;
- scenario changes propagate through a controlled and explainable chain;
- and model boundaries remain clear.

This framework is what allows Module 1 to operate as a governed, scenario-capable synthetic decisioning simulator rather than a static synthetic data generator.

9. SCENARIO FRAMEWORK & COMPARISON ARCHITECTURE

9.1 PURPOSE

This section defines the scenario architecture used by Module 1 to support controlled comparison of alternative strategy assumptions.

The purpose of the scenario framework is to allow users to evaluate how changes in model assumptions, product structure, rate environment, or future decision strategy parameters affect portfolio outcomes while holding the synthetic borrower population constant.

This capability transforms Module 1 from a static synthetic data generator into a **scenario-capable decisioning simulation framework**.

The central scenario question is:

What changes in portfolio outcomes when the assumptions change, while the borrower population remains the same?

This is critical because without matched scenario comparison, outcome differences can be difficult to interpret. Differences may reflect actual strategy or assumption changes, or they may simply reflect random population variation. Module 1 is designed to eliminate that ambiguity.

9.2 DESIGN OBJECTIVE

The scenario framework is designed to support five primary objectives:

1. **Preserve matched borrower populations**
   Scenarios intended for comparison should use the same `population_id` so that borrower-level attributes remain aligned across runs.
2. **Isolate parameter-driven effects**
   Outcome differences should be attributable to controlled changes in assumptions, not stochastic variation.
3. **Enable row-level attribution**
   The framework should support comparison of the same synthetic application across multiple scenarios.
4. **Support portfolio and segment-level scenario summaries**
   Users should be able to compare outcomes by portfolio, product, score band, and product × score band.
5. **Preserve auditability and scenario lineage**
   Scenario outputs should be labeled, archived, and traceable to a scenario family and specific scenario name.

This framework enables controlled experimentation without requiring real customer data.

9.3 CORE SCENARIO DESIGN CONCEPTS

Module 1 introduces several scenario-design concepts to support governed comparison.

Scenario Concept	Purpose	Design Interpretation
scenario_set_name	Groups related scenario runs into a comparison family	Used to define which scenarios are intended to be compared together
scenario_name	Identifies the specific scenario or assumption set	Used to label baseline, rate-up, stress, or alternative strategy runs
population_id	Anchors deterministic borrower generation	Used to preserve the same synthetic borrower population across comparable scenarios
rate_shift_bps	Applies a controlled rate environment shift	Used to test how rate changes flow through APR, payment, PTI, estimated risk, and Expected Loss
archive_reset_mode	Controls scenario archive replacement or reset behavior	Used to manage repeatable scenario output persistence
scenario archive	Stores completed scenario outputs	Enables portfolio, segment, and row-level comparison across runs
matched row comparison	Compares the same application across scenarios	Supports attribution of outcome differences to assumption changes

These concepts are intentionally separated. Scenario names identify **what is being tested**; `population_id` identifies **who is being tested**.

9.4 SCENARIO SET NAME

Business Purpose

`scenario_set_name` identifies a family of related scenario runs that are intended to be compared.

Target Behavior

All scenarios within the same scenario set should represent related analytical tests. Examples may include:

- baseline versus rate-up scenario,
- baseline versus tighter credit mix,
- baseline versus alternative product mix,
- baseline versus affordability-threshold adjustment.

Design Requirement

`scenario_set_name` should remain consistent across runs that belong to the same comparison family.

Business Interpretation

A scenario set allows reviewers to understand the comparison context. It answers:

Which scenarios belong together?

Without a scenario set, scenario runs may be difficult to organize, compare, or audit.

9.5 SCENARIO NAME

Business Purpose

`scenario_name` identifies the specific scenario or assumption configuration applied within a scenario set.

Target Behavior

Each scenario should have a clear, interpretable name that reflects the scenario being tested.

Examples include:

- `BASELINE`
- `RATE_UP_200BP`
- `TIGHTER_CREDIT_MIX`
- `HIGHER_REVOLVING_SENSITIVITY`
- `ALT_PRODUCT_MIX`

Design Requirement

`scenario_name` should be unique within a scenario set so that outputs remain clearly attributable.

Business Interpretation

The scenario name answers:

What assumption set was applied?

This enables scenario outputs to be reviewed without reverse-engineering parameter values from downstream results.

9.6 POPULATION ID & MATCHED BORROWER DESIGN

Business Purpose

`population_id` anchors the deterministic synthetic borrower population.

Target Behavior

When the user wants direct scenario comparison, `population_id` should remain unchanged across scenarios.

This ensures that:

- the same synthetic applications are generated,
- borrower-level attributes remain identical,
- product and score assignment remain stable where intended,
- and observed differences reflect scenario assumptions rather than population variation.

Design Requirement

Users should only change `population_id` when they intentionally want to generate a different synthetic population.

Business Interpretation

`population_id` answers:

Are we comparing the same borrowers under different assumptions, or a different synthetic population?

This distinction is foundational to scenario analysis.

9.7 SCENARIO ARCHIVE PERSISTENCE

Business Purpose

The scenario archive allows completed scenario outputs to be stored and compared after execution.

Target Behavior

Each archived scenario should retain:

- scenario set name,
- scenario name,
- population ID,
- application-level outputs,
- product and score information,
- affordability metrics,
- estimated-PD proxy,
- LGD,
- Expected Loss,
- and scenario timestamp or archive metadata.

Design Requirement

Archived outputs should support both aggregate and row-level comparison.

Business Interpretation

Scenario archive persistence answers:

Can we compare completed runs without rerunning or manually reconstructing outputs?

This supports repeatability, auditability, and controlled scenario review.

9.8 SCENARIO COMPARISON LEVELS

Module 1 should support scenario comparison at multiple levels of aggregation.

Comparison Level	Purpose	Example Questions
Portfolio-level comparison	Evaluate overall movement in APR, payment, PTI, estimated-PD proxy, and Expected Loss	Did the rate-up scenario increase average payment burden?
Product-level comparison	Evaluate differential scenario impact by product	Which products are most sensitive to rate changes?
Score-band comparison	Evaluate scenario impact by borrower risk tier	Did weaker score bands experience larger estimated-risk movement?
Product × score-band comparison	Evaluate interaction effects across product and borrower risk	Which product/score combinations are most sensitive?
Matched row-level comparison	Compare the same application across scenarios	Which specific rows experienced the largest increase in payment, PTI, estimated-PD proxy, or Expected Loss?

This layered comparison structure allows users to move from executive-level portfolio summaries to targeted diagnostic review.

9.9 SCENARIO TRANSMISSION LOGIC

Scenario changes should propagate through the model only where structurally appropriate.

For example, a rate-up scenario should generally affect:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

However, not every row should necessarily experience the same level of movement. Scenario effects should be:

- directional,
- explainable,
- segment-sensitive,
- product-sensitive,
- and dependent on whether the parameter change affects downstream thresholds.

This is important because a realistic scenario framework should not mechanically move every record by the same amount.

9.10 RATE ENVIRONMENT SCENARIO EXAMPLE

The configurable rate environment is a core scenario lever in Module 1.

A rate-up scenario can be used to evaluate how higher APR assumptions affect affordability and downstream estimated risk.

9.10 RATE ENVIRONMENT SCENARIO EXAMPLE - CONTINUED

Expected directional behavior:

Scenario Effect	Expected Direction
Rate environment increases	APR increases where applicable
APR increases	Payment burden may increase
Payment burden increases	PTI may increase
PTI increases	Estimated-PD proxy may increase for affected rows
Estimated-PD proxy increases	Expected Loss may increase

The purpose of this scenario is not to forecast actual market rate behavior. It is to evaluate sensitivity under a controlled assumption set.

Interpretation Boundary

A rate scenario answers:

What happens to this synthetic portfolio if rate assumptions are changed?

It does not answer:

What will happen to future real-world rates or customer outcomes?

9.11 MATCHED ROW-LEVEL ATTRIBUTION

Matched row-level comparison is one of the most important design features of the V2.0 scenario framework.

When `population_id` remains constant, the same synthetic application can be compared across scenarios.

This allows users to observe:

- baseline APR versus scenario APR,
- baseline payment versus scenario payment,
- baseline PTI versus scenario PTI,
- baseline estimated-PD proxy versus scenario estimated-PD proxy,
- baseline Expected Loss versus scenario Expected Loss.

Business Purpose

Matched row-level attribution helps answer:

Which specific synthetic applications were most sensitive to the scenario change, and why?

This is useful for:

- targeted strategy review,
- sensitivity analysis,
- affordability diagnostics,
- risk translation review,
- and scenario governance.

9.12 SCENARIO GOVERNANCE REQUIREMENTS

Scenario analysis must be governed to remain interpretable.

The following requirements should apply:

Requirement	Rationale
Use clear scenario names	Prevents ambiguity in interpretation
Preserve population ID for matched comparisons	Eliminates population noise
Archive scenario outputs	Enables repeatable comparison and audit review
Document changed parameters	Supports explainability and governance
Review Section 9 QA outputs after scenario changes	Confirms realism and stability
Avoid interpreting scenario outputs as forecasts	Maintains correct model boundary
Use matched row comparisons for attribution	Identifies where and why outcomes changed

These requirements ensure scenario analysis remains controlled, transparent, and useful.

9.13 SCENARIO OUTPUTS & INTERPRETATION

Scenario outputs should be interpreted as **controlled simulation results**, not production forecasts.

Valid interpretations include:

- rate-up scenario increased average payment burden,
- a product segment showed higher sensitivity to rate changes,
- a score band experienced greater estimated-risk movement,
- specific rows showed meaningful Expected Loss increases,
- scenario changes were attributable to parameter settings rather than new population generation.

Invalid interpretations include:

- the scenario predicts future defaults,
- the scenario estimates real-world macroeconomic outcomes,
- the scenario provides production credit policy,
- the scenario outputs should be used for real customer decisions.

This distinction is essential for maintaining the integrity of the framework.

9.14 RELATIONSHIP TO VALIDATION SUMMARY

The companion Validation Summary validates that the scenario architecture functions as designed.

Validation evidence confirms that:

- baseline and rate-stress scenarios each retained 50,000 rows;
- compared scenarios used the same `population_id`;
- borrower-level attributes remained identical across scenario runs;
- scenario outputs changed in an interpretable direction;
- matched row-level comparison was supported;
- and estimated-PD proxy movement was targeted rather than uniformly applied.

The BRD defines the scenario design requirements. The Validation Summary provides evidence that those requirements are met.

9.15 CONCLUSION

The scenario framework is a core V2.0 capability.

It enables Module 1 to support controlled, repeatable, and auditable strategy experimentation. By preserving matched borrower populations and isolating parameter-driven effects, the scenario architecture allows users to evaluate how assumptions affect affordability, estimated risk, and Expected Loss without introducing population noise.

This capability elevates Module 1 from a static synthetic application generator to a **scenario-capable decisioning simulation framework**.

The framework is fit for:

- baseline versus stress comparison,
- rate sensitivity review,
- product and score-band scenario analysis,
- matched row-level attribution,
- pre-production decision framework design,
- and governed strategy testing.

It is not intended to produce forecasts or production decisions. Its purpose is to support controlled experimentation under transparent, documented assumptions.

10. VALIDATION, QA, & DIAGNOSTIC GOVERNANCE FRAMEWORK

10.1 PURPOSE

This section defines the validation, QA, and diagnostic governance framework required to support Module 1 as a controlled, explainable, and audit-ready synthetic decisioning simulation engine.

The purpose of the QA framework is to ensure that outputs are not merely generated successfully, but are:

- structurally valid;
- directionally realistic;
- internally consistent;
- reproducible;
- explainable;
- bounded;
- and appropriate for pre-production strategy simulation.

A successful script run is not, by itself, sufficient evidence that the system is behaving correctly. Module 1 therefore includes a structured QA framework designed to evaluate portfolio construction, distribution behavior, product differentiation, score-band relationships, affordability translation, estimated-PD proxy behavior, scenario comparability, and edge-case realism.

10.2 DESIGN OBJECTIVE

The validation and QA framework is designed to support six objectives:

1. **Confirm structural integrity**
   Validate that product mix, score-band distribution, row counts, and core configuration assumptions are preserved.
2. **Confirm deterministic reproducibility**
   Ensure that identical parameter settings and population controls produce identical outputs.
3. **Confirm behavioral realism**
   Validate that borrower, product, affordability, and estimated-risk relationships behave in economically coherent ways.
4. **Confirm workstream-specific enhancements**
   Validate that V2.0 refinements—mortgage realism, estimated-PD proxy dispersion, scenario framework design, and revolving sensitivity—function as intended.
5. **Confirm tail and edge-case governance**
   Ensure that extreme observations remain present but bounded, explainable, and non-dominant.
6. **Support auditability and reviewer interpretation**
   Provide diagnostic outputs that allow business, risk, validation, and technical reviewers to understand why outputs behave as they do.

This framework ensures that Module 1 remains a governed simulation engine rather than an opaque synthetic-data generator.

10.3 QA FRAMEWORK PRINCIPLES

The QA framework follows the same design philosophy as the engine itself.

QA Principle	Design Meaning
Repeatable	QA outputs must be reproducible under identical configuration and population controls
Layered	QA must evaluate portfolio-level, segment-level, product-level, and row-level behavior
Diagnostic	QA must help identify why outputs behave as observed, not merely report summary metrics
Workstream-aware	QA must directly validate targeted V2.0 enhancements
Boundary-aware	QA must interpret outputs within the system's intended-use boundaries
Evidence-based	Validation conclusions must be supported by observable outputs, not narrative assertion alone

These principles are essential because Module 1 is intended to be modified through parameters and scenario assumptions. QA must therefore help users distinguish between:

- expected scenario movement;
- parameter-driven calibration changes;
- realistic tail behavior;
- and genuine logic or calibration concerns.

10.4 CORE QA LAYERS

Module 1 validation should be organized across several QA layers.

QA Layer	Purpose	Example Review Questions
Record-count validation	Confirms expected population size	Did the run generate the expected number of applications?
Product mix validation	Confirms product assignment aligns to configured assumptions	Does the observed mix match the intended product distribution?
Score-band validation	Confirms borrower segmentation is balanced and usable	Are all score bands adequately represented?
Variable distribution validation	Reviews central tendency, spread, and tails	Are income, requested amount, PTI, and estimated-PD proxy distributions plausible?
Product-level validation	Confirms product-specific behavior	Do products differ appropriately in exposure, APR, payment, PTI, LGD, and Expected Loss?
Score-band profile validation	Confirms risk gradients	Do borrower quality and estimated-risk metrics worsen directionally across score bands?
PTI bucket validation	Confirms affordability-to-risk translation	Do higher PTI buckets show higher estimated-risk and loss outcomes?
Product × score-band validation	Confirms interaction realism	Do product and borrower-risk effects interact coherently?
Mixed-signal validation	Confirms realistic overlap	Do strong-credit/high-PTI and weak-credit/manageable-PTI cases exist?
Scenario validation	Confirms matched scenario comparison	Are row-level differences attributable to parameter changes rather than population drift?
Tail-frequency validation	Quantifies extreme affordability prevalence	Are high-PTI rows present but non-dominant?
Diagnostic trace validation	Reviews intermediate transformation fields	Can mortgage and estimated-PD proxy behavior be explained from intermediate outputs?

This layered structure ensures that validation captures both broad portfolio behavior and targeted diagnostic detail.

10.5 REQUIRED BASELINE QA CHECKS

At minimum, each material run should review the following baseline QA checks.

QA Area	Required Review
Population count	Confirm total application count matches the configured population size
Product mix	Confirm observed product shares align with configured product weights
Score-band mix	Confirm observed score-band shares align with configured score assumptions
Variable profile statistics	Review median, mean, percentiles, and maximum values for core variables
Score-band profile summary	Confirm directional risk and affordability gradients
Product profile summary	Confirm product-specific exposure, APR, payment, PTI, LGD, and Expected Loss behavior
PTI bucket summary	Confirm affordability burden translates into downstream estimated risk
Product × score-band matrix	Confirm interaction effects and risk differentiation
Mixed-signal spot checks	Confirm realistic overlap and non-deterministic borrower behavior
Tail-frequency checks	Confirm high-affordability-stress observations remain controlled
Scenario comparison outputs	Confirm matched scenario comparison where applicable

These checks should be treated as the standard QA workflow for Module 1.

10.6 V2.0 WORKSTREAM-SPECIFIC QA REQUIREMENTS

V2.0 introduces targeted enhancements that require specific validation checks.

Workstream A — Mortgage Realism

QA must confirm that mortgage exposure and affordability tails are governed without eliminating realistic large-balance mortgage behavior.

Required review areas include:

- mortgage candidate amount;
- amount-to-income behavior;
- mortgage tail smoothing factor;
- provisional mortgage payment;
- provisional mortgage PTI;
- PTI dampener zone;
- PTI dampener factor;
- final mortgage requested amount;
- mortgage PTI distribution;
- high-PTI mortgage observations.

Successful behavior means:

- upper-tail exposure is reduced;
- central mortgage behavior remains stable;
- high-PTI cases remain present but explainable;
- mortgage realism improves without broad compression.

Workstream B — Estimated-PD Proxy Dispersion

QA must confirm that estimated-PD proxy behavior remains directionally coherent while reducing excessive cap crowding.

Required review areas include:

- raw estimated-PD proxy before saturation;
- saturation zone;
- saturation factor;
- estimated-PD proxy after saturation;
- final governed estimated-PD proxy;
- capped-row frequency;
- score-band monotonicity;
- high-risk segment differentiation.

Successful behavior means:

- estimated-PD proxy ordering remains directionally correct;
- weaker segments continue to show higher estimated risk;
- high-risk observations remain bounded;
- fewer rows collapse mechanically at the governance cap.

Workstream C — Scenario Framework

QA must confirm that scenario comparison is matched, traceable, and attributable.

Required review areas include:

- scenario set name;
- scenario name;
- archive persistence;
- population ID consistency;
- row counts by scenario;
- borrower attribute matching;
- portfolio-level scenario deltas;
- product and score-band scenario deltas;
- matched row-level comparison outputs.

Successful behavior means:

- compared scenarios use the same synthetic borrower population;
- borrower-level attributes remain aligned;
- output differences are attributable to parameter changes;
- scenario results are interpretable and reproducible.

Workstream D — Revolving Sensitivity

QA must confirm that revolving payment responds directionally to APR movement while preserving the simplified proxy boundary.

Required review areas include:

- revolving APR movement;
- revolving payment movement;
- revolving PTI movement;
- product × score-band deltas;
- rate-up scenario impact;
- row-level sensitivity where applicable.

Successful behavior means:

- revolving payment deltas become positive under rate-up scenarios;
- PTI responds directionally where payment burden changes;
- downstream estimated risk and Expected Loss respond modestly and plausibly;
- the revolving proxy remains bounded and does not become a full billing-cycle model.

10.7 TAIL-FREQUENCY QA

Tail-frequency QA is required because percentile and maximum-value review alone cannot fully describe the prevalence of high-affordability-stress observations.

Module 1 should explicitly measure the frequency of observations exceeding defined PTI thresholds, including:

- PTI > 40%;
- PTI > 50%;
- PTI > 75%.

The purpose is to confirm that high-PTI observations remain:

- present;
- explainable;
- bounded;
- and non-dominant.

High-PTI rows should not be eliminated entirely because realistic portfolios contain affordability-stressed borrowers. However, excessive concentration in high-PTI ranges may indicate unrealistic exposure scaling, payment miscalibration, or insufficient affordability governance.

Tail-frequency QA therefore provides a quantitative complement to distributional statistics and row-level edge-case review.

10.8 MIXED-SIGNAL & EDGE-CASE QA

A realistic synthetic credit engine must preserve borrower overlap and mixed-signal profiles.

QA should confirm that the final population includes examples such as:

- strong-credit borrowers with elevated PTI;
- weaker-credit borrowers with manageable PTI;
- large-exposure borrowers with manageable affordability;
- lower-risk borrowers with isolated adverse characteristics;
- higher-risk borrowers with manageable new-payment burden.

These cases are not validation failures. They are necessary for realism.

The QA objective is to confirm that such observations are:

- plausible;
- traceable;
- non-dominant;
- bounded;
- and consistent with product structure and borrower attributes.

This prevents the system from collapsing into deterministic segmentation.

10.9 SCENARIO QA & MATCHED COMPARISON REVIEW

Scenario QA should confirm that scenario outputs are comparable and interpretable.

When comparing scenarios, reviewers should confirm:

- each scenario is assigned the correct `scenario_set_name`;
- each run has a meaningful `scenario_name`;
- compared scenarios use the same `population_id`;
- row counts match across compared scenarios;
- borrower-level attributes remain identical where intended;
- output differences occur only where parameter changes propagate through the model;
- matched row-level comparison is available for attribution.

Scenario QA should focus on the following question:

Did the scenario change produce explainable outcome differences without introducing population noise?

This QA layer is critical because scenario analysis is only meaningful when population composition is held constant.

10.10 DIAGNOSTIC FIELD GOVERNANCE

Diagnostic fields are intentionally included to support validation, explainability, and debugging.

Diagnostic fields may include:

- mortgage candidate amount;
- mortgage amount-to-income ratio;
- mortgage tail zone;
- mortgage smoothing factor;
- provisional mortgage payment;
- provisional mortgage PTI;
- mortgage PTI dampener zone;
- mortgage PTI dampener factor;
- raw estimated-PD proxy before saturation;
- estimated-PD proxy saturation zone;
- estimated-PD proxy saturation factor;
- estimated-PD proxy after saturation;
- scenario archive metadata;
- scenario comparison deltas.

These fields should be interpreted as intermediate diagnostic outputs, not independent borrower attributes or underwriting inputs.

Their purpose is to help reviewers answer:

- Why did the requested amount change?
- Why did a mortgage observation receive dampening?
- Why did a row approach the estimated-PD proxy cap?
- Why did scenario results differ from baseline?
- Which part of the transmission chain drove the outcome?

This diagnostic transparency is central to auditability.

10.11 QA ACCEPTANCE CRITERIA

A Module 1 run should be considered QA-acceptable when the following conditions are met:

QA Acceptance Area	Acceptance Expectation
Row count	Generated population count matches configured assumptions
Product mix	Product shares align with configured product weights
Score-band mix	Score-band shares align with configured score assumptions
Reproducibility	Identical parameters and population ID produce identical outputs
Product behavior	Products show distinct, economically plausible profiles
Score-band behavior	Borrower and risk metrics move directionally across score bands
Affordability behavior	PTI behaves as a meaningful bridge between payment burden and estimated risk
Estimated-PD proxy behavior	Estimated risk remains directionally coherent, bounded, and differentiated
Expected Loss behavior	Expected Loss reflects exposure, estimated-PD proxy, and LGD interaction
Tail behavior	Extreme observations are present but non-dominant
Mixed-signal realism	Realistic exceptions and overlap remain visible
Scenario comparability	Compared scenarios preserve matched borrower populations
Diagnostic traceability	Intermediate outputs explain final behavior

Failure in any of these areas does not automatically mean the system logic is defective. It indicates that the output requires review, parameter adjustment, calibration refinement, or root-cause analysis.

10.12 QA WORKFLOW FOR USERS

The recommended QA workflow is:

1. **Run the full script using the intended parameters.**
2. **Confirm row count and product mix.**
3. **Confirm score-band distribution.**
4. **Review variable profile statistics.**
5. **Review product-level and score-band summaries.**
6. **Review PTI bucket behavior.**
7. **Review product × score-band interaction matrices.**
8. **Review mixed-signal and edge-case examples.**
9. **Review workstream-specific diagnostics where relevant.**
10. **Review tail-frequency metrics.**
11. **Review scenario archive and matched comparison outputs when scenarios are used.**
12. **Document any unexpected behavior and determine whether it reflects parameter configuration, scenario design, or a true logic concern.**

This workflow supports repeatable review and reduces the risk of relying on isolated summary statistics.

10.13 BOUNDARY-AWARE INTERPRETATION

QA conclusions must remain within the Module 1 boundary.

Validation may conclude that:

- the synthetic population is realistic;
- product behavior is directionally coherent;
- affordability relationships are functioning;
- estimated-PD proxy behavior is bounded and useful for relative risk analysis;
- Expected Loss is useful for scenario comparison;
- scenario deltas are attributable under controlled assumptions.

Validation should not conclude that:

- the engine is ready for production underwriting;
- `estimated_pd` is a calibrated probability of default;
- APR is a production offer price;
- Expected Loss is a production loss forecast;
- the revolving proxy models contractual credit-card billing;
- scenario outputs forecast real macroeconomic outcomes;
- or QA outputs replace formal regulatory model validation.

This boundary discipline preserves the credibility of the system.

10.14 RELATIONSHIP TO COMPANION VALIDATION SUMMARY

The companion Validation Summary provides evidence that the QA and diagnostic framework functions as intended.

The BRD defines:

- what should be validated;
- why each QA layer matters;
- what outputs should be reviewed;
- how diagnostics should be interpreted;
- and what boundaries apply.

The Validation Summary demonstrates:

- portfolio construction stability;
- deterministic reproducibility;
- scenario comparability;
- variable distribution realism;
- product-level behavior;
- score-band behavior;
- affordability-to-risk translation;
- workstream-specific validation;
- mixed-signal realism;
- edge-case governance;
- and final acceptance.

Together, the BRD and Validation Summary form a complete design-and-validation package.

10.15 CONCLUSION

The validation, QA, and diagnostic governance framework is a foundational part of Module 1.

It ensures that the system is not only technically executable, but also reviewable, explainable, reproducible, and fit for governed strategy simulation.

The framework supports:

- regression testing;
- parameter governance;
- workstream validation;
- scenario comparison;
- tail governance;
- diagnostic transparency;
- and audit-ready interpretation.

This QA architecture is what allows Module 1 to remain credible as users modify assumptions, run scenarios, and evaluate strategy tradeoffs.

In short:

The QA framework turns synthetic output generation into governed simulation evidence.

11. PARAMETER GOVERNANCE & CONFIGURATION FRAMEWORK

11.1 PURPOSE

This section defines the parameter governance and configuration framework for Module 1.

Module 1 is intentionally designed as a **parameterized simulation engine**, not a hard-coded static data generator. Key assumptions are centralized near the top of the SQL implementation so that users can modify portfolio size, scenario labels, product mix, score mix, rate environment, exposure bounds, and severity assumptions without rewriting core generation logic.

The purpose of this framework is to ensure that parameter changes remain:

- transparent;
- controlled;
- reproducible;
- auditable;
- explainable;
- and validated through the embedded QA framework.

Parameter governance is central to the system's credibility. Without governed assumptions, scenario results cannot be interpreted reliably.

11.2 DESIGN OBJECTIVE

The parameter framework is designed to support five objectives:

1. **Centralize key assumptions**
   Important configuration values should be visible and adjustable in one controlled parameter layer.
2. **Separate assumptions from logic**
   Users should adjust parameters before modifying downstream SQL logic.
3. **Enable scenario testing**
   Scenario names, population identifiers, and rate-shift assumptions should allow controlled comparison across runs.
4. **Preserve reproducibility**
   Population and deterministic seed controls should allow identical outputs to be regenerated under identical inputs.
5. **Support validation and governance**
   Parameter changes should be checked through validation gates and post-run QA outputs.

This ensures Module 1 remains a governed decisioning simulation framework rather than an informal modeling script.

11.3 CORE PARAMETER CATEGORIES

Module 1 parameters can be grouped into several governed categories.

Parameter Category	Purpose	Example Controls
Run identity parameters	Identify the scenario, scenario set, population, and execution context	`scenario_set_name`, `scenario_name`, `population_id`, archive controls
Population parameters	Define synthetic population size and generation scope	application count, simulation window
Product mix parameters	Define portfolio composition across product types	mortgage %, revolving %, auto %, personal loan %, HELOC %
Score mix parameters	Define credit-score segment composition	super prime, prime, near prime, subprime, deep subprime weights
Rate environment parameters	Define macro / pricing sensitivity assumptions	`rate_shift_bps`
Product amount bounds	Define product-specific exposure floors and ceilings	mortgage min/max, auto min/max, revolving min/max
APR assumptions	Define product-level base pricing and rate behavior	product APR ranges, score/risk adjustments
Payment assumptions	Define product-specific payment mechanics	installment amortization, HELOC interest-only proxy, revolving APR-sensitive proxy
Estimated-risk assumptions	Define estimated-PD proxy baselines, multipliers, saturation, and caps	score-band base risk, behavioral multipliers, governance cap
LGD assumptions	Define product-level severity assumptions	product-level LGD rates
QA / archive controls	Define scenario persistence and validation behavior	archive reset mode, scenario archive population

This structure makes it clear where users should make controlled changes and where downstream logic should remain stable.

11.4 RUN IDENTITY & SCENARIO CONTROLS

Run identity parameters govern how a simulation run is labeled, regenerated, archived, and compared.

Key controls include:

Control	Purpose	Governance Expectation
scenario_set_name	Groups related scenario runs into a comparison family	Should remain consistent across scenarios intended to be compared
scenario_name	Identifies the specific scenario configuration	Should clearly describe the assumption set being tested
population_id	Anchors deterministic borrower generation	Should remain unchanged for matched scenario comparison
archive_reset_mode	Controls archive replacement / clearing behavior	Should be used deliberately to avoid accidental loss or duplication of scenario outputs

These controls are not borrower attributes. They are system-level configuration controls that enable reproducibility, scenario organization, and audit traceability.

Interpretation

`scenario_name` answers:

What scenario is being tested?

`population_id` answers:

Which synthetic population is being used?

This distinction is essential. Changing the scenario name should not necessarily change the borrower population. Changing the population ID intentionally creates a different synthetic population.

11.5 PRODUCT MIX GOVERNANCE

Product mix is a top-of-script portfolio assumption and must be governed carefully.

The product mix controls the relative share of synthetic applications assigned to each supported product type. Because product type materially affects requested amount, APR, payment behavior, PTI, estimated-PD proxy behavior, LGD, and Expected Loss, product mix changes can materially alter portfolio outcomes.

Default Product Mix

Module 1 ships with a representative default mix for a diversified consumer-retail bank portfolio. This default is designed for demonstration and strategy-simulation purposes. It is not intended to represent a universal institution-level truth.

Product	Default Mix
Mortgage	30%
Revolving Line	25%
Auto Loan	20%
Unsecured Personal Loan	15%
HELOC	10%

Governance Requirements

Product mix parameters must satisfy the following requirements:

- Product weights must sum to **100%**.
- Each product should remain within a plausible range for the intended scenario.
- Users should document any product mix changes before comparing outcomes.
- Product mix changes should be reviewed through product-level QA outputs.
- Scenario comparisons should distinguish between product-mix-driven changes and other assumption-driven changes.

Business Interpretation

Product mix changes are not cosmetic. They alter the structural composition of the portfolio and can affect:

- exposure distribution;
- payment burden;
- PTI;
- estimated-PD proxy distribution;
- LGD mix;
- Expected Loss;
- scenario sensitivity.

For this reason, product mix must be treated as a governed assumption.

11.6 SCORE MIX GOVERNANCE

Score mix parameters govern the distribution of borrower credit-quality segments.

Score band composition affects borrower attributes, pricing pressure, estimated-PD proxy behavior, and Expected Loss. It also affects the statistical usability of segment-level validation.

Governance Requirements

Score mix parameters should:

- sum to **100%**;
- preserve adequate density across all score bands;
- avoid unrealistic concentration in one segment unless explicitly testing a stress scenario;
- support downstream score-band validation;
- and be reviewed after material changes.

Business Interpretation

Score mix controls the risk composition of the synthetic population.

Changing score mix can materially alter:

- average credit score;
- income distribution;
- utilization distribution;
- delinquency and derogatory rates;
- estimated-PD proxy distribution;
- Expected Loss;
- and product × score-band matrices.

Score mix changes should therefore be documented as strategy or scenario assumptions, not treated as incidental tuning.

11.7 RATE ENVIRONMENT PARAMETER

The rate environment parameter allows users to evaluate how broad rate shifts affect the portfolio.

The primary rate environment control is:

rate_shift_bps

This parameter allows the system to simulate a rate-up or rate-down environment by shifting applicable APR assumptions.

Purpose

The rate environment parameter supports sensitivity testing across the transmission chain:

Rate Environment → APR → Payment → PTI → estimated-PD proxy → Expected Loss

Governance Requirements

Rate environment changes should be:

- clearly labeled through `scenario_name`;
- archived under a meaningful `scenario_set_name`;
- compared against a matched baseline using the same `population_id`;
- reviewed through scenario comparison outputs;
- and interpreted as controlled sensitivity analysis, not macroeconomic forecasting.

Rate Environment Parameter Reference

Parameter	Meaning	Example / Interpretation
`rate_shift_bps`	Uniform basis-point shift applied to applicable APR assumptions	`0` = baseline; `+200` = 200bp rate-up scenario
Product base APR assumptions	Product-specific baseline rate structure before borrower risk adjustment	Mortgage lower; revolving and unsecured higher
Risk premium logic	Borrower risk and behavioral factors influence APR pressure	Weaker score / higher-risk profiles generally receive higher APR pressure
Deterministic pricing noise	Small deterministic dispersion prevents artificial clustering	Preserves realistic within-segment pricing variation

Boundary

The rate environment parameter is not a macroeconomic forecasting model. It answers:

What happens under this rate assumption?

It does not answer:

What will happen to actual future market rates?

11.8 PRODUCT AMOUNT BOUND GOVERNANCE

Product amount bounds define the minimum and maximum requested amount ranges by product.

These parameters are important because requested amount directly influences:

- exposure;
- payment burden;
- PTI;
- amount-to-income ratio;
- Expected Loss;
- and affordability tail behavior.

Governance Requirements

Product amount bounds should:

- remain product-specific;
- reflect plausible consumer-credit ranges;
- avoid unrealistic exposure scaling;
- preserve meaningful variation within each product;
- and be validated through product-level and tail-frequency QA.

V2.0 Mortgage Governance

Mortgage amount behavior receives additional governance in V2.0 because mortgage is the largest-balance product and can dominate affordability tails if not controlled.

Mortgage amount parameters should be reviewed alongside:

- amount-to-income behavior;
- mortgage tail smoothing;
- provisional mortgage PTI;
- PTI-aware dampening;
- max requested amount;
- and high-PTI tail frequency.

The objective is to govern mortgage tail behavior without removing realistic high-capacity mortgage borrowers.

11.9 APR & PRICING ASSUMPTION GOVERNANCE

APR assumptions are used to estimate payment burden before downstream strategy layers are introduced.

APR in Module 1 is a **burden-estimation proxy**, not a production offer price.

Purpose

APR assumptions support:

- product-level pricing differentiation;
- score/risk-sensitive pricing pressure;
- rate environment sensitivity;
- payment calculation;
- PTI generation;
- and downstream estimated-risk behavior.

Governance Requirements

APR parameters should:

- remain product-specific;
- preserve realistic pricing separation across secured and unsecured products;
- respond to score/risk characteristics where appropriate;
- incorporate rate-shift assumptions in scenario testing;
- and remain bounded within plausible ranges.

Boundary

APR should not be interpreted as:

- production pricing;
- optimized offer rate;
- profitability-maximizing rate;
- compliance-approved pricing;
- or customer-level pricing recommendation.

It is a simulation input used to estimate payment burden and scenario sensitivity.

11.10 PAYMENT ASSUMPTION GOVERNANCE

Payment assumptions determine how requested amount and APR translate into monthly payment burden.

Payment behavior must remain product-aware because products impose repayment burden differently.

Payment Method Governance

Product Type	Payment Method	Governance Interpretation
Mortgage	Amortizing proxy	Long-term secured installment burden
Auto Loan	Amortizing proxy	Medium-term secured installment burden
Unsecured Personal Loan	Amortizing proxy	Shorter-term unsecured installment burden
HELOC	Interest-only proxy	Lower immediate payment burden for Module 1
Revolving Line	APR-sensitive minimum-payment proxy	Simplified revolving burden responsive to APR

V2.0 Revolving Governance

Workstream D introduced APR sensitivity into revolving payment behavior. This allows rate scenarios to flow into revolving payment burden and PTI.

However, revolving payment remains a proxy. It does not model:

- statement balances;
- contractual minimum payment rules;
- fees;
- grace periods;
- promotional APRs;
- transaction behavior;
- or full amortization.

The governance objective is directional sensitivity, not billing-system precision.

11.11 ESTIMATED-PD PROXY PARAMETER GOVERNANCE

The estimated-PD proxy framework converts borrower characteristics, behavioral signals, affordability stress, and structural factors into a governed synthetic risk proxy.

Purpose

The estimated-PD proxy supports:

- relative risk ordering;
- segmentation;
- affordability sensitivity;
- scenario comparison;
- and Expected Loss calculation.

Governance Requirements

Estimated-PD proxy parameters should:

- preserve directional risk ordering by score band;
- respond to behavioral risk signals;
- incorporate affordability stress;
- remain bounded by governed caps;
- avoid excessive upper-bound compression;
- and be validated through score-band, PTI-bucket, product × score-band, and saturation diagnostics.

V2.0 Soft-Saturation Governance

Workstream B introduces soft saturation to improve high-risk dispersion before the final governance cap.

The cap is a simulation guardrail, not a calibrated production default-rate ceiling.

Users should monitor:

- capped observation frequency;
- saturation zones;
- saturation factors;
- score-band monotonicity;
- high-risk segment differentiation.

Boundary

The estimated-PD proxy is not a calibrated probability-of-default model. Parameter changes should not be interpreted as production risk-model calibration unless separately designed, validated, and governed.

11.11 LGD PARAMETER GOVERNANCE

LGD assumptions define product-level severity used in Expected Loss calculation.

LGD is intentionally simpler than the estimated-PD proxy framework in Module 1.

Purpose

LGD supports:

- product-level loss differentiation;
- expected-loss comparison;
- secured versus unsecured contrast;
- and scenario analysis.

Governance Requirements

LGD assumptions should:

- remain product-specific;
- reflect simplified severity distinctions;
- be changed only when testing explicit severity assumptions;
- and be reviewed through Expected Loss summaries.

Boundary

LGD is not a row-level recovery model in Module 1. It is a product-level severity assumption designed for transparent simulation.

11.13 EXPECTED LOSS GOVERNANCE

Expected Loss is calculated as:

Expected Loss = Exposure × estimated-PD proxy × LGD

Because Expected Loss depends on multiple upstream components, parameter changes can affect it through several pathways.

Drivers of Expected Loss

Expected Loss may change because of:

- requested amount changes;
- product mix changes;
- score mix changes;
- APR changes;
- payment changes;
- PTI changes;
- estimated-PD proxy changes;
- LGD assumption changes;
- scenario settings.

Governance Requirements

Expected Loss should be interpreted as a comparative simulation output.

It is appropriate for:

- strategy tradeoff analysis;
- scenario comparison;
- product-level contrast;
- affordability stress review;
- and portfolio segmentation.

It is not a production loss forecast, reserve estimate, capital estimate, or profitability measure.

11.14 VALIDATION GATEKEEPERS

The parameter framework includes validation gatekeepers to prevent invalid or unstable configurations.

Examples include:

Validation Gatekeeper	Purpose
Product mix sum check	Ensures product weights sum to 100%
Score mix sum check	Ensures score-band weights sum to 100%
Positive population count	Prevents empty or invalid population generation
Valid product bounds	Ensures product amount ranges are logically ordered
Valid APR ranges	Prevents impossible pricing values
Valid LGD values	Ensures severity assumptions remain between 0 and 1
Valid estimated-PD proxy cap	Prevents unbounded estimated-risk output
Scenario metadata checks	Ensures scenario outputs are interpretable and comparable

These gatekeepers are intended to fail fast when assumptions are invalid.

11.15 PARAMETER CHANGE GOVERNANCE WORKFLOW

Users should follow a disciplined workflow when modifying parameters.

Recommended workflow:

1. **Define the business question**
   Identify what the change is intended to test.
2. **Determine whether the change is a parameter change or logic change**
   Parameter changes should be attempted before rewriting downstream logic.
3. **Update only the relevant top-of-script parameter values**
   Avoid changing downstream CASE logic unless necessary.
4. **Assign clear scenario labels**
   Use meaningful `scenario_set_name` and `scenario_name`.
5. **Preserve `population_id` for matched comparisons**
   Change it only when intentionally generating a new synthetic population.
6. **Run the script and review QA outputs**
   Confirm product mix, score mix, distributions, product profiles, PTI buckets, and edge cases.
7. **Review scenario comparison outputs if applicable**
   Confirm that differences are attributable to parameter changes.
8. **Document observed behavior**
   Record whether changes behaved as expected.

This workflow supports repeatability, auditability, and clean scenario interpretation.

11.16 CHANGE INTERPRETATION FRAMEWORK

Parameter changes should be interpreted according to the type of assumption being modified.

Parameter Change Type	Expected Interpretation
Product mix change	Changes portfolio composition and product-level weighting
Score mix change	Changes borrower risk composition
Rate shift change	Tests APR, payment, PTI, estimated-risk, and loss sensitivity
Amount bound change	Alters exposure and affordability distributions
APR assumption change	Alters payment burden and rate sensitivity
Payment assumption change	Alters PTI and downstream estimated risk
Estimated-PD proxy parameter change	Alters relative estimated-risk behavior and cap dynamics
LGD change	Alters Expected Loss severity component
Population ID change	Creates a different synthetic population

This framework helps reviewers distinguish between expected output changes and potential validation concerns.

11.17 PARAMETER GOVERNANCE BOUNDARIES

Parameter governance does not mean unrestricted model use.

Parameter changes should not be used to claim:

- production underwriting readiness;
- calibrated default-rate estimation;
- pricing optimization;
- real customer decisioning;
- regulatory model validation;
- macroeconomic forecasting;
- or credit-card billing accuracy.

Any parameter change that moves the system beyond the Module 1 boundary should be treated as a future module or separate validation scope.

11.18 RELATIONSHIP TO VALIDATION SUMMARY

The companion Validation Summary demonstrates that the current V2.0 parameter framework produces stable, realistic, and governed outputs.

Validation evidence confirms:

- product mix aligns with configured assumptions;
- score-band distribution remains stable;
- deterministic population generation preserves reproducibility;
- scenario comparison supports matched attribution;
- mortgage exposure and PTI tails are governed;
- estimated-PD proxy cap crowding is reduced;
- revolving payment responds to APR movement;
- and tail-frequency behavior remains present but non-dominant.

The BRD defines how parameters should be governed. The Validation Summary demonstrates that the final V2.0 configuration behaves as intended.

11.19 CONCLUSION

The parameter governance and configuration framework is essential to Module 1.

It ensures that users can modify assumptions without compromising reproducibility, interpretability, or auditability.

The framework supports:

- controlled scenario testing;
- transparent assumption management;
- matched population comparison;
- validation gatekeeping;
- product and score mix governance;
- rate environment sensitivity;
- affordability and estimated-risk review;
- and Expected Loss comparison.

This framework is what allows Module 1 to remain flexible without becoming uncontrolled.

In short:

Parameter governance turns configuration flexibility into controlled, auditable strategy simulation.

12. OUTPUT DATA MODEL & CONSUMPTION FRAMEWORK

12.1 PURPOSE

This section defines the output data model and consumption framework for Module 1.

Module 1 produces a deterministic synthetic application-level dataset designed to support pre-production strategy testing, scenario analysis, affordability review, estimated-risk proxy evaluation, Expected Loss comparison, and validation diagnostics.

The purpose of the output framework is to ensure that users understand:

- what outputs are produced;
- how outputs are organized;
- which fields are borrower-like attributes versus system metadata;
- which fields are final analytical outputs versus intermediate diagnostics;
- how outputs should be used;
- and which interpretations remain outside the Module 1 boundary.

The output dataset is intended to be analytical, explainable, and auditable. It is not intended to represent real customer data or production decision output.

12.2 DESIGN OBJECTIVE

The output data model is designed to support six objectives:

1. **Application-level analysis**
   Each row represents a synthetic application-level observation suitable for portfolio, segment, product, and scenario review.
2. **Traceable risk translation**
   Outputs should allow users to understand how borrower attributes, product structure, pricing, affordability, estimated-PD proxy, LGD, and Expected Loss relate.
3. **Scenario comparability**
   Outputs should support matched comparison across scenario runs using scenario metadata and deterministic population controls.
4. **QA and validation review**
   Outputs should support embedded SQL QA, distribution checks, product summaries, score-band summaries, PTI-bucket summaries, and edge-case review.
5. **Downstream strategy design**
   Outputs should provide a synthetic foundation for future policy, approval, segmentation, line assignment, or decision strategy modules.
6. **Boundary-aware interpretation**
   Outputs should be interpreted as synthetic simulation results, not real customer outcomes, production decisions, or calibrated default forecasts.

12.3 PRIMARY OUTPUT TABLE

The primary Module 1 output is the synthetic application-level dataset:

credit_decisioning_sim.synthetic_applications

This table is the main analytical output of the engine.

It is designed to support:

- portfolio summaries;
- product-level analysis;
- score-band segmentation;
- affordability analysis;
- estimated-risk proxy review;
- Expected Loss comparison;
- QA validation;
- and scenario comparison.

Each record represents one synthetic application generated under a defined scenario configuration and deterministic population framework.

12.4 OUTPUT FIELD CATEGORIES

The final output dataset should be interpreted through several field categories.

Field Category	Purpose	Examples
Row identity fields	Identify each synthetic application record	application ID, population ID
Scenario metadata	Identify the scenario context for the row	scenario set name, scenario name, archive timestamp
Product structure fields	Define the product and structural loan terms	product type, requested amount, term months
Borrower profile fields	Represent synthetic borrower characteristics	credit score, score band, income, DTI, utilization
Behavioral risk fields	Represent credit-risk signals	delinquency count, derogatory flag, bankruptcy flag, inquiries, tradeline count
Pricing fields	Represent APR assumptions used for burden estimation	base APR for payment calculation, rate-shift effects
Payment and affordability fields	Translate product structure into borrower burden	monthly payment proxy, PTI, amount-to-income ratio
Estimated-risk fields	Represent synthetic estimated risk	estimated-PD proxy and related diagnostic fields
Severity fields	Represent simplified product-level loss severity	LGD
Loss fields	Represent comparative forward-looking loss proxy	Expected Loss amount
Diagnostic fields	Support validation and traceability	mortgage smoothing fields, saturation zones, dampener factors
Archive / comparison fields	Support scenario persistence and comparison	scenario archive metadata, scenario comparison outputs

This categorization helps users distinguish between input-like characteristics, derived outputs, diagnostic fields, and scenario metadata.

12.5 FINAL ANALYTICAL OUTPUTS

The most important final analytical outputs are:

Output	Business Meaning	Interpretation Boundary
requested_amount	Synthetic requested exposure amount	Simulated exposure proxy, not actual booked balance
base_apr_for_payment_calculation	APR proxy used to estimate payment burden	Burden-estimation input, not production pricing
monthly_payment_proxy	Estimated payment burden for affordability analysis	Simplified proxy, not contractual payment
payment_to_income_ratio / PTI	Affordability burden relative to income	Forward-looking affordability stress metric
estimated_pd	Synthetic estimated-PD proxy	Not a calibrated production probability of default
estimated_lgd	Product-level LGD assumption	Simplified severity assumption, not recovery model
expected_loss_amount	Comparative expected-loss proxy	Scenario / strategy comparison output, not production loss forecast

These outputs form the core risk and affordability framework of Module 1.

The core conceptual relationship is:

Expected Loss = Exposure × estimated-PD proxy × LGD

Where exposure is represented by requested amount.

12.6 SCENARIO OUTPUT FIELDS

Scenario outputs are essential to V2.0 because the system supports matched scenario comparison.

Scenario-related fields should allow users to identify:

- which scenario family a row belongs to;
- which specific scenario was executed;
- which population was used;
- when the scenario output was archived;
- and how the row compares to its matched baseline or comparison scenario.

Key scenario output concepts include:

Scenario Output Concept	Purpose
scenario_set_name	Groups comparable scenario runs
scenario_name	Identifies the specific scenario
population_id	Confirms matched synthetic population
scenario archive timestamp	Supports run traceability
baseline values	Preserve comparison reference values where applicable
scenario values	Represent outputs under the comparison scenario
delta fields	Quantify scenario-driven change
row-level match keys	Enable same-row comparison across scenarios

Scenario outputs should be interpreted as controlled simulation results. They support sensitivity analysis, not forecasting.

12.7 DIAGNOSTIC OUTPUT FIELDS

Diagnostic fields are included to support validation, auditability, and root-cause review.

They are not intended to represent borrower attributes or underwriting inputs.

Diagnostic field families include:

Diagnostic Family	Purpose
Mortgage affordability diagnostics	Explain mortgage tail smoothing and PTI-aware dampening
Estimated-PD proxy saturation diagnostics	Explain raw risk pressure, saturation behavior, and cap interaction
Revolving APR-payment sensitivity diagnostics	Validate that revolving payment responds directionally to APR
Scenario comparison diagnostics	Explain row-level and segment-level scenario deltas
Tail-frequency diagnostics	Quantify the prevalence of high-affordability-stress cases

Diagnostic fields help reviewers answer:

- Why was a mortgage amount dampened?
- Why did PTI change?
- Why did a row approach the estimated-PD proxy cap?
- Why did Expected Loss move under a scenario?
- Which segment or product drove the change?

These fields are essential for audit-readiness because they reduce the need to reverse-engineer model behavior from final outputs alone.

12.8 OUTPUT CONSUMPTION USE CASES

Module 1 outputs are designed for controlled analytical use.

Valid consumption use cases include:

Use Case	Description
Portfolio profiling	Review total population, product mix, score mix, and distribution behavior
Product analysis	Compare exposure, APR, payment, PTI, estimated risk, LGD, and Expected Loss by product
Score-band analysis	Evaluate borrower-quality gradients and estimated-risk ordering
Affordability analysis	Review PTI buckets, high-burden cases, and affordability-to-risk behavior
Scenario comparison	Compare matched baseline and stress outputs using the same population
Strategy prototyping	Use synthetic outputs to test future decision rules or segmentation logic
Validation review	Run QA summaries, diagnostic queries, and edge-case checks
Portfolio artifact demonstration	Show governed SQL-based decision-system architecture without exposing PII

These use cases align with Module 1's role as a pre-production synthetic strategy simulation engine.

12.9 INVALID OUTPUT USES

Module 1 outputs should not be used for:

- real customer credit decisions;
- production underwriting;
- production pricing;
- production default prediction;
- account booking;
- servicing or billing activity;
- regulatory model validation;
- fair lending adjudication;
- macroeconomic forecasting;
- reserve estimation;
- capital modeling;
- profitability optimization;
- or any operational customer treatment.

These limitations apply even when outputs appear realistic.

Realism supports strategy testing and educational interpretation. It does not convert synthetic outputs into production decisioning assets.

12.10 AGGREGATION & REPORTING DESIGN

Outputs should support multiple levels of reporting.

Reporting Level	Purpose
Portfolio-level	Executive summary of total exposure, PTI, estimated risk, and Expected Loss
Product-level	Product structure and risk comparison
Score-band level	Borrower-quality and estimated-risk gradient review
PTI-bucket level	Affordability-to-risk relationship review
Product × score-band level	Interaction and segmentation review
Scenario-level	Baseline versus comparison scenario analysis
Row-level diagnostic	Deep-dive review of edge cases, transformations, and deltas

This reporting structure supports both executive interpretation and technical review.

The same output dataset should allow a reviewer to move from high-level portfolio performance to specific row-level explanations.

12.11 DATA LINEAGE & TRACEABILITY

Output lineage is central to the design of Module 1.

Every final output should be traceable to:

- configuration parameters;
- deterministic population controls;
- borrower profile generation;
- product assignment;
- requested amount logic;
- APR assumptions;
- payment proxy logic;
- PTI calculation;
- estimated-PD proxy framework;
- LGD assumptions;
- Expected Loss calculation;
- and scenario metadata.

This lineage supports auditability, debugging, and validation.

A reviewer should be able to answer:

- What parameter assumptions generated this output?
- Which product structure influenced this row?
- Why did this row have this payment burden?
- Why did this row have this estimated-PD proxy value?
- Why did Expected Loss change under a scenario?
- Was this difference caused by parameter change or population change?

The output framework is designed so that those questions can be answered without relying on undocumented model behavior.

12.12 OUTPUT QUALITY EXPECTATIONS

The final output dataset should satisfy the following quality expectations:

Output Quality Requirement	Description
Complete row population	All expected synthetic applications are generated
No unintended nulls in required fields	Core analytical outputs are populated
Valid product values	Product type values remain within supported categories
Valid score-band values	Score-band values remain within supported categories
Positive financial values where required	Requested amount, payment, and Expected Loss remain non-negative
Bounded ratios	PTI, estimated-PD proxy, LGD, and related ratios remain within governed ranges
Scenario metadata populated where applicable	Scenario outputs remain traceable
Diagnostic fields interpretable	Intermediate fields support validation rather than create ambiguity
QA outputs stable under identical inputs	Deterministic reproducibility is preserved

These expectations should be reviewed as part of the Section 9 QA process.

12.13 OUTPUT INTERPRETATION PRINCIPLES

Users should interpret Module 1 outputs according to the following principles:

1. Synthetic does not mean arbitrary

The data is synthetic, but it is governed by structured assumptions, deterministic generation, and economic relationships.

2. Realistic does not mean production-calibrated

Outputs are designed to be directionally realistic, not calibrated to an institution's production models or observed performance.

3. Scenario-comparable does not mean predictive

Scenario outputs support controlled sensitivity review, not forecasting.

4. Estimated-PD proxy does not mean calibrated PD

The estimated-PD proxy supports relative risk ordering and Expected Loss comparison, not production probability-of-default prediction.

5. Expected Loss is comparative

Expected Loss should be used for relative comparison across products, segments, strategies, and scenarios—not as a financial reporting estimate.

6. Diagnostics are explanatory

Diagnostic fields support validation and auditability; they should not be treated as independent decisioning variables unless explicitly designed into a downstream module.

12.14 EXAMPLE SCENARIO INTERPRETATIONS

Scenario 1: Strong Credit, High Structural Burden

Profile	**Score ~720; Personal Loan; 36-month term; elevated APR proxy; otherwise, reasonable borrower profile.**
Interpretation	Despite stronger credit, the shorter term and elevated APR raise monthly payment burden. PTI increases, and estimated-PD proxy moves upward accordingly. This demonstrates why structure must be modeled explicitly rather than inferred from score alone.

Scenario 2: Weaker Credit, Managed Affordability

Profile	**Score ~620; Auto Loan; 72-month term; moderate requested amount; weaker score but controlled structure.**
Interpretation	Longer term structure lowers the monthly payment enough to keep PTI more manageable than the score alone might suggest. The borrower is still weaker from a baseline-risk perspective, but structure moderates affordability stress and stabilizes estimated-PD proxy relative to a shorter-term design.

Scenario 3: Large Exposure, Structured Stability

Profile	**Large Mortgage exposure; long amortization horizon; strong or stable income; sizable balance but structurally manageable payment burden.**
Interpretation	A large balance does not automatically imply high risk. When amortized over a long horizon, payment burden can remain reasonable and PTI can stay controlled. This scenario demonstrates the central thesis of the module: exposure size becomes risk only after it is translated through structure and affordability.

Scenario 4: High APR, Low Exposure

Profile	**Revolving or personal loan borrower with high APR but small balance**
Interpretation	High APR can increase payment pressure, but total affordability impact depends on exposure and payment method.

Scenario 5: Low LGD, High Expected Loss

Profile	**Mortgage borrower with low LGD but very large exposure**
Interpretation	Expected Loss reflects exposure × estimated-PD proxy × LGD; low LGD does not always mean low dollar loss.

Scenario 6: Scenario-sensitive revolving exposure

Profile	**Revolving line under rate-up scenario**
Interpretation	APR-sensitive payment logic allows rate changes to flow into payment burden, PTI, estimated-PD proxy, and Expected Loss without becoming a billing engine.

12.15 RELATIONSHIP TO FUTURE MODULES

Module 1 outputs are designed to serve as the foundation for downstream modules.

Potential future modules may include:

- decision policy simulation;
- approval / decline / review strategies;
- line assignment or exposure strategy;
- pricing scenario layers;
- advanced loss simulation;
- account lifecycle simulation;
- performance transition modeling;
- or governance reporting dashboards.

Module 1 does not perform those functions directly. It creates the synthetic application and risk foundation needed for them.

This separation is intentional and consistent with the system's modular architecture.

12.16 RELATIONSHIP TO VALIDATION SUMMARY

The companion Validation Summary demonstrates that the output data model behaves as intended.

Validation evidence confirms that:

- row count is stable;
- product mix aligns to configured assumptions;
- score-band distribution is well formed;
- variables exhibit realistic distributions;
- product-level behavior is differentiated;
- score-band gradients are directionally coherent;
- PTI functions as the affordability bridge;
- estimated-PD proxy behavior is bounded and better dispersed in V2.0;
- Expected Loss reflects exposure, estimated-PD proxy, and LGD;
- mixed-signal cases remain present;
- and edge cases remain governed.

The BRD defines what the outputs are intended to represent. The Validation Summary confirms that the final V2.0 outputs behave accordingly.

12.17 CONCLUSION

The Module 1 output data model is designed to be analytical, interpretable, scenario-capable, and audit-ready.

It provides:

- synthetic application-level records;
- borrower and product attributes;
- pricing and payment proxies;
- affordability metrics;
- estimated-risk proxy outputs;
- LGD assumptions;
- Expected Loss outputs;
- diagnostic traceability;
- and scenario comparison support.

This output framework enables users to evaluate credit strategy assumptions in a controlled, reproducible, and explainable environment without exposing real customer data or making production decisions.

In short:

The output data model converts governed synthetic generation into usable decision-system evidence.

13. MODEL BOUNDARIES, INTENDED USE & GOVERNANCE GUARDRAILS

13.1 PURPOSE

This section defines the intended use boundaries and governance guardrails for Module 1.

Module 1 is designed as a **synthetic application and risk simulation engine** for pre-production strategy analysis. It produces governed, deterministic, explainable, and scenario-comparable outputs that allow users to evaluate how borrower attributes, product structure, pricing assumptions, affordability, estimated-risk proxy behavior, LGD assumptions, and Expected Loss interact.

This section is included to prevent misuse or over-interpretation of the system.

Module 1 is intentionally designed to support:

- synthetic portfolio generation;
- strategy simulation;
- affordability review;
- estimated-risk proxy analysis;
- scenario comparison;
- Expected Loss comparison;
- QA and validation review;
- and future module development.

13.1 PURPOSE - CONTINUED

It is not designed to support real customer decisioning, production model deployment, regulatory validation, or calibrated forecasting.

Boundary Area	V2.0 Position
Data	Synthetic, not real customer data
Estimated-PD proxy	Directional synthetic proxy, not calibrated PD
APR	Burden-estimation proxy, not production pricing
Expected Loss	Comparative simulation metric, not forecast
Revolving payment	APR-sensitive proxy, not billing engine
Scenario outputs	Sensitivity analysis, not prediction
Compliance	Not fair lending / regulatory validation

13.2 INTENDED USE

Module 1 is appropriate for controlled analytical use within the following boundaries.

Intended Use	Description
Synthetic application generation	Generate realistic, non-PII application-level records for analysis and strategy testing
Pre-production strategy simulation	Evaluate how product structure, score mix, affordability, and estimated-risk assumptions affect portfolio outcomes
Scenario analysis	Compare baseline and alternative assumptions using matched synthetic populations
Affordability analysis	Evaluate PTI behavior and payment burden across products, score bands, and scenarios
Estimated-risk proxy analysis	Review relative risk ordering and directional risk behavior
Expected Loss comparison	Compare relative loss outcomes across products, segments, and scenarios
QA and validation review	Support structured testing of realism, stability, tail behavior, and edge cases
Portfolio demonstration artifact	Demonstrate governed SQL-based decision-system architecture without using real customer data

These uses are consistent with Module 1's purpose as a governed simulation environment.

13.3 EXPLICIT NON-USES

Module 1 should not be used for the following purposes.

Non-Use	Explanation
Production underwriting decisions	The engine does not contain approved operational credit policy, production governance, or real applicant data
Customer-level credit approvals or declines	All records are synthetic and should not drive actual customer decisions
Calibrated probability-of-default modeling	`estimated_pd` is a synthetic estimated-PD proxy, not an empirically calibrated default model
Production pricing or offer assignment	APR values are burden-estimation proxies, not approved offer prices
Credit-card billing or servicing simulation	Revolving payment logic is a simplified APR-sensitive proxy, not a statement-cycle model
Macroeconomic forecasting	`rate_shift_bps` supports sensitivity testing, not future rate or macro forecasting
Regulatory model validation	The system is explainable and governed but is not a substitute for formal model validation
Fair lending or compliance adjudication	The system does not perform protected-class analysis, disparate-impact testing, or compliance review
Financial reporting, reserves, or capital estimation	Expected Loss is a comparative simulation metric, not a production loss forecast
Real customer segmentation or treatment	No output should be applied to actual customers

These exclusions are intentional design boundaries, not system weaknesses.

13.4 ESTIMATED-PD PROXY BOUNDARY

The `estimated_pd` field is one of the most important interpretive boundaries in Module 1.

In the implementation, `estimated_pd` represents a **synthetic estimated-PD proxy** generated through transparent score-band baselines, documented multipliers, affordability effects, product effects, and V2.0 soft-saturation logic.

It is designed to support:

- relative risk ordering;
- segmentation analysis;
- affordability sensitivity;
- scenario comparison;
- high-risk dispersion review;
- and Expected Loss simulation.

It is not designed to support:

- production default prediction;
- calibrated probability-of-default estimation;
- model deployment;
- regulatory capital estimation;
- reserve modeling;
- or empirical performance forecasting.

Governance Interpretation

The estimated-PD proxy should be interpreted as:

a governed synthetic risk translation measure

not:

a calibrated production default probability

This distinction is critical because Module 1 prioritizes explainability, scenario comparability, and directional coherence over empirical calibration.

13.5 ESTIMATED-PD PROXY GOVERNANCE CAP

The estimated-PD proxy framework includes a governed upper cap.

This cap functions as a **simulation guardrail**, not a calibrated production default-rate ceiling.

Its purpose is to:

- prevent runaway multiplier behavior;
- preserve bounded outputs;
- create a severe-risk region;
- support scenario comparability;
- and keep high-risk behavior explainable.

The cap should not be interpreted as an observed or institution-calibrated default-rate threshold.

V2.0 improves this boundary through soft-saturation logic, which reduces excessive cap crowding while preserving severe-risk observations.

Interpretation

The correct interpretation is:

The cap bounds synthetic estimated-risk behavior for simulation purposes.

The incorrect interpretation is:

The cap represents a calibrated probability that real customers will default.

13.6 EXPECTED LOSS BOUNDARY

Expected Loss is calculated conceptually as:

Expected Loss = Exposure × estimated-PD proxy × LGD

Expected Loss in Module 1 is a **comparative simulation output**, not a production loss forecast.

It is appropriate for:

- relative product comparison;
- score-band analysis;
- scenario sensitivity review;
- strategy tradeoff analysis;
- and portfolio-level loss pattern review.

It is not appropriate for:

- reserve estimation;
- capital modeling;
- financial reporting;
- production profitability analysis;
- pricing optimization;
- or regulatory stress testing.

Expected Loss should be interpreted as an analytical construct that supports strategy comparison under controlled assumptions.

13.7 APR & PRICING BOUNDARY

APR in Module 1 is a **payment-burden proxy**, not a production price.

APR is used to support:

- product differentiation;
- payment calculation;
- PTI generation;
- rate environment sensitivity;
- and downstream estimated-risk behavior.

APR should not be interpreted as:

- an approved customer offer rate;
- a production pricing model output;
- a profitability-optimized price;
- a regulatory pricing decision;
- or a customer-level pricing recommendation.

The rate environment parameter, `rate_shift_bps`, is similarly a sensitivity control, not a macroeconomic forecast.

13.8 REVOLVING PAYMENT BOUNDARY

Revolving payment logic in Module 1 is intentionally simplified.

V2.0 introduces APR-sensitive revolving payment behavior so that rate changes can flow directionally into:

APR → Payment → PTI → estimated-PD proxy → Expected Loss

However, the revolving payment proxy is not a credit-card billing model.

It does not model:

- statement balances;
- actual utilization-based balance behavior;
- contractual minimum-payment hierarchies;
- transactor versus revolver behavior;
- fees;
- grace periods;
- promotional APRs;
- interest accrual timing;
- or full balance amortization.

This boundary is intentional. The purpose is to model directional affordability sensitivity, not product servicing mechanics.

13.9 SCENARIO BOUNDARY

The scenario framework is designed for **controlled comparison**, not prediction.

Valid scenario interpretation:

What changes under this controlled assumption set?

Invalid scenario interpretation:

What will happen in the real world?

Scenario outputs should be interpreted as sensitivity results. They support analysis of how assumptions affect the same synthetic borrower population.

They do not forecast:

- macroeconomic conditions;
- real customer behavior;
- real default rates;
- pricing outcomes;
- or operational performance.

Scenario validity depends on preserving the same `population_id` when comparing scenario runs. If `population_id` changes, the user is comparing a different synthetic population rather than the same borrowers under different assumptions.

13.10 SYNTHETIC DATA BOUNDARY

Module 1 produces synthetic data only.

The data is designed to be realistic, but it does not represent:

- real customers;
- real applications;
- real bureau files;
- real income records;
- real tradelines;
- real repayment behavior;
- or observed account performance.

Synthetic realism supports safe strategy testing and education. It does not create real-world customer insight by itself.

Governance Principle

The correct interpretation is:

The data is synthetic but economically structured.

The incorrect interpretation is:

The data is real or directly predictive of real customers.

13.11 FAIR LENDING & COMPLIANCE BOUNDARY

Module 1 does not perform fair lending, bias testing, disparate-impact analysis, or compliance adjudication.

The engine does not include:

- protected-class attributes;
- proxy protected-class analysis;
- adverse action reason logic;
- regulatory compliance testing;
- policy approval rules;
- or customer treatment logic.

Future modules could add compliance-oriented analysis, but such work would require separate design, governance, and validation.

Module 1 should therefore not be used to make any claim regarding fair lending compliance, disparate impact, or regulatory acceptability.

13.12 PRODUCTION DEPLOYMENT BOUNDARY

Module 1 is not a production deployment artifact.

Before any related system could support production decisioning, it would require additional capabilities, including:

- approved credit policy logic;
- empirical model development or calibration where applicable;
- production data governance;
- model validation;
- fair lending review;
- compliance review;
- monitoring and controls;
- implementation testing;
- operational documentation;
- and change-management governance.

Module 1 intentionally stops before that boundary.

It provides a synthetic strategy simulation foundation, not an operational credit-decisioning system.

13.13 GOVERNANCE REQUIREMENTS FOR PROPER USE

Any proper use of Module 1 should follow these governance requirements:

- Use the parameter layer for controlled changes.
- Preserve `population_id` for matched scenario comparison.
- Use meaningful `scenario_set_name` and `scenario_name` values.
- Review Section 9 QA outputs after material parameter or logic changes.
- Monitor tail-frequency behavior when changing amount, rate, or payment assumptions.
- Monitor estimated-PD proxy cap behavior when changing risk multipliers.
- Interpret APR, estimated-PD proxy, LGD, and Expected Loss as simulation constructs.
- Avoid production-use claims.
- Document scenario assumptions and outputs.
- Treat unusual outputs as prompts for diagnostic review, not automatic defects.

These requirements preserve the interpretability and credibility of the framework.

13.14 BOUNDARY-AWARE COMMUNICATION

When communicating Module 1 outputs, users should use boundary-aware language.

Avoid Saying	Prefer Saying
"The model predicts default."	"The engine produces an estimated-PD proxy for synthetic risk comparison."
"This is the approved APR."	"This is an APR proxy used for payment burden estimation."
"This customer would be declined."	"This synthetic applicant would trigger a strategy outcome in a future decision module."
"Expected Loss is the forecasted loss."	"Expected Loss is a comparative simulation metric."
"The scenario predicts rate stress outcomes."	"The scenario evaluates sensitivity under a controlled rate assumption."
"The data represents real borrowers."	"The data represents synthetic borrower-like profiles."

This communication discipline is important because it prevents strong simulation outputs from being misinterpreted as production-grade forecasts or decisions.

13.15 RELATIONSHIP TO VALIDATION SUMMARY

The companion Validation Summary confirms that Module 1 V2.0 behaves as intended within these boundaries.

Validation evidence demonstrates that the system is:

- stable;
- deterministic;
- product-aware;
- affordability-sensitive;
- scenario-comparable;
- diagnostically transparent;
- and portfolio-realistic.

Validation does not expand the system boundary beyond the intended use defined in this BRD.

The BRD defines what the system is designed to be. The Validation Summary confirms that it behaves accordingly.

13.16 CONCLUSION

Module 1 is a governed synthetic simulation framework.

Its outputs are realistic, explainable, reproducible, and useful for strategy testing, but they remain synthetic and bounded.

The system is fit for:

- pre-production strategy analysis;
- scenario comparison;
- affordability review;
- estimated-risk proxy evaluation;
- Expected Loss comparison;
- and decision-system design.

The system is not fit for:

- real customer decisioning;
- production underwriting;
- production pricing;
- calibrated probability-of-default modeling;
- credit-card billing;
- regulatory validation;
- fair lending adjudication;
- or operational deployment.

This boundary discipline is what allows Module 1 to be both powerful and credible.

In short:

Module 1 is designed to inform strategy thinking, not to execute production decisions.

14. FUTURE ENHANCEMENT ROADMAP & MODULAR EXPANSION STRATEGY

14.1 PURPOSE

This section defines the future enhancement roadmap for Module 1 and establishes how future functionality should be added without compromising the validated V2.0 design.

Module 1 V2.0 is complete within its current scope. It provides a governed, deterministic, product-aware, scenario-capable synthetic application and risk modeling engine suitable for pre-production strategy simulation.

Future enhancements are therefore not required to make V2.0 usable. Instead, they represent logical extensions that could deepen realism, expand strategy-testing capability, or support downstream modules.

This roadmap is intended to preserve three principles:

- **V2.0 completeness** — the current system is validated and fit for its intended purpose.
- **Modular expansion** — future functionality should be added in controlled layers.
- **Separate validation** — each enhancement should be independently designed, governed, and validated.

14.2 ROADMAP DESIGN PHILOSOPHY

Future enhancements should follow the same design philosophy used in Module 1 V2.0:

- Parameterized over hard-coded
- Explainable over opaque
- Deterministic over non-reproducible
- Product-aware over generic
- Scenario-comparable over isolated
- Governed over ad hoc
- Modular over monolithic

Future work should not convert Module 1 into an all-purpose production decisioning system. Instead, enhancements should preserve the clean separation between:

- synthetic population generation,
- product structure,
- affordability translation,
- estimated-risk proxy behavior,
- Expected Loss comparison,
- scenario analysis,
- and downstream decision strategy logic.

This modular discipline protects the system from scope creep.

14.3 ROADMAP SUMMARY

Potential future enhancements can be grouped into several design categories.

Roadmap Area	Current V2.0 Boundary	Future Design Opportunity
Decision strategy layer	Module 1 produces synthetic application and risk outputs but does not execute approval rules	Add downstream approval, decline, review, and segmentation strategy logic
Line assignment / exposure strategy	Requested amount is generated but no approval or assigned-line strategy is applied	Add strategy logic to determine approved exposure, line amount, or counteroffer amount
Pricing strategy layer	APR is used as an affordability proxy, not production pricing	Add pricing-policy simulation or risk-based offer logic in a separate module
Enhanced LGD framework	LGD is product-level and intentionally simplified	Add row-level or scenario-sensitive LGD variation
Expanded macro sensitivity	`rate_shift_bps` supports broad rate sensitivity only	Add income stress, unemployment stress, collateral value stress, or recession overlays
Revolving behavior enhancement	Revolving payment is an APR-sensitive proxy, not billing logic	Add utilization-sensitive balance behavior or more detailed payment proxy mechanics
Account lifecycle simulation	Module 1 is application-level only	Add booked-account behavior, vintage curves, delinquency transitions, or performance paths
Empirical benchmarking	Estimated-PD proxy is directional, not calibrated	Compare synthetic outputs to external benchmarks or historical patterns where available

Roadmap Area	Current V2.0 Boundary	Future Design Opportunity
Automated reporting layer	QA outputs exist as SQL queries and validation documents	Add automated validation reports, scenario dashboards, or executive summaries
Governance documentation	Parameters and QA are documented in code and artifacts	Add formal parameter dictionary, change log, and scenario governance register

These enhancements should be treated as future design modules, not as required corrections to V2.0.

14.4 NATURAL NEXT MODULE: DESIGN STRATEGY SIMULATOR

The most natural next extension is a downstream **Decision Strategy Simulator**.

Module 1 already produces the core inputs needed for strategy testing:

- product type,
- credit score and score band,
- borrower capacity variables,
- requested amount,
- APR proxy,
- monthly payment proxy,
- PTI,
- estimated-PD proxy,
- LGD,
- Expected Loss,
- scenario metadata.

A future decision strategy module could use these fields to simulate:

- approval / decline / review outcomes,
- affordability thresholds,
- score-band policy cutoffs,
- exposure limits,
- product-specific decision rules,
- manual review routing,
- counteroffer logic,
- and strategy comparison outputs.

Design Boundary

A future decision strategy module should still be framed as pre-production simulation unless separately governed for production use.

The purpose would be to answer:

How would a proposed strategy behave on a governed synthetic population?

not:

What production decision should be made for a real applicant?

14.5 FUTURE EXPOSURE / LINE ASSIGNMENT MODULE

A future exposure strategy module could extend requested amount logic into simulated approval exposure.

Current V2.0 behavior generates a synthetic requested amount. It does not determine:

- approved amount,
- counteroffer amount,
- credit limit,
- line assignment,
- partial approval,
- or exposure cap policy.

A future module could introduce:

Potential Feature	Purpose
Approved amount logic	Simulate approved exposure below requested amount
Line assignment rules	Assign revolving or HELOC line amounts based on capacity and risk
Counteroffer framework	Simulate reduced approved amount when requested exposure exceeds strategy limits
Exposure cap rules	Apply product, score, income, or PTI-based exposure caps
Strategy comparison	Compare risk and loss outcomes across exposure strategies

This would allow the system to move from application-risk simulation into controlled decision-strategy testing.

14.6 FUTURE PRICING STRATEGY MODULE

V2.0 uses APR as a burden-estimation proxy. It does not optimize price, assign customer offers, or model profitability.

A future pricing strategy module could add:

- risk-based pricing logic,
- product-specific pricing grids,
- rate tiers,
- margin assumptions,
- cost of funds,
- revenue estimates,
- risk-adjusted return,
- profitability comparison,
- and scenario sensitivity.

Governance Boundary

Any future pricing module would require clear documentation that pricing logic remains simulated unless separately approved for production use.

Pricing extensions should avoid implying:

- production pricing approval,
- profitability optimization,
- regulatory pricing compliance,
- or customer-level offer assignment.

14.7 FUTURE LGD & SEVERITY ENHANCEMENT

In V2.0, LGD is intentionally product-level and simpler than the estimated-PD proxy framework.

A future enhancement could introduce row-level LGD variation based on:

- product type,
- collateral proxy,
- amount-to-income ratio,
- score band,
- exposure size,
- scenario stress,
- or recovery assumptions.

Potential design benefits include:

- more nuanced Expected Loss behavior,
- stronger secured versus unsecured differentiation,
- scenario-sensitive severity analysis,
- and better product-level loss decomposition.

Validation Requirement

Any row-level LGD enhancement would require separate validation to ensure:

- LGD remains bounded,
- severity relationships are explainable,
- Expected Loss behavior remains coherent,
- and LGD does not become an opaque second risk model.

14.8 FUTURE MACROECONOMIC SENSITIVITY EXPANSION

V2.0 includes `rate_shift_bps` as a controlled rate environment parameter.

Future macro-sensitivity extensions could include:

- income stress factors,
- unemployment stress proxies,
- inflation / expense burden stress,
- collateral value stress,
- product-specific rate sensitivity,
- delinquency stress multipliers,
- or recession-style scenario overlays.

Design Boundary

These extensions should remain sensitivity tools unless supported by formal macroeconomic modeling.

They should answer:

How does the portfolio behave under this assumption?

not:

What macroeconomic outcome will occur?

This distinction is essential for preserving the simulation boundary.

14.9 FUTURE REVOLVING BEHAVIOR ENHANCEMENT

Workstream D restored APR-sensitive revolving payment behavior while preserving the boundary that Module 1 is not a billing engine.

Future revolving enhancements could include:

- utilization-sensitive balance proxies,
- statement balance approximation,
- minimum-payment rule variants,
- payment-rate scenarios,
- revolver / transactor segmentation,
- promotional APR assumptions,
- or payoff / amortization approximations.

These extensions would improve revolving realism but should remain modular.

Governance Boundary

A future revolving enhancement should not blur into full credit-card servicing unless explicitly designed as a separate servicing simulation module.

Module 1’s current boundary should remain:

directional affordability proxy, not contractual billing calculation.

14.10 FUTURE ACCOUNT LIFECYCLE SIMULATION

Module 1 operates at the application level. It does not model booked-account performance over time.

A future lifecycle module could simulate:

- account booking,
- monthly performance,
- delinquency transitions,
- cure behavior,
- charge-off timing,
- prepayment,
- attrition,
- utilization changes,
- or vintage-level performance.

This would be a major scope expansion and should be treated as a separate module.

Design Implication

Lifecycle simulation would require:

- time-series architecture,
- transition logic,
- state management,
- performance assumptions,
- and separate validation standards.

It should not be embedded casually into Module 1.

14.11 FUTURE EMPIRICAL BENCHMARKING OR CALIBRATION

V2.0 does not claim empirical calibration. The estimated-PD proxy is directionally validated, bounded, and scenario-useful.

A future enhancement could benchmark outputs against:

- external industry ranges,
- public benchmark data,
- synthetic benchmark portfolios,
- or institution-specific historical data where governance permits.

Potential objectives include:

- assessing reasonableness of absolute estimated-risk levels,
- benchmarking score-band gradients,
- reviewing product-level loss patterns,
- testing alternative estimated-PD proxy caps,
- and comparing Expected Loss distributions.

Governance Boundary

Benchmarking should not be confused with production model calibration unless the process includes:

- historical data,
- outcome labels,
- calibration methodology,
- validation metrics,
- stability testing,
- and model governance review.

14.12 FUTURE AUTOMATED QA & REPORTING LAYER

The current system includes rich SQL-based QA outputs and a companion Validation Summary.

A future reporting layer could automate:

- QA summaries,
- product mix checks,
- score-band distribution checks,
- tail-frequency monitoring,
- scenario comparison dashboards,
- workstream diagnostic summaries,
- row-level exception reports,
- and executive scenario reports.

This would improve usability and reviewer access without changing the underlying model.

Design Requirement

Automated reporting should preserve:

- traceability to SQL outputs,
- reproducibility,
- scenario metadata,
- and clear distinction between metrics and interpretation.

14.13 FUTURE GOVERNANCE DOCUMENTATION ENHANCEMENTS

Future governance artifacts could include:

- formal parameter dictionary,
- scenario register,
- change log,
- validation checklist,
- QA result archive,
- known limitations register,
- model-boundary statement,
- and user operating guide.

These artifacts would strengthen the system as an enterprise-grade simulation platform.

Design Benefit

Governance documentation would make it easier for users and reviewers to understand:

- what changed,
- why it changed,
- which scenario produced which output,
- what assumptions were active,
- and which validation checks were performed.

14.14 ROADMAP PRIORITIZATION

Future enhancements should be prioritized based on business value, complexity, and validation burden

Priority Tier	Enhancement Type	Rationale
Tier 1 — Near-term extensions	Decision strategy simulator, scenario library expansion, automated QA reporting, parameter dictionary	Builds directly on V2.0 architecture
Tier 2 — Analytical deepening	Row-level LGD, enhanced revolving behavior, macro sensitivity expansion, empirical benchmarking	Adds realism but requires stronger validation
Tier 3 — Separate modules	Account lifecycle simulation, survival modeling, fair lending analysis, production decisioning, pricing optimization	Materially expands scope and governance requirements

This prioritization helps preserve the current system while enabling thoughtful expansion.

14.15 ROADMAP GOVERNANCE REQUIREMENTS

Any future enhancement should follow the same governance discipline as V2.0.

Required design steps should include:

1. **Define the business objective**
2. **State the module boundary**
3. **Identify affected parameters and outputs**
4. **Preserve deterministic reproducibility**
5. **Document expected behavior**
6. **Add or update QA checks**
7. **Validate distributional effects**
8. **Validate segment-level effects**
9. **Review edge cases**
10. **Confirm scenario comparability where applicable**
11. **Document limitations**
12. **Update BRD and Validation Summary artifacts**

This ensures future enhancements remain controlled and reviewable.

14.16 RELATIONSHIP TO VALIDATION SUMMARY

The companion Validation Summary confirms that Module 1 V2.0 is fit for its current purpose. Future enhancements are not required for V2.0 acceptance.

The roadmap should therefore be interpreted as:

a forward expansion strategy

not:

a list of unresolved validation defects.

This distinction matters. V2.0 is validated as complete within its intended scope.

Future roadmap items should be separately scoped, implemented, and validated.

14.17 CONCLUSION

The future roadmap establishes a clear path for expanding Module 1 while preserving the integrity of the validated V2.0 system.

The most natural future direction is to build downstream decision strategy modules that use the validated synthetic application, affordability, estimated-risk proxy, and Expected Loss outputs generated by Module 1.

Additional enhancements may deepen realism, expand scenario coverage, automate reporting, or introduce new analytical modules.

However, future work should remain modular, governed, and boundary-aware.

In short:

V2.0 is a validated foundation. Future enhancements should extend it, not dilute it.

15. FINAL DESIGN STATEMENT

15.1 PURPOSE

This section provides the final design statement for Module 1 — Synthetic Application & Risk Modeling Engine V2.0.

The purpose of this statement is to confirm that the BRD defines a complete, governed, and internally coherent system design for a synthetic credit strategy simulation engine.

Module 1 V2.0 is designed to generate deterministic synthetic application-level portfolios that support pre-production analysis of product structure, affordability, estimated-risk proxy behavior, Expected Loss, and scenario sensitivity without relying on real customer data.

15.1 PURPOSE - CONTINUED

This document defines:

- what the system is intended to do;
- how the system is structured;
- how core variables behave;
- how product structure translates into affordability and downstream estimated risk;
- how scenarios are compared;
- how outputs should be governed and consumed;
- what boundaries apply;
- and how future enhancements should be modularly designed.

15.2 FINAL SYSTEM DESIGN POSITION

Module 1 V2.0 is best understood as a:

portfolio-grade, deterministic, scenario-capable synthetic credit decisioning simulation engine

It is not merely a synthetic data generator. It is a governed decision-system design artifact that demonstrates how credit strategy inputs can be translated into structured, explainable portfolio outcomes.

The system is intentionally built around the following design chain:

Application → Product Structure → APR → Payment → PTI → estimated-PD proxy → LGD → Expected Loss

This chain ensures that modeled outcomes are not assigned arbitrarily. Instead, outcomes emerge through interactions between borrower attributes, product structure, pricing assumptions, affordability burden, estimated-risk translation, and severity assumptions.

15.3 DESIGN MATURITY ACHIEVED IN V2.0

V2.0 represents a material maturity step beyond the initial validated V1.0 foundation.

The V2.0 design strengthens the system across four major dimensions:

Design Dimension	V2.0 Maturity Achieved
Mortgage realism	Upper-tail mortgage exposure and affordability behavior are governed through tail smoothing and PTI-aware logic
Estimated-risk differentiation	Estimated-PD proxy behavior is better dispersed through soft-saturation logic while preserving governance caps
Scenario comparability	Scenario set, scenario name, archive persistence, and matched row-level comparison support controlled strategy testing
Revolving sensitivity	Revolving payment behavior responds directionally to APR movement while preserving simplified proxy boundaries

These enhancements do not change the core purpose of Module 1. They strengthen its realism, comparability, and auditability within the existing simulation boundary.

15.4 DESIGN PRINCIPLES CONFIRMED

The final V2.0 design reflects the system's core principles:

Design Principle	Final Design Confirmation
Deterministic reproducibility	Synthetic populations are anchored through deterministic generation and `population_id`
Full auditability	Logic, parameters, diagnostics, QA queries, and scenario outputs are documented and reviewable
Parameter governance	Key assumptions are centralized and subject to validation gatekeepers
Separation of concerns	Population generation, product structuring, payment logic, affordability, estimated-risk translation, LGD, and Expected Loss are staged distinctly
Scenario-based analysis	Matched scenario comparison enables controlled evaluation of assumption changes
Product-aware realism	Product type materially influences exposure, APR, payment, PTI, LGD, and Expected Loss
Boundary discipline	Outputs are explicitly framed as synthetic simulation results, not production decisions or forecasts

Together, these principles ensure that Module 1 is flexible without becoming uncontrolled, explainable without becoming simplistic, and realistic without claiming production calibration.

15.5 INTENDED USE CONFIRMATION

Module 1 V2.0 is intended for:

- synthetic application portfolio generation;
- pre-production credit strategy analysis;
- affordability sensitivity review;
- product and score-band segmentation analysis;
- estimated-risk proxy evaluation;
- Expected Loss comparison;
- matched scenario comparison;
- QA and validation review;
- and future downstream decision-strategy module development.

Within these boundaries, the system is fit for purpose.

The correct interpretation is:

Module 1 allows users to explore how credit strategy assumptions affect a governed synthetic portfolio.

The incorrect interpretation is:

Module 1 produces production credit decisions or calibrated real-world forecasts.

15.6 BOUNDARY CONFIRMATION

Module 1 V2.0 is not designed or approved for:

- production underwriting;
- real customer approvals or declines;
- calibrated probability-of-default modeling;
- production pricing;
- credit-card billing or servicing;
- macroeconomic forecasting;
- fair lending adjudication;
- regulatory model validation;
- financial reporting;
- reserve estimation;
- capital modeling;
- or operational deployment.

These boundaries are intentional.

The system's value comes from its ability to provide a safe, explainable, deterministic simulation environment where strategy concepts can be explored before production design, governance, and implementation.

15.7 RELATIONSHIP TO VALIDATION SUMMARY

This BRD defines the system's design intent, architecture, and governance requirements.

The companion Validation Summary confirms that the final V2.0 implementation behaves as intended.

Together, the two artifacts serve complementary purposes:

Artifact	Primary Role
BRD	Defines what the system is, how it is designed, why it works, and how it should be governed
Validation Summary	Demonstrates that the implementation behaves as intended using QA evidence, workstream validation, and scenario testing

The BRD is therefore the **design artifact**.
The Validation Summary is the **evidence artifact**.

Together, they establish Module 1 V2.0 as a complete, reviewable, and portfolio-grade system package.

15.8 FINAL STATEMENT

Module 1 V2.0 is a complete and governed synthetic application and risk modeling engine.

It demonstrates:

- deterministic population generation;
- product-aware synthetic credit structure;
- transparent affordability modeling;
- estimated-PD proxy risk translation;
- Expected Loss calculation;
- scenario comparison architecture;
- QA and diagnostic governance;
- parameterized design;
- clear model boundaries;
- and a modular roadmap for future expansion.

The system is intentionally designed to support strategy thinking before production decisioning.

It is realistic enough to support meaningful analysis, transparent enough to be reviewed, deterministic enough to be reproduced, and bounded enough to remain credible.

Final design conclusion:

Module 1 V2.0 provides a governed, explainable, scenario-capable foundation for synthetic credit decisioning strategy simulation. It is complete within its intended scope and ready to serve as the foundation for downstream decision-strategy modules.

APPENDIX A — DEFAULT CONFIGURATION & ESTIMATION-RISK REFERENCE TABLES

A.1 PURPOSE

This appendix summarizes the default configuration values and estimated-risk reference assumptions used in Module 1 V2.0.

These values are included to support:

- implementation transparency;
- parameter governance;
- audit review;
- validation traceability;
- and controlled scenario interpretation.

The values below are **default V2.0 simulation assumptions**. They are not universal credit-risk standards, production underwriting policy, calibrated default-rate estimates, or institution-specific model parameters.

A.2 DEFAULT RUN CONFIGURATION

Configuration Item	V2.0 Default Value	Interpretation
`scenario_set_name`	`MODULE1_V2_BASELINE_SET`	Default scenario comparison family
`scenario_name`	`BASELINE_V2`	Default baseline scenario label
`archive_run_flag`	`TRUE`	Baseline run is archived for comparison
`archive_reset_mode`	`SCENARIO_ONLY`	Replaces the same scenario within the same scenario set / population
`population_id`	`BASE_POPULATION_V1`	Deterministic borrower population anchor
`application_count`	`50,000`	Full validation-scale synthetic portfolio
`portfolio_start_date`	`2025-01-01`	Start of simulated application window
`portfolio_end_date`	`2025-12-31`	End of simulated application window
`anchor_date`	`2025-12-31`	Reporting reference date

Interpretation

The default run configuration is designed for a full-scale, reproducible baseline scenario. The `population_id` should remain unchanged when comparing baseline and stress scenarios against the same synthetic borrower population.

A.4 DEFAULT PRODUCT MIX

Product	V2.0 Default Mix
Mortgage	30%
Revolving Line	25%
Auto Loan	20%
Unsecured Personal Loan	15%
HELOC	10%

Interpretation

The product mix is intended to represent a diversified consumer-retail bank portfolio for strategy simulation. It is not intended to represent a universal industry benchmark.

A.4 DEFAULT SCORE-BAND MIX

Score Band	V2.0 Default Mix
Super Prime	18%
Prime	28%
Near Prime	24%
Subprime	18%
Deep Subprime	12%

Interpretation

The default score mix is weighted toward prime and near-prime borrowers while preserving meaningful subprime and deep-subprime representation for estimated-risk and Expected Loss contrast.

A.5 CORE HARD BOUNDS

Parameter	V2.0 Default Value	Interpretation
Minimum credit score	500	Lower modeled bureau-style score floor
Maximum credit score	850	Upper bureau-style score ceiling
Minimum annual income	$20,000	Entry-level but plausible borrower income floor
Maximum annual income	$300,000	High-income ceiling with affluent tail preserved
Minimum DTI	5%	Allows very low leverage without forcing zero-debt rows
Maximum DTI	70%	High-stress but bounded obligation ratio
Minimum utilization	0%	Allows unused revolving capacity
Maximum utilization	95%	Allows near-max utilization without forcing literal 100%

A.6 CREDIT SCORE RANGES

Score Band	V2.0 Score Range
Super Prime	760–850
Prime	700–759
Near Prime	640–699
Subprime	580–639
Deep Subprime	500–579

Interpretation

Score ranges anchor segmentation and base estimated-risk behavior. They also support downstream product, pricing, affordability, and estimated-PD proxy relationships.

A.7 PRODUCT AMOUNT BOUNDS

Product	Minimum Amount	Maximum Amount
Mortgage	$75,000	$1,500,000
Revolving Line	$1,000	$30,000
Auto Loan	$5,000	$80,000
Unsecured Personal Loan	$1,000	$50,000
HELOC	$10,000	$250,000

Interpretation

These bounds define the first-stage requested amount anchors by product. V2.0 mortgage requested amounts are further governed by mortgage tail smoothing and PTI-aware overlay logic before final output.

A.8 PRODUCT TERM STRUCTURES

Product	V2.0 Term Options
Mortgage	180, 240, 360 months
Auto Loan	36, 48, 60, 72, 84 months
Unsecured Personal Loan	24, 36, 48, 60 months
HELOC	120, 180, 240 months
Revolving Line	Not applicable / no fixed amortizing term

Interpretation

Term structures are product-specific. Revolving products are handled through a payment proxy rather than a fixed amortizing term.

A.9 BASE APR ASSUMPTIONS

Product	Base APR Proxy
Mortgage	6.00%
Auto Loan	6.50%
Unsecured Personal Loan	11.00%
Revolving Line	20.00%
HELOC	8.00%

Interpretation

Base APR values are payment-burden anchors, not production pricing offers. Final APR used for payment calculation also reflects score-band risk premium, deterministic dispersion, rate environment shift, and product-specific floors / ceilings.

A.10 APR RISK PREMIUM BY SCORE BAND

Score Band	V2.0 APR Risk Premium Range
Super Prime	0.00% to 1.00%
Prime	1.00% to 3.00%
Near Prime	3.00% to 6.00%
Subprime	6.00% to 10.00%
Deep Subprime	10.00% to 15.00%

Additional APR dispersion:

Component	V2.0 Value
Deterministic pricing noise	Approximately ±1.00 percentage point
Rate environment shift	`rate_shift_bps / 10,000`

Interpretation

APR increases directionally with weaker score bands while retaining within-band variation. The rate environment parameter allows scenario shifts such as `+200 bps`.

A.11 PRODUCT APR FLOORS & CEILINGS

Product	APR Floor	APR Ceiling
Mortgage	4.00%	10.00%
Auto Loan	5.00%	22.00%
Unsecured Personal Loan	8.00%	35.00%
Revolving Line	15.00%	35.00%
HELOC	5.00%	15.00%

Interpretation

APR floors and ceilings prevent unrealistic pricing extremes. They are simulation bounds, not production pricing rules.

A.12 RATE ENVIRONMENT PARAMETER

Parameter	V2.0 Default	Example Interpretation
`rate_shift_bps`	0	Baseline rate environment
`+100`	N/A	+1.00 percentage point rate shift
`+200`	N/A	+2.00 percentage point rate shift
`-100`	N/A	-1.00 percentage point rate shift

Interpretation

`rate_shift_bps` supports scenario sensitivity testing. It is not a macroeconomic forecast.

A.13 REVOLVING PAYMENT PARAMETERS

Parameter	V2.0 Default Value	Interpretation
`revolving_base_min_payment_pct`	2.00%	Baseline monthly payment percentage applied to revolving requested amount
`revolving_interest_passthrough_factor`	0.50	Share of monthly APR pressure flowing into payment proxy

Conceptual Formula

Revolving payment proxy = requested amount × [base minimum-payment percentage + monthly APR pressure × passthrough factor]

Where:

monthly APR pressure = APR / 12

Interpretation

This restores APR-sensitive revolving payment behavior while preserving the boundary that Module 1 is not a credit-card billing engine.

A.14 PRODUCT LGD ASSUMPTIONS

Product	V2.0 LGD
Mortgage	20%
HELOC	25%
Auto Loan	45%
Unsecured Personal Loan	70%
Revolving Line	85%

Interpretation

LGD is intentionally product-level in Module 1. It supports secured versus unsecured severity contrast without introducing a full recovery model.

A.15 BASE ESTIMATED-PD PROXY BY SCORE BAND

Score Band	Score Range	Base Estimated-PD Proxy
Super Prime	760–850	1.00%
Prime	700–759	2.50%
Near Prime	640–699	6.00%
Subprime	580–639	14.00%
Deep Subprime	500–579	26.00%

Interpretation

These are base synthetic estimated-risk anchors before behavioral, affordability, product, and structural multipliers are applied. They are not calibrated production default probabilities.

A.16.1 BEHAVIORAL RISK MULTIPLIERS – DELINQUENCY COUNT

Delinquency Count	Multiplier
0	1.00
1	1.20
2	1.45
3	1.75
4+	2.10

A.16.2 BEHAVIORAL RISK MULTIPLIERS – MAJOR DEROGATORY & BANKRUPTCY

Driver	Condition	Multiplier
Major derogatory flag	No	1.00
Major derogatory flag	Yes	1.60
Bankruptcy flag	No	1.00
Bankruptcy flag	Yes	1.85

A.16.3 BEHAVIORAL RISK MULTIPLIERS – UTILIZATION

Utilization Bucket	Multiplier
<10%	0.95
10%–29%	1.00
30%–49%	1.10
50%–74%	1.25
75%+	1.45

A.16.4 BEHAVIORAL RISK MULTIPLIERS – DTI

DTI Bucket	Multiplier
<20%	0.95
20%–34%	1.00
35%–44%	1.10
45%–54%	1.25
55%+	1.45

A.16.5 BEHAVIORAL RISK MULTIPLIERS – RECENT INQUIRY COUNT

Recent Inquiry Count	Multiplier
0–1	1.00
2–3	1.05
4–5	1.12
6+	1.20

A.17.1 CREDIT DEPTH MULTIPLIERS – TRADELINE COUNT

Tradeline Count	Multiplier	Interpretation
1–2	1.12	Thin-file / limited credit depth
3–5	1.05	Somewhat thin but acceptable
6–12	1.00	Neutral / mature baseline
13–18	1.03	Mild complexity
19+	1.08	Heavy file / complex account profile

A.17.2 CREDIT DEPTH MULTIPLIERS – MONTHS SINCE OLDEST TRADE

Months Since Oldest Trade	Multiplier	Interpretation
<12 months	1.18	Very young file
12–23 months	1.10	Early-stage file
24–59 months	1.05	Moderately seasoned
60+ months	1.00	Fully seasoned baseline

A.18.1 AFFORDABILITY & PRODUCT MULTIPLIERS – PTI

PTI Bucket	Multiplier
<5%	0.95
5%–9%	1.00
10%–14%	1.10
15%–19%	1.25
20%+	1.45

A.18.2 AFFORDABILITY & PRODUCT MULTIPLIERS – AMOUNT-TO-INCOME

Amount-to-Income Bucket	Multiplier
<0.10	1.00
0.10–0.29	1.03
0.30–0.59	1.08
0.60–0.99	1.15
1.00+	1.25

A.18.3 AFFORDABILITY & PRODUCT MULTIPLIERS – PRODUCT

Product	Multiplier	Interpretation
Mortgage	0.90	Secured product; modest downward adjustment
Auto Loan	1.00	Baseline secured installment product
Unsecured Personal Loan	1.10	Unsecured installment risk adjustment
Revolving Line	1.12	Behavior-sensitive unsecured revolving exposure
HELOC	0.95	Secured revolving exposure; mild downward adjustment

A.18.4 AFFORDABILITY & PRODUCT MULTIPLIERS – DETERMINISTIC IDIOSYNCRATIC ESTIMATED-RISK NOISE

Component	V2.0 Range	Interpretation
Deterministic risk noise factor	0.95 to 1.05	Adds controlled row-level dispersion without stochastic instability

Implementation Concept

risk noise factor = 0.95 + deterministic uniform draw × 0.10

This prevents identical-looking rows from clustering too tightly while preserving deterministic reproducibility.

A.19.1 ESTIMATED-PD PROXY SOFT-SATURATION & CAP GOVERNANCE – RAW ESTIMATED-PD PROXY FORMULA

The raw estimated-PD proxy is calculated conceptually as:

raw estimated-PD proxy = base estimated-PD proxy × all applicable multipliers × deterministic risk noise

Where applicable multipliers include:

- delinquency count;
- major derogatory flag;
- bankruptcy flag;
- utilization;
- DTI;
- recent inquiries;
- tradeline count;
- file age;
- PTI;
- amount-to-income;
- product type;
- deterministic risk noise.

A.19.2 ESTIMATED-PD PROXY SOFT-SATURATION & CAP GOVERNANCE – SOFT-SATURATION ZONES

Raw Estimated-PD Proxy Zone	Threshold	Saturation Zone
Normal	≤30%	`NORMAL`
High	>30% and ≤55%	`HIGH`
Extreme	>55%	`EXTREME`

A.19.3 ESTIMATED-PD PROXY SOFT-SATURATION & CAP GOVERNANCE – SOFT-SATURATION FORMULA

Raw Estimated-PD Proxy Range	V2.0 Saturated Value
≤30%	unchanged
>30% and ≤55%	`0.30 + ((raw_pd - 0.30) × 0.50)`
>55%	`0.425 + ((raw_pd - 0.55) × 0.15)`

Interpretation

V2.0 soft saturation reduces high-end compression before the final governance cap. It is designed to reduce excessive clustering near the 45% cap while preserving severe-risk differentiation.

A.19.4 ESTIMATED-PD PROXY SOFT-SATURATION & CAP GOVERNANCE – FINAL ESTIMATED-PD PROXY FLOOR & CAP

Control	V2.0 Value	Interpretation
Floor	0.50%	Prevents unrealistically tiny estimated-PD proxy values
Cap	45.00%	Simulation guardrail to prevent runaway tail values

Final Governance Concept

final estimated-PD proxy = min(45%, max(0.50%, saturated estimated-PD proxy))

Boundary

The 45% cap is a simulation guardrail, not a calibrated production default-rate ceiling.

A.20.1 MORTGAGE V2.0 AFFORDABILITY GOVERNANCE – A2 MORTGAGE AMOUNT-TO-INCOME TAIL SMOOTHING

Let: **r = mortgage candidate amount ÷ annual income**

Mortgage Candidate Amount-to-Income Ratio	Zone	Smoothing Factor
r ≤ 3.75	`NORMAL`	1.00
3.75 < r ≤ 4.75	`UPPER_MID_TAIL`	`1.00 - ((r - 3.75) × 0.08)`
r > 4.75	`EXTREME_TAIL`	`max(0.85, 0.92 - ((r - 4.75) × 0.04))`

Interpretation

This layer reduces unrealistic large-balance mortgage exposure relative to income while preserving valid high-capacity borrowers.

A.20.2 MORTGAGE V2.0 AFFORDABILITY GOVERNANCE – A3 MORTGAGE PTI-AWARE DAMPENER

Provisional Mortgage PTI	Zone	Dampener Factor
PTI ≤ 45%	`NORMAL`	`1.00`
45% < PTI ≤ 60%	`HOT`	`1.00 - (((PTI - 0.45) / 0.15) × 0.03)`
PTI > 60%	`EXTREME_HOT`	`max(0.94, 0.97 - ((PTI - 0.60) × 0.05))`

Interpretation

This layer is intentionally light. It is designed to clean up the hottest mortgage affordability edge cases after the A2 amount-to-income smoothing layer has already governed the main exposure tail.

A.21 FINAL APPENDIX A NOTE

The values in Appendix A represent the default V2.0 configuration and estimated-risk reference assumptions used by Module 1. They are included for transparency, auditability, and implementation traceability. They should not be interpreted as production underwriting policy, empirical default calibration, pricing guidance, or institution-specific risk appetite. Any change to these assumptions should be treated as a governed parameter change and reviewed through the QA and validation framework described in Section 10.

APPENDIX B — SQL IMPLEMENTATION GUIDANCE & SECTION MAP

B.1 PURPOSE

This appendix documents the SQL implementation structure for Module 1 V2.0.

The purpose is to make the SQL script reviewable as a governed system artifact, not merely as executable code. Module 1 is intentionally written as a **teaching, audit, and implementation artifact** with extensive commentary, staged transformations, deterministic controls, validation gates, scenario archiving, and embedded QA queries.

This appendix supports:

- implementation transparency;
- reviewer onboarding;
- audit traceability;
- future maintenance;
- parameter-change governance;
- scenario comparison;
- and alignment between the SQL code, BRD, and Validation Summary.

The SQL implementation is expected to remain readable, explainable, and directly traceable to the system design documented in this BRD.

B.2 IMPLEMENTATION PHILOSOPHY

The SQL implementation follows the same design principles as the broader Module 1 system.

Implementation Principle	Meaning
Parameterized over hard-coded	Core assumptions are centralized in the parameter table rather than embedded throughout downstream logic
Deterministic over stochastic	Synthetic randomness is generated through deterministic hash-based pseudo-random values, not session-level `random()`
Staged over monolithic	The build is broken into clear temporary materialization stages rather than one opaque query
Product-aware over generic	Product structure drives amount, term, APR, payment, PTI, LGD, and downstream estimated-risk behavior
Diagnostic over opaque	Intermediate fields are surfaced to explain mortgage dampening, estimated-PD proxy saturation, and scenario movement
Scenario-comparable over isolated	Scenario metadata and archive persistence allow matched row-level comparison
QA-integrated over post-hoc	QA and validation queries are embedded directly in the script as part of the implementation artifact
Boundary-aware over overclaimed	Code commentary explicitly distinguishes simulation outputs from production underwriting, pricing, billing, or calibrated default modeling

This philosophy ensures the code can be reviewed by technical, business, validation, and governance stakeholders.

B.3 SQL OBJECT INVENTORY

The implementation uses a controlled set of schemas, temporary tables, permanent tables, archive tables, and QA queries.

Object / Object Family	Type	Purpose
`credit_decisioning_sim`	Schema	Dedicated namespace for Module 1 outputs
`tmp_module1_params`	Temporary parameter table	Centralized configuration layer for scenario identity, product mix, score mix, bounds, APR, LGD, and payment assumptions
`tmp_m1_stage_1_population`	Temporary stage table	Base synthetic application skeleton and deterministic seed values
`tmp_m1_stage_2_borrower_profile`	Temporary stage table	Borrower characteristics and behavioral risk signals
`tmp_m1_stage_3_product_structuring`	Temporary stage table	Requested amount, APR, term, payment proxy, PTI, and affordability structure
`tmp_m1_stage_4_risk`	Temporary stage table	Estimated-PD proxy, LGD, and Expected Loss inputs

Object / Object Family	Type	Purpose
`credit_decisioning_sim.synthetic_applications`	Permanent latest-run output table	Final application-level synthetic output for the current run
`credit_decisioning_sim.synthetic_applications_scenario_archive`	Permanent scenario archive table	Stores completed scenario runs for comparison
Section 9 QA queries	Review query suite	Validates single-run realism, structure, distribution, workstream behavior, and edge cases
Section 10 scenario comparison queries	Review query suite	Compares archived scenarios at portfolio, segment, and row levels

This object structure supports both latest-run analysis and multi-scenario archive comparison.

B.4 SQL SECTION MAP

The SQL implementation is organized into major sections. Each section has a defined purpose and governance role. This map should remain current as the SQL evolves.

SQL Section	Section Name	Purpose	Primary Outputs / Effects
0	Schema Setup	Creates and sets the dedicated simulation schema	`credit_decisioning_sim` schema and search path
1	Reset / Clean Rerun	Drops latest-run output and temporary stage tables for clean rerun behavior	Removes prior latest-run outputs and intermediate temp objects
2	User Parameters	Defines centralized configuration assumptions	`tmp_module1_params`
3	Validation Gatekeeper	Performs fail-fast parameter validation before data generation	Halts execution if invalid assumptions are detected
4	Base Application Skeleton + Deterministic Seed Values	Creates synthetic application IDs and deterministic uniform values	`tmp_m1_stage_1_population`
5	Product & Score Segment Assignment	Assigns product type and score band using configured mix weights	Product and score-band segmentation
6	Borrower Profile Feature Generation	Generates borrower attributes and behavioral risk signals	`tmp_m1_stage_2_borrower_profile`
7	Product Structuring — Amount, APR, Payment, PTI	Generates requested amount, APR, payment proxy, PTI, and affordability fields	`tmp_m1_stage_3_product_structuring`
8	Risk Modeling — Estimated-PD Proxy, LGD, Expected Loss	Generates estimated-PD proxy, LGD, and expected-loss inputs	`tmp_m1_stage_4_risk`
8.5	Scenario Archive / Run Persistence	Stores latest run in governed scenario archive if enabled	`synthetic_applications_scenario_archive`
9	QA / Review Queries	Provides single-run validation and diagnostic review	Portfolio, product, score, PTI, tail, and workstream QA outputs
10	Scenario Comparison QA	Compares archived scenario outputs	Scenario inventory, deltas, and matched row-level comparison

B.5 SECTION 0 — SCHEMA SETUP

Purpose

Establish a dedicated schema for Module 1 simulation outputs.

Implementation Standard

The script should create and use:

`credit_decisioning_sim`

Governance Rationale

A dedicated schema:

- isolates simulation objects from unrelated database objects;
- avoids naming collisions;
- makes cleanup and review easier;
- supports repeatable execution;
- and gives reviewers a clear namespace for Module 1 artifacts.

Expected Behavior

The script should safely create the schema if it does not already exist and set the session search path to use the simulation schema first.

B.6 SECTION 1 — RESET / CLEAN RERUN

Purpose

Ensure each script execution starts from a clean latest-run state.

Implementation Standard

The reset section should remove:

- the latest-run output table;
- temporary stage tables;
- and any other transient execution objects used by the build.

Expected Reset Targets

Object	Reset Behavior
`credit_decisioning_sim.synthetic_applications`	Dropped and recreated each run
`tmp_m1_stage_1_population`	Dropped if present
`tmp_m1_stage_2_borrower_profile`	Dropped if present
`tmp_m1_stage_3_product_structuring`	Dropped if present
`tmp_m1_stage_4_risk`	Dropped if present

Governance Rationale

The latest-run table should represent only the current completed execution. This prevents silent mixing of outputs across different parameter sets or scenarios.

Scenario history should be preserved through the scenario archive, not through repeated accumulation in the latest-run output table.

B.7 SECTION 2 — USER PARAMETERS

Purpose

Provide a centralized configuration layer for all major simulation assumptions.

Implementation Standard

User-adjustable assumptions should be inserted into:

`tmp_module1_params`

Parameter Families

Parameter Family	Purpose
Scenario archive / comparison controls	Define scenario set, scenario name, archive behavior, and comparison grouping
Run identity / reproducibility	Define `population_id` and related deterministic controls
Population size / time window	Define application count and simulated application period
Product mix	Define product composition
Score mix	Define borrower quality composition
Structural limits / baseline economics	Define score bounds, income bounds, DTI bounds, utilization bounds, amount bounds, base rates, and LGD
Macro / rate environment	Define `rate_shift_bps`
Revolving payment assumptions	Define APR-sensitive revolving payment behavior

Design Standard

Users should first attempt controlled changes through the parameter layer before modifying downstream SQL logic.

Governance Guidance

Parameter changes should be made one family at a time when possible. After each material change, users should rerun Section 9 QA before making additional logic changes.

B.8 SECTION 3 — VALIDATION GATEKEEPER

Purpose

Stop the script before data generation if parameters are invalid, inconsistent, or outside governed bounds.

Implementation Standard

The validation gatekeeper should use fail-fast logic to prevent misleading outputs from being produced under invalid assumptions.

Required Validation Checks

Validation Gate	Purpose
tmp_module1_params contains exactly one row	Ensures each run has one and only one active parameter set
archive_run_flag is not null	Prevents ambiguous archive persistence behavior
scenario_set_name is not null or blank	Prevents unrelated scenario families from being mixed or unlabeled
scenario_name is not null or blank	Ensures each archived run has a usable scenario label
archive_reset_mode is valid	Ensures archive clearing behavior is intentional and governed
Application count > 0	Prevents empty or invalid population generation
Portfolio start date ≤ end date	Ensures valid simulated application window
Product mix sums to 1.00	Ensures valid portfolio composition
Score mix sums to 1.00	Ensures valid borrower quality composition
Min credit score < max credit score	Ensures valid score range
Income floor > 0 and < income cap	Ensures plausible income bounds
DTI lower / upper bounds valid	Prevents impossible DTI values
Utilization lower / upper bounds valid	Prevents impossible utilization values
Product amount min < max	Ensures valid product amount ranges
Product base APRs > 0	Prevents invalid pricing inputs
LGD values between 0 and 1	Ensures valid severity assumptions
Revolving base payment percentage between 0 and 10%	Prevents unrealistic minimum-payment assumptions
Revolving APR passthrough factor between 0 and 1	Prevents APR sensitivity from exceeding modeled interest pressure

Governance Rationale

The validation gatekeeper protects the system from silent parameter failure. A failed run is preferable to a completed run with distorted economics.

V2.0 Governance Hardening

The final V2.0 implementation extends the validation gatekeeper beyond traditional portfolio parameter checks. In addition to validating mix weights, bounds, rates, LGD assumptions, and revolving payment controls, Section 3 now validates scenario archive controls and parameter-table cardinality.

This ensures that each run has exactly one active parameter set and that scenario archive behavior is explicit, governed, and resistant to silent configuration error. This is especially important because Workstream C introduced persistent scenario archive outputs used for matched row-level comparison.

B.9 SECTION 4 — BASE APPLICATION SKELETON & DETERMINISTIC SEED VALUES

Purpose

Create the base synthetic application population and deterministic uniform values used throughout the simulation.

Implementation Standard

The engine should not use session-randomized functions such as `random()` for core generation.

Instead, deterministic pseudo-random values should be generated using stable hash logic tied to:

- application identity;
- `population_id`;
- and deterministic key suffixes.

Design Rationale

This enables:

- identical outputs under identical inputs;
- stable QA results;
- row-level debugging;
- matched scenario comparison;
- and reproducible validation evidence.

Expected Output

This section should produce the first stage table:

`tmp_m1_stage_1_population`

Governance Interpretation

This is the reproducibility foundation of the engine. If this section is not stable, scenario comparison and QA traceability are not reliable.

B.10 SECTION 4/5 — STAGE MATERIALIZATION

Purpose

Materialize the first stage of the pipeline into a temporary table before continuing downstream transformations.

Implementation Standard

V2.0 uses staged temporary tables rather than one very large monolithic query.

Rationale

Staged materialization improves:

- readability;
- maintainability;
- debugging;
- PostgreSQL planner stability;
- performance;
- and reviewer comprehension.

Governance Interpretation

Staging is not merely a performance choice. It is an auditability choice. Each stage can be inspected if outputs behave unexpectedly.

B.11 SECTION 5 — PRODUCT & SCORE SEGMENT ASSIGNMENT

Purpose

Assign product type and score band to each synthetic application using configured product and score mix assumptions.

Implementation Standard

Product and score assignment should be deterministic and driven by configured cumulative mix thresholds.

Expected Behavior

Assignment Area	Expected Behavior
Product type	Reflects configured product weights
Score band	Reflects configured score-band weights
Assignment stability	Remains identical under identical `population_id` and parameters
Scenario comparability	Remains stable when scenarios are intended to compare the same borrower population

Governance Interpretation

This section establishes the foundational portfolio structure. Workstream enhancements should not unintentionally alter product or score distributions unless a scenario explicitly changes those assumptions.

B.12 SECTION 6 — BORROWER PROFILE FEATURE GENERATION

Purpose

Generate borrower-like attributes and behavioral risk signals.

Expected Output

This section materializes:

`tmp_m1_stage_2_borrower_profile`

Generated Feature Families

Feature Family	Examples
Credit quality	credit score, score band
Capacity	income, DTI
Revolving behavior	utilization
Credit history depth	tradeline count, months since oldest trade
Recent activity	inquiry count
Adverse performance signals	delinquency count, major derogatory flag, bankruptcy flag
Customer relationship signals	prequalified flag, returning customer flag

Design Standard

Borrower characteristics should be correlated with score band directionally, but not deterministically.

Governance Interpretation

The system should preserve realistic overlap. Stronger score bands should generally look stronger, but mixed-signal cases must remain possible.

B.13 SECTION 7 — PRODUCT STRUCTURING: AMOUNT, APR, PAYMENT, PTI

Purpose

Translate borrower and product context into requested amount, APR proxy, payment burden, PTI, and affordability structure.

Expected Output

This section materializes:

`tmp_m1_stage_3_product_structuring`

Core Sub-Layers

Sub-Layer	Purpose
Amount and APR base	Establish product-conditioned requested amount and APR anchor
Mortgage tail smoothing	Apply Workstream A2 amount-to-income tail governance
Mortgage pre-floor	Preserve post-A2 mortgage amount before PTI overlay
APR bounding	Apply product-level APR floors and ceilings
Mortgage provisional payment	Calculate provisional mortgage payment for PTI overlay logic
Mortgage provisional PTI	Convert provisional payment into affordability stress
Mortgage PTI overlay	Apply Workstream A3 PTI-aware dampener for hot mortgage cases
Mortgage finalized	Apply final mortgage amount floors / ceilings
Payment and PTI finalization	Generate final payment proxy and PTI

Governance Interpretation

Section 7 is where product structure becomes affordability burden.

This is the operational implementation of the BRD chain:

Product Structure → APR → Payment → PTI

Workstream A Implementation Role

This section implements the V2.0 mortgage realism enhancements:

- A2 amount-to-income tail smoothing;
- A3 PTI-aware overlay;
- mortgage diagnostic fields;
- final requested amount governance.

Workstream D Implementation Role

This section also implements the APR-sensitive revolving payment proxy, restoring:

APR → Payment → PTI

for revolving products.

B.14 SECTION 8 — RISK MODELING: ESTIMATED-PD PROXY, LGD, EXPECTED LOSS

Purpose

Translate borrower characteristics, behavioral risk signals, product structure, and affordability stress into:

- estimated-PD proxy;
- LGD;
- Expected Loss.

Expected Output

This section materializes:

`tmp_m1_stage_4_risk`

Risk Translation Logic

The estimated-PD proxy framework should:

1. assign a base estimated-PD proxy by score band;
2. apply interpretable behavioral and structural multipliers;
3. apply V2.0 soft-saturation logic;
4. apply final floor and cap controls;
5. attach product-level LGD;
6. calculate Expected Loss.

Boundary

`estimated_pd` is a synthetic estimated-PD proxy, not a calibrated production default model.

Workstream B Implementation Role

Section 8 implements Workstream B:

- raw estimated-PD proxy before saturation;
- saturation zone;
- saturation factor;
- estimated-PD proxy after saturation;
- final governed estimated-PD proxy.

Expected Loss Formula

Expected Loss = requested_amount × estimated-PD proxy × LGD

Governance Interpretation

This section is the risk translation layer. It should remain transparent, bounded, and auditable.

B.15 FINAL OUTPUT SHAPE

Purpose

Create the final latest-run output table.

Final Output Table

`credit_decisioning_sim.synthetic_applications`

Required Output Categories

Output Category	Examples
Version and scenario metadata	model version, scenario set, scenario name, population ID, anchor date
Row identity	application sequence, application ID, applicant ID, application date
Product and borrower attributes	product type, score band, credit score, income, DTI, utilization
Behavioral signals	tradelines, file age, inquiries, delinquencies, derogatory, bankruptcy
Product structure	requested amount, term, APR proxy
Mortgage diagnostics	candidate amount, smoothing zone, smoothing factor, provisional payment, provisional PTI, dampener zone, dampener factor
Affordability outputs	monthly payment proxy, PTI, amount-to-income ratio
Estimated-risk components	base estimated-PD proxy, multipliers, raw estimated-PD proxy, saturation zone, saturation factor, final estimated-PD proxy
Severity and loss	LGD, Expected Loss
Audit fields	created timestamp

Indexing Standard

The final output table should include indexes supporting common QA patterns:

Index Purpose	Fields
Product-level QA	`product_type`
Score-band QA	`score_band`
Product × score-band QA	`product_type, score_band`

Governance Interpretation

The final output table should expose enough detail to support both executive summaries and row-level diagnostic review.

B.16 SECTION 8.5 — SCENARIO ARCHIVE / RUN PERSISTENCE

Purpose

Preserve completed runs so users can compare multiple scenarios after execution.

Archive Table

`credit_decisioning_sim.synthetic_applications_scenario_archive`

Archive Design

The archive table should mirror the latest-run output structure and add an archive timestamp.

Archive Controls

Control	Purpose
`archive_run_flag`	Determines whether the current run is archived
`archive_reset_mode = NONE`	Preserves existing archive rows
`archive_reset_mode = SCENARIO_ONLY`	Replaces only the current scenario within the scenario set and population
`archive_reset_mode = SET_ONLY`	Clears all scenarios in the current scenario set for the current population
`archive_reset_mode = ALL`	Clears the full archive table

Recommended Default

For ordinary reruns of a scenario:

`SCENARIO_ONLY`

Scenario Archive Indexes

Index Purpose	Fields
Scenario inventory and retrieval	scenario set, scenario name, population ID
Matched row comparison	scenario set, population ID, application ID
Segment scenario comparison	scenario set, scenario name, population ID, product type, score band

Governance Interpretation

The latest-run table is intentionally simple and overwritten each run. The scenario archive is the governed history layer.

B.17 SECTION 9 — SINGLE-RUN QA / REVIEW QUERIES

Purpose

Provide the embedded QA framework for validating one completed scenario output.

Core Principle

A successful SQL run is not sufficient. Section 9 exists to prove that the output behaves plausibly.

Section 9 QA Map

QA Section	QA Name	Purpose
9.1	Record Count	Confirms expected row count
9.2	Product Mix	Confirms product assignment aligns with configured weights
9.3	Score Band Distribution	Confirms score-band population structure
9.4	Variable Profile Statistics	Reviews central tendency, spread, and tails
9.5	Score Band Profile Summary	Confirms borrower and estimated-risk gradients
9.6	Product Profile Summary	Confirms product-level differentiation
9.7	PTI Bucket Summary	Confirms affordability-to-estimated-risk relationship
9.8	Product × Score Band Matrix	Confirms interaction effects
9.9	Mixed-Signal Spot Checks	Confirms realistic overlap and edge-case plausibility
9.10	Distribution Sanity by Product and Score Band	Provides quick cross-sectional realism check
9.11	Optional Quick Profile Sample	Supports row-level inspection
9.12	Estimated-PD Proxy Saturation Diagnostics	Validates Workstream B cap-crowding behavior
9.13	Mortgage A3 Overlay Diagnostics	Validates Workstream A PTI-aware dampening
9.14	Mortgage Transformation Trace	Shows mortgage amount evolution through A2 and A3
9.15	Workstream D Revolving Payment vs APR Relationship	Validates revolving APR-to-payment sensitivity
9.16	Revolving Payment by APR Bucket	Checks directional payment behavior across APR buckets
9.17	PTI Tail Frequency Check	Quantifies high-PTI prevalence

Governance Interpretation

Section 9 should be reviewed after any material parameter or logic change.

It answers:

Did the generated portfolio behave as intended?

B.18 SECTION 10 — SCENARIO COMPARISON QA

Purpose

Compare archived scenarios within a scenario set.

Core Principle

Section 9 validates one scenario. Section 10 compares multiple archived scenarios.

Section 10 QA Map

QA Section	QA Name	Purpose
10.1	Scenario Archive Inventory	Lists archived scenarios and row counts
10.2	Portfolio-Level Scenario Summary	Compares portfolio metrics across scenarios
10.3	Product × Score Scenario Summary	Compares scenario movement by product and score band
10.4	Matched Row-Level Scenario Comparison	Defines matched-row comparison concept
10.4A	Matched Row-Level Delta Sample	Shows row-level baseline versus scenario deltas
10.4B	Matched Row-Level Delta Summary	Summarizes how many rows changed and by how much
10.4C	Matched Row-Level Delta Summary by Product and Score Band	Identifies which segments drove scenario movement

Scenario Alias Boundary

Compact aliases such as `avg_pd`, `delta_pd`, and `rows_pd_increased` may be retained for readable QA output naming, but they refer to the **synthetic estimated-PD proxy**, not calibrated production default probabilities.

Governance Interpretation

Section 10 answers:

What changed between scenarios, where did it change, and can those changes be attributed to controlled assumptions rather than population drift?

B.19 STAGED MATERIALIZATION STANDARD

The V2.0 implementation intentionally uses staged temporary tables.

Stage Table	Role	Why It Matters
`tmp_m1_stage_1_population`	Application skeleton and deterministic seeds	Reproducibility foundation
`tmp_m1_stage_2_borrower_profile`	Borrower characteristics and behavioral risk signals	Enables borrower realism review
`tmp_m1_stage_3_product_structuring`	Amount, APR, payment, PTI, affordability	Translates product structure into burden
`tmp_m1_stage_4_risk`	Estimated-PD proxy, LGD, Expected Loss	Final risk and loss translation

Implementation Rationale

Staged materialization improves:

- performance;
- readability;
- error isolation;
- reviewer comprehension;
- auditability;
- and future extensibility.

Governance Rule

Future enhancements should preserve staged materialization unless there is a compelling performance or design reason to change it.

B.20 SQL COMMENTING & DOCUMENTATION STANDARDS

The SQL script should remain heavily commented because it serves as both implementation and teaching artifact.

Required Comment Types

Comment Type	Purpose
Executive purpose	Defines what the script is and is not
Terminology notes	Clarifies estimated-PD proxy and other bounded concepts
Parameter guidance	Explains when and why to change assumptions
Section purpose comments	Explains each implementation block
Teaching notes	Clarifies SQL patterns and modeling logic
Boundary notes	Prevents production-model or forecast misinterpretation
Workstream notes	Identifies V2.0 enhancements and diagnostics
QA interpretation notes	Explains what each QA output means

Commenting Standard

Comments should explain:

- business purpose;
- modeling rationale;
- implementation logic;
- expected behavior;
- interpretation boundary;
- and QA relevance.

Governance Interpretation

The code should be readable by a reviewer who understands credit strategy but does not want to reverse-engineer SQL logic line by line.

B.21 NAMING CONVENTIONS

Consistent naming improves readability and auditability.

Naming Area	Standard
Final output table	`credit_decisioning_sim.synthetic_applications`
Scenario archive table	`credit_decisioning_sim.synthetic_applications_scenario_archive`
Parameter table	`tmp_module1_params`
Stage tables	`tmp_m1_stage_[n]_[description]`
Mortgage V2 diagnostic fields	suffix _v2
Estimated-PD proxy diagnostics	prefix / stem pd_ retained for compactness, interpreted as estimated-PD proxy
Scenario controls	`scenario_set_name`, `scenario_name`, `population_id`
QA aliases	Compact aliases may be used if interpretation is documented

Estimated-PD Proxy Naming Boundary

Implementation fields may retain compact names such as:

- `base_pd`
- `estimated_pd`
- `pd_raw_pre_saturation_v2`
- `pd_saturation_zone_v2`
- `pd_saturation_factor_v2`
- `pd_after_saturation_v2`

These names are implementation shorthand. They should be interpreted as part of the **synthetic estimated-PD proxy framework**, not as calibrated production PD fields.

B.22 EXECUTION WORKFLOW

The recommended execution workflow is:

1. **Review the parameter block in Section 2.**
2. **Modify only controlled parameter values where possible.**
3. **Set or confirm `scenario_set_name`, `scenario_name`, and `population_id`.**
4. **Run the script end to end.**
5. **Confirm validation gatekeepers pass.**
6. **Review the final output table.**
7. **Run and review Section 9 QA queries.**
8. **If scenario comparison is required, ensure the run is archived.**
9. **Run and review Section 10 scenario comparison queries.**
10. **Document any unexpected behavior.**
11. **Determine whether behavior reflects expected parameter movement, scenario effect, or a true logic concern.**

Best Practice

Change one parameter family at a time when testing. This makes scenario effects easier to attribute.

B.23 SCENARIO EXECUTION WORKFLOW

A governed scenario comparison should follow this process:

Step	Action
1	Run baseline with `archive_run_flag = TRUE`
2	Use a clear `scenario_set_name`
3	Use `scenario_name = BASELINE_V2` or equivalent
4	Preserve `population_id`
5	Modify scenario parameter, such as `rate_shift_bps`
6	Change `scenario_name` to reflect the new scenario
7	Rerun and archive scenario
8	Review Section 10 archive inventory
9	Review portfolio-level scenario summary
10	Review product × score scenario summary
11	Review matched row-level deltas
12	Interpret differences as parameter-driven only if population is matched

Scenario Interpretation Boundary

Scenario outputs are controlled simulation results, not forecasts.

B.24 QA OUTPUT CONSUMPTION STANDARD

QA outputs should be used as validation evidence.

QA Output Type	Consumption Purpose
Record count	Confirms run completeness
Product mix	Confirms configured portfolio structure
Score mix	Confirms borrower quality distribution
Variable profile statistics	Confirms central tendency and tail behavior
Product summaries	Confirms product differentiation
Score-band summaries	Confirms risk ordering
PTI summaries	Confirms affordability-to-risk translation
Mixed-signal outputs	Confirms realistic overlap
Saturation diagnostics	Confirms estimated-PD proxy cap governance
Mortgage diagnostics	Confirms Workstream A behavior
Revolving diagnostics	Confirms Workstream D behavior
Tail-frequency output	Confirms high-PTI rows remain present but non-dominant
Scenario deltas	Confirms matched parameter-driven movement

Governance Interpretation

QA outputs should not be treated as decorative afterthoughts. They are the evidence layer that supports the Validation Summary.

B.25 IMPLEMENTATION ACCEPTANCE CRITERIA

The SQL implementation should be considered acceptable when it satisfies the following conditions.

Acceptance Area	Acceptance Requirement
Schema isolation	Outputs are created in the dedicated simulation schema
Clean rerun	Latest-run outputs are reset and regenerated cleanly
Parameter centralization	Key assumptions are controlled through `tmp_module1_params`
Gatekeeper validation	Invalid assumptions fail fast before generation
Deterministic generation	Identical inputs reproduce identical outputs
Staged transformations	Major logic layers are materialized and inspectable
Product-aware behavior	Product structure influences amount, APR, payment, PTI, LGD, and loss
Estimated-PD proxy transparency	Risk proxy logic is explainable and bounded
Scenario archiving	Completed runs can be archived for comparison
QA integration	Section 9 review queries are available and interpretable
Scenario comparison	Section 10 supports matched row-level attribution
Boundary language	Comments prevent production-model overinterpretation

B.26 MAINTENANCE & CHANGE-CONTROL GUIDANCE

Future changes should follow a disciplined change-control process.

Before Changing Code

The developer should identify:

- business objective;
- parameter versus logic change;
- affected section;
- expected output movement;
- required QA checks;
- scenario comparison needs;
- and documentation updates.

After Changing Code

The developer should:

- rerun the full script;
- confirm gatekeepers pass;
- review Section 9 QA;
- review Section 10 if scenarios are affected;
- compare against prior baseline if applicable;
- document observed differences;
- update BRD sections if design intent changes;
- update Validation Summary if validation evidence changes.

Governance Principle

Do not change downstream transformation logic casually when the desired behavior can be achieved through governed parameters.

B.27 FUTURE IMPLEMENTATION EXTENSION STANDARDS

Future modules should preserve the architectural discipline established in Module 1.

Potential future modules may include:

- decision strategy logic;
- approval / decline / review simulation;
- exposure assignment;
- pricing strategy;
- enhanced LGD;
- macro overlays;
- lifecycle simulation;
- or reporting dashboards.

B.27 FUTURE IMPLEMENTATION EXTENSION STANDARDS - CONTINUED

Each future module should include:

- purpose header;
- parameter layer;
- validation gatekeepers;
- deterministic controls where applicable;
- staged transformation logic;
- QA section;
- output interpretation boundaries;
- and validation artifact linkage.

Design Principle

Future enhancements should extend the architecture without weakening reproducibility, auditability, or boundary discipline.

B.28 RELATIONSHIP TO BRD & VALIDATION SUMMARY

This appendix connects the implementation artifact to the broader documentation set.

Artifact	Role
BRD	Defines design intent, architecture, parameters, outputs, governance, and boundaries
SQL Implementation	Executes the governed synthetic application and risk modeling engine
Validation Summary	Demonstrates that the implementation behaves as intended through QA evidence

Together, these artifacts create a complete system package:

Design → Implementation → Validation

This structure is important because the system is not only a coding exercise. It is a governed, reviewable, and scenario-capable simulation framework.

B.29 FINAL IMPLEMENTATION GUIDANCE

The SQL implementation should remain:

- readable;
- parameterized;
- deterministic;
- staged;
- auditable;
- scenario-aware;
- QA-integrated;
- and boundary-aware.

The script should be maintained as a **portfolio-grade implementation artifact**. It should demonstrate not only that the system can run, but that the system can be understood, reviewed, validated, and governed.

In short:

The SQL implementation is the executable expression of the BRD. Its structure must remain as transparent and governed as the simulation logic it produces.

APPENDIX C — REQUIREMENTS TRACEABILITY MATRIX

C.1 PURPOSE

This appendix provides a concise requirements traceability matrix for Module 1 V2.0.

The matrix connects:

- business and design requirements;
- BRD design sections;
- SQL implementation / QA anchors;
- validation evidence;
- and final requirement status.

The purpose is to confirm that Module 1 V2.0 is traceable from:

requirement → design → implementation → validation

This appendix is intentionally concise. Detailed field behavior is documented in Section 7, detailed implementation guidance is documented in Appendix B, and detailed validation evidence is documented in the companion Validation Summary.

C.2 TRACEABILITY STATUS DEFINITIONS

Status	Meaning
Satisfied	Requirement is implemented and validated within V2.0 scope
Boundary-Controlled	Requirement is intentionally limited by documented scope boundaries
Future Scope	Requirement is intentionally deferred to a future module or enhancement
Not Applicable	Requirement is outside Module 1 design intent

C.3 FUNCTIONAL REQUIREMENTS TRACEABILITY

ID	Requirement	BRD Anchor	SQL / QA Anchor	Validation Acceptance Signal	Status
FR-01	Generate deterministic synthetic application-level population	BRD Sections 3, 5, 7, and 12	SQL Sections 4–5; QA 9.1	50,000 synthetic applications generated and validated	Satisfied
FR-02	Preserve deterministic reproducibility under identical inputs	BRD Sections 3.1, 5.2, 7B.2, and 11	SQL deterministic seed framework; reproducibility QA	100% identical outputs across repeated runs	Satisfied
FR-03	Assign product type using governed product mix	BRD Sections 6, 7A.1, and 11.5	SQL Section 5; QA 9.2	Observed product mix aligned within approximately 0.2 percentage points	Satisfied
FR-04	Assign score bands using governed score mix	BRD Sections 7A.2 and 11.6	SQL Section 5; QA 9.3	Score bands fully populated and stable across V2.0 runs	Satisfied
FR-05	Generate borrower profile and behavioral attributes	BRD Sections 7A.2–7A.11	SQL Section 6; QA 9.5 and 9.9	Score-band gradients and mixed-signal behavior validated	Satisfied

ID	Requirement	BRD Anchor	SQL / QA Anchor	Validation Acceptance Signal	Status
FR-06	Generate product-specific requested amount	BRD Sections 7A.12, 8, and 11.8	SQL Section 7; QA 9.6 and 9.14	Mortgage tail controlled; max requested amount reduced from approximately \$1.15M to \$1.05M	Satisfied
FR-07	Generate APR proxy for payment calculation	BRD Sections 7A.14, 8, and 11.9	SQL Section 7; scenario QA 10.2	Rate-up scenario moved average APR from approximately 14.56% to 16.14%	Satisfied
FR-08	Generate product-specific monthly payment proxy	BRD Sections 7A.15, 8, and 11.10	SQL Section 7; QA 9.6, 9.15, and 9.16	Rate-up scenario moved average payment from approximately \$1,511 to \$1,612; revolving deltas became positive	Satisfied
FR-09	Calculate PTI as primary affordability bridge	BRD Sections 7A.16 and 8	SQL Section 7; QA 9.7 and 9.17	PTI bucket relationship validated; PTI greater than 50% reduced from 1.77% to 1.20%	Satisfied
FR-10	Generate estimated-PD proxy using transparent bounded framework	BRD Sections 7A.18, 8.8, 8.9, and 11.11	SQL Section 8; QA 9.5 and 9.12	Estimated-PD proxy monotonic by score band; cap crowding reduced from 17.72% to 12.86%	Satisfied
FR-11	Generate LGD using product-level severity assumptions	BRD Sections 7A.19, 8.10, and 11.12	SQL Section 8; QA 9.6	LGD varies by product and supports secured versus unsecured severity contrast	Satisfied
FR-12	Calculate Expected Loss	BRD Sections 7A.20, 8.10, and 12	SQL Section 8; QA 9.6 and 9.7	Expected Loss responds to exposure × estimated-PD proxy × LGD	Satisfied
FR-13	Support scenario metadata and archive persistence	BRD Sections 7B.1, 9, and 12	SQL Section 8.5; QA 10.1	Baseline and rate-stress scenarios archived with 50,000 rows each	Satisfied
FR-14	Support matched row-level scenario comparison	BRD Sections 5.3, 7B.2, and 9.11	SQL Section 10; QA 10.4	Same `population_id`; 100% identical borrower-level attributes across scenarios	Satisfied
FR-15	Surface diagnostic fields for validation and auditability	BRD Sections 7B.3, 10.10, and 12.7	SQL Sections 7–8; QA 9.12–9.17	Mortgage, saturation, revolving, and tail diagnostics used in validation	Satisfied
FR-16	Preserve mixed-signal borrower realism	BRD Sections 3.4, 7, and 10.8	QA 9.9; Validation Summary Section 10	Strong-credit/high-PTI, weaker-credit/manageable-PTI, and large-exposure/manageable-affordability cases validated	Satisfied

C.4 NON-FUNCTIONAL & GOVERNANCE REQUIREMENTS TRACEABILITY

ID	Requirement	BRD Anchor	SQL / QA Anchor	Validation Acceptance Signal	Status
NFR-01	Deterministic reproducibility	BRD Sections 3.1, 5.2, and 11.4	SQL deterministic hash design	100% identical outputs under identical inputs	Satisfied
NFR-02	Auditability and traceability	BRD Sections 3.7, 7B, 10, and 12	Diagnostic fields; QA Sections 9–10	Outputs explainable through parameters, diagnostics, and QA evidence	Satisfied
NFR-03	Parameter governance	BRD Sections 3.2 and 11	SQL parameter table; validation gatekeepers	Product mix, score mix, bounds, rate shift, LGD, and caps centrally controlled	Satisfied
NFR-04	Separation of concerns	BRD Sections 3.5, 5, and 8	SQL staged transformation architecture	Population, product, borrower, payment, risk, and loss layers distinct	Satisfied
NFR-05	Scenario comparability	BRD Sections 3.6, 5.3, and 9	Scenario archive; QA Section 10	50,000 matched rows per scenario; same `population_id`	Satisfied
NFR-06	Tail governance	BRD Sections 8.7, 10.7, and 11.8	QA 9.17; mortgage diagnostics	PTI greater than 40%, 50%, and 75% measured; severe PTI tail reduced	Satisfied
NFR-07	Estimated-risk dispersion governance	BRD Sections 7A.18, 8.9, and 11.11	QA 9.12	Capped estimated-PD proxy observations reduced from 17.72% to 12.86%	Satisfied
NFR-08	Product-aware modeling	BRD Sections 6 and 8.4	QA 9.6	Products show distinct exposure, APR, payment, PTI, LGD, and Expected Loss profiles	Satisfied
NFR-09	Maintainability and reviewability	BRD Section 10 and Appendix B	Staged SQL sections; embedded comments	SQL structured as documented implementation artifact	Satisfied

C.5 BOUNDARY REQUIREMENTS TRACEABILITY

ID	Boundary Requirement	BRD Anchor	Validation / Governance Evidence	Status
BND-01	System must not expose or rely on real customer data	BRD Sections 1, 4, and 13.10	Synthetic-only application population	Boundary-Controlled
BND-02	System must not be used for production underwriting	BRD Sections 4.2, 13.2, and 13.3	Explicit non-use boundary in BRD and Validation Summary	Boundary-Controlled
BND-03	`estimated_pd` must be interpreted as an estimated-PD proxy, not calibrated PD	BRD Sections 7A.18, 8.8, and 13.4	Validation emphasizes directional coherence, not calibration	Boundary-Controlled
BND-04	APR must be interpreted as burden-estimation proxy, not production pricing	BRD Sections 7A.14, 11.9, and 13.7	APR supports payment/PTI modeling and rate sensitivity only	Boundary-Controlled
BND-05	Revolving payment must remain a proxy, not a billing engine	BRD Sections 8.5, 11.10, and 13.8	Workstream D restores APR sensitivity while excluding servicing mechanics	Boundary-Controlled
BND-06	Expected Loss must be interpreted as comparative simulation output	BRD Sections 8.10, 12.5, and 13.6	Expected Loss used for comparison, not reserves, capital, or forecasting	Boundary-Controlled
BND-07	Scenario outputs must be interpreted as sensitivity analysis, not forecasts	BRD Sections 9 and 13.9	Scenario framework uses matched synthetic populations and controlled assumptions	Boundary-Controlled
BND-08	Fair lending and regulatory validation are out of Module 1 scope	BRD Sections 4.2 and 13.11	Future separate governance required for compliance analysis	Boundary-Controlled

C.6 FUTURE-SCOPE REQUIREMENTS

ID	Future Capability	Current V2.0 Position	BRD Roadmap Anchor	Status
FUT-01	Approval / decline / review strategy layer	Not included in Module 1	BRD Section 14.4	Future Scope
FUT-02	Approved amount / line assignment	Requested amount generated; approval amount not assigned	BRD Section 14.5	Future Scope
FUT-03	Production pricing or profitability strategy	APR is payment-burden proxy only	BRD Section 14.6	Future Scope
FUT-04	Row-level LGD enhancement	LGD is product-level in V2.0	BRD Section 14.7	Future Scope
FUT-05	Expanded macro-overlays	Rate shift only in V2.0	BRD Section 14.8	Future Scope
FUT-06	Account lifecycle / performance simulation	Application-level only in V2.0	BRD Section 14.10	Future Scope
FUT-07	Empirical calibration / benchmarking	Directional proxy validation only	BRD Section 14.11	Future Scope
FUT-08	Automated QA dashboard / reporting layer	SQL QA outputs and documents currently support review	BRD Section 14.12	Future Scope

C.7 TRACEABILITY CONCLUSION

The requirements traceability matrix confirms that Module 1 V2.0 requirements are covered across the design, implementation, validation, and governance artifacts.

The system demonstrates traceability across four levels:

Traceability Layer	Confirmation
Design	Requirements are defined in the BRD
Implementation	Requirements are implemented through staged SQL logic and parameter controls
Validation	Requirements are tested through QA outputs and the companion Validation Summary
Governance	Boundaries and future-scope items are explicitly documented

Final traceability conclusion:

Module 1 V2.0 requirements are satisfied within the documented system boundary. Future capabilities are clearly separated from current validated functionality.

APPENDIX D — V1.0 to V2.0 DESIGN EVOLUTION SUMMARY

D.1 PURPOSE

This appendix summarizes how Module 1 evolved from V1.0 to V2.0.

The purpose is to show that V2.0 is a direct, governed evolution of the original Module 1 design—not a separate or unrelated redesign.

V1.0 established the core synthetic application and risk modeling foundation. V2.0 preserves that foundation while strengthening realism, scenario comparability, diagnostic transparency, and governance.

D.2 DESIGN EVOLUTION SUMMARY

Design Area	V1.0 Foundation	V2.0 Enhancement	Design Impact
System positioning	Synthetic application and risk modeling engine for pre-production strategy testing	Reframed as a governed, scenario-capable decisioning simulation framework	Moves the system from static generation toward controlled experimentation
Core transmission chain	Application → Product Structure → APR → Payment → PTI → PD → Expected Loss	Application → Product Structure → APR → Payment → PTI → estimated-PD proxy → Expected Loss	Preserves structure-to-risk logic while clarifying proxy-model boundaries
Deterministic reproducibility	Deterministic generation anchored to population controls	Elevated through `population_id`, scenario labels, archive persistence, and matched row comparison	Enables true apples-to-apples scenario testing
Product-aware design	Product type drives amount, APR, payment, PTI, LGD, and risk treatment	Product behavior further refined through mortgage realism and revolving sensitivity workstreams	Strengthens product-level realism and downstream risk differentiation

Design Area	V1.0 Foundation	V2.0 Enhancement	Design Impact
Mortgage behavior	Mortgage treated as large-balance secured installment product	Workstream A adds tail smoothing and PTI-aware affordability overlay	Reduces unrealistic upper-tail behavior without collapsing the core mortgage distribution
Revolving behavior	Revolving payment represented through simplified minimum-payment proxy	Workstream D adds APR-sensitive revolving payment behavior	Restores APR → Payment → PTI → estimated-risk transmission
Risk framework	Transparent multiplier-based PD proxy framework	Workstream B formalizes estimated-PD proxy terminology and adds soft-saturation logic	Reduces cap crowding while preserving directional score-band ordering
Scenario capability	Primarily single-run simulation with deterministic reproducibility	Workstream C introduces `scenario_set_name`, `scenario_name`, archive persistence, and matched row-level comparison	Enables controlled scenario attribution and repeatable strategy testing
QA framework	Core QA checks for row count, product mix, score distribution, PTI, PD, Expected Loss, and edge cases	Expanded QA framework with workstream diagnostics, tail-frequency validation, saturation diagnostics, scenario QA, and matched comparison	Converts validation from static checks into governed evidence
Tail behavior	Outlier realism acknowledged and reviewed through distribution / spot checks	Tail-frequency validation explicitly measures PTI >40%, >50%, and >75%	Quantifies whether tails remain present but non-dominant
Model boundaries	System identified as synthetic and non-production	Boundaries expanded for estimated-PD proxy, APR proxy, revolving proxy, Expected Loss, scenario outputs, and non-use cases	Strengthens auditability and prevents overclaiming
Implementation documentation	SQL guidance included for parameterization, QA, and structured implementation	V2.0 adds formal appendices for configuration values, SQL section map, traceability, and design evolution	Makes the BRD more implementation-ready and reviewable

D.3 WORKSTREAM EVOLUTION SUMMARY

Workstream	Initial Design Gap	V2.0 Design Response	Final Design Outcome
A — Mortgage Realism	Mortgage tail behavior could create excessive affordability stress in upper-balance cases	Added amount-to-income tail smoothing and PTI-aware overlay	Mortgage tail governed while median mortgage behavior remains stable
B — Estimated-PD Proxy Dispersion	High-risk observations could cluster mechanically near the governance cap	Added soft-saturation logic before final cap application	Reduced cap crowding and improved high-risk differentiation
C — Scenario Framework	Deterministic runs were reproducible, but scenario comparison was not yet formally archived or matched	Added scenario set/name, archive persistence, population anchoring, and row-level comparison	Scenario deltas become attributable to parameters rather than population noise
D — Revolving Sensitivity	Revolving payment did not sufficiently respond to APR movement	Added APR-sensitive revolving payment proxy	Restored rate-to-affordability transmission while preserving billing-model boundary

D.4 WHAT V2.0 PRESERVES FROM V1.0

V2.0 preserves the core V1.0 design foundation:

- synthetic application-level generation;
- multi-product portfolio construction;
- score-band-driven borrower segmentation;
- product-conditioned requested amount, APR, payment, PTI, LGD, and Expected Loss;
- deterministic reproducibility;
- transparent estimated-risk logic;
- mixed-signal borrower realism;
- pre-production strategy-testing scope;
- and clear separation from production underwriting.

V2.0 does not discard the V1.0 architecture. It matures it.

D.5 WHAT V2.0 ADDS BEYOND V1.0

V2.0 adds four major design capabilities:

1. **Greater realism**
   Mortgage exposure and affordability tails are governed more effectively.
2. **Greater differentiation**
   Estimated-PD proxy behavior is less compressed near the upper governance cap.
3. **Greater comparability**
   Scenarios can be archived and compared using matched synthetic populations.
4. **Greater sensitivity**
   Revolving products now transmit APR movement into payment burden and PTI.

Together, these additions transform Module 1 from a strong synthetic application engine into a more complete scenario-capable decisioning simulation framework.

D.6 FINAL EVOLUTION STATEMENT

Module 1 V2.0 is a structured evolution of V1.0.

V1.0 established the foundational simulation architecture.

V2.0 strengthens that architecture through targeted realism, dispersion, scenario, and sensitivity enhancements.

Final design conclusion:

V2.0 preserves the validated V1.0 foundation while adding the governance, diagnostics, and scenario-comparison capabilities required for a portfolio-grade synthetic credit strategy simulator.